FOUR YEARS UNDERGRADUATE PROGRAMME (FYUGP) IN ECONOMICS
(under the Framework of Honours School System)

B.A. (Hons.)/B.A.(Hons.)with Research in Economics Under the Framework of Honours School System

1st to 8th Semester for the Examination 2026-27

Department of Economics
Panjab University Campus, Chandigarh-160014.

Template of FYUGP Programme in Economics
Under the Framework of Honours School System

Semester-I			
Course	**Course Code**	**Paper**	**Credits**
Economics Core Course 1	ECO (HS) DSC 1	Introductory Microeconomics	4
Economics Core Course 2	ECO (HS) DSC 2	Mathematical Methods for Economics-I	4
Skill Enhancement Course 1	ECO (HS) SEC 1	Data Collection and Handling Techniques	3
Multidisciplinary Course 1	MDSC 1	Fundamentals of Sociology	3
Ability Enhancement Compulsory Course 1	AECC 1	English	2
Value Added Course 1	ECO (HS) VAC 1	Indian Economic Thought	2
Value Added Course 2	ECO (HS) VAC 2	Introduction to data Analysis Using Excel-I	2
MDC 1: Mathematics Course	MAT-DSC-101	Calculus	4
Total Credits			**24**
Semester-II			
Economics Core Course 3	ECO (HS) DSC 3	Introductory Macroeconomics	4
Economics Core Course 4	ECO (HS) DSC 4	Mathematical Methods for Economics-II	4
Skill Enhancement Course 2	ECO (HS) SEC 2	Data Analysis for Beginners	3
Multidisciplinary Course 2	PUB MDSC 2	Basics of Indian Administration	3
Ability Enhancement Compulsory Course 2	AECC 2	Course offered by Department of English	2
Value Added Course 3	ECO (HS) VAC 3	Introduction to Environmental Economics	2
Value Added Course 4	ECO (HS) VAC 4	Introduction to data Analysis Using Excel-II	2
MDC 2: Mathematics Course 2	MAT-DSC 151	Algebra and Ordinary Differential Equations	4
Total Credits			**24**

Semester-III			
Economics Core Course 5	ECO (HS) DSC 5	Intermediate Microeconomics-I	4
Economics Core Course 6	ECO (HS) DSC 6	Introductory Statistical Methods for Economic Analysis	4
Economics Core Course 7	ECO (HS) DSC 7	History of Economic Thought	4
Skill Enhancement Course 3	ECO (HS) -SEC-3	Working with Open-Source Software (Python and R)	3
Multidisciplinary Course 3	MDSC 3	Social Stratification	3
Ability Enhancement Compulsory Course 3	AECC 3	Course offered by Department of Punjabi/Hindi	2
MDC 3: Mathematics Course 3	MAT-M-3	Partial Differential Equations and Numerical Analysis	4
Total Credits			**24**
Semester-IV			
Economics Core Course 8	ECO (HS) DSC 8	Intermediate Microeconomics- II	4
Economics Core Course 9	ECO (HS) DSC 9	Intermediate Macroeconomics	4
Economics Core Course 10	ECO (HS) DSC 10	Introductory Econometrics	4
Economics Core Course 11	ECO (HS) DSC 11	Growth Models	4
Ability Enhancement Compulsory Course 4	AECC 4	Course offered by Punjabi/Hindi Department	2
MDC 4: Statistics Course 4	STAT-M-162	Introductory Probability	4
Internship/CW/FP		Internship/CW/FP	2
Total Credits			**24**

Semester-V			
Economics Core Course 1	ECO (HS) DSC 12	Intermediate Econometrics	4
Economics Core Course 2	ECO (HS) DSC 13	International Economics	4
Economics Core Course 3	ECO (HS) DSC 14	Indian Economy-I	4
Economics Core Course 4	ECO (HS) DSC 15	Introductory Public Finance	4
Minor Disciplinary Course 5	MDC5	Statistics- STAT-M APPLIED STATISTICS Or Mathematics-Offered by Dept. of Mathematics	4
Value Added Course 5	ECO (HS) VAC 5 (a) Or ECO (HS) VAC 5 (b)	Financial Literacy Or Introduction to Research	2
Internship/CW/FP		Internship/CW/FP	2
Total Credits			**24**
Semester-VI			
EconomicsCoreCourse1	ECO (HS) DSC 16	Economics of Development & Planning	4
Economics CoreCourse 2	ECO (HS) DSC 17	Indian Economy-II	4
EconomicsCoreCourse3	ECO (HS) DSC 18	Introductory Agriculture Economics	4
EconomicsCoreCourse4	ECO (HS) DSC 19	Introductory Industrial Economics	4
EconomicsCore Course 5	ECO (HS) DSC 20:	Basics of Financial Markets	4
Minor DisciplinaryCourse6	MDC6	Statistics- STAT-M BASICS OF STATISTICS INFERENCE Or Mathematics: Offered by Dept. of Mathematics	4
Total Credits			**24**

Semester-VII			
For a degree of B.A. (Hons.) under the framework of Hons. School System			
EconomicsCoreCourse1	ECO (HS) DSC 21	International Trade and Trading Systems	4
Economics CoreCourse 2	ECO (HS) DSC 22	Fundamental Issues in Political Economy	4
EconomicsCoreCourse3	ECO (HS) DSC 23	Optimisation Techniques and Game Theory	4
EconomicsResearch Based CoreCourse1	ECO (HS)-RBCRM(A)	Research Methodology	4
Economics Discipline Specific Elective Course	ECO (HS) DSE (A) **or** ECO (HS) DSE (B)	Economics of Population **or** Labour Economics	4
Minor DisciplinaryCourse7	MDC7	Statistics- Offered by Dept. of Statistics Or Mathematics: Offered by Dept. of Mathematics	4
Total Credits			**24**
For a degree of B.A. (Hons.) with Research under the framework of Hons. School System			
EconomicsCoreCourse1	ECO (HS) DSC 21	International Trade and Trading Systems	4
Economics CoreCourse 2	ECO (HS) DSC 22	Fundamental Issues in Political Economy	4
EconomicsCoreCourse3	ECO (HS) DSC 23	Optimisation Techniques and Game Theory	4
EconomicsResearch Based Core Course1	ECO (HS)-RBCRM(A)	Research Methodology	4
Dissertation Course	ECO (HS)-RBCDST(A)	Dissertation	4
Minor DisciplinaryCourse7	MDC7	Statistics- Offered by Dept. of Statistics Or Mathematics: Offered by Dept. of Mathematics	4
Total Credits			**24**

Semester-VIII			
For a degree of B.A. (Hons.) under the framework of Hons. School System			
EconomicsCoreCourse1	ECO (HS) DSC 24	Advanced Public Finance	4
Economics CoreCourse 2	ECO (HS) DSC 25 ECO (HS) DSE (E)	Environmental Economics	4
EconomicsCoreCourse3	ECO (HS) DSC 26 ECO (HS) DSE (F)	National Income Accounting	4
Economics Research Based CoreCourse1	ECO (HS)-RBCRM(B)	Research Publication and Ethics	4
Economics Discipline Specific Elective Course	ECO (HS) DSE (C) **or** ECO (HS) DSE (D)	Economics of Money and Banking **or** Health Economics	4
Minor DisciplinaryCourse8	MDC8	Statistics- Offered by Dept. of Statistics Or Mathematics: Offered by Dept. of Mathematics	4
Total Credits			**24**
For a degree of B.A. (Hons.) with Research under the framework of Hons. School System			
EconomicsCoreCourse1	ECO (HS) DSC 24	Advanced Public Finance	4
Economics Research Based CoreCourse1	ECO (HS)-RBCRM(B)	Research Publication and Ethics	4
Economics Discipline Specific Elective Course	ECO (HS) DSE (C) **or** ECO (HS) DSE (D) **or** ECO (HS) DSE (E) **or** ECO (HS) DSE (F)	Economics of Money and Banking **or** Health Economics **or** Environment Economics **or** National Income Accounting	4
Dissertation Course	ECO (HS)-RBCDST(B)	Dissertation	8
Minor DisciplinaryCourse8	MDC8	Statistics- Offered by Dept. of Statistics Or Mathematics: Offered by Dept. of Mathematics	4
Total Credits			**24**

Note:

1. A6creditminor(150marks) paper offered by another department to the students of the Department of Economics as either MDC or a Minor course would betreatedatparwith a4 creditminor (100marks)and150markstobenormalizedto100forBA (Hons.) Programme offered under the framework of Honours School system.
2. The students who are opting for Economics as a Minor and belong to the Science stream (NEP Sciences) of Panjab University, the minor papers will include a practical of 02 credits in each semester. The total credit for minor (only for NEP Sciences) will be 06. The practical will be conducted by the teacher concerned as per the requirements of the particular subject, and a practical score out of 50 will be submitted by them in the office. The result for Economics minor for NEP Sciences will be prepared by Department of Economics following the practice of 06 credits.
3. The students of B.A. (Hons.) Economics (FYUGP) under NEP will be offered only Mathematics and Statistics as a Minor as passed in draft regulations (2023-24), and for the first three semesters, they will study the Mathematics Minor. Fourth semester onwards, they can either opt for Statistics Minor or they can continue with Mathematics Minor.
4. Those who scored at least 75 percent marks in B.A. 3rd year shall be eligible to opt for B.A. (Honours) with Research.
5. A synopsis of at most 3000 words shall be prepared by the candidate and a presentation before faculty members of department shall be arranged immediately after the end of odd semester examination to evaluate the performance of candidate. Marks out of 100 shall be awarded to assess candidate's performance.
6. The dissertation submitted to the Department, before a pre-fixed date announced by the Department, shall be evaluated by two subject experts. The supervisor shall be one of these two experts. The second expert shall be an external examiner appointed from an institution beyond the administrative jurisdiction of Panjab University, Chandigarh (i.e., expert from another university or research institute not affiliated with the Panjab University). The viva-voce shall be conducted by the external expert in the presence of the members of RAC. Marks out of 200 shall be awarded during the viva-voce examination with the consensus of the members of RAC and the external expert. The bifurcation of 200 marks shall be as follows:

   i) Marks out of 150 shall be awarded for the quality of dissertation work; and
   ii) Marks out of 50 shall be awarded on the basis of candidate's performance in viva-voce.

   Hence, the dissertation Course in 4th semester shall carry 8 credits (200 marks) and the grade in this Course shall be decided on the basis of the percentage marks obtained out of 200 in the viva-voce examination. Passing marks for a dissertation paper shall be 40% (i.e., 80 out of 200) with a grade letter 'P'.

   Note: It is not necessary to score passing marks/grade separately but a sum of marks obtained from dissertation evaluation (marks out of 150) and viva-voce examination (marks out of 50) shall be considered to fulfill passing grade criterion.

Outlines of Syllabi and Course Structure for FYUGP Courses in the Department of Economics, Panjab University, Chandigarh (2026-27)

The Four-Year Undergraduate Programme shall be offered under the framework of the Honours School system by the Department of Economics, Panjab University, Chandigarh. The courses offered shall be "B.A. (Honours)/B.A.(Honours) with Research under the Framework of Honours School System" with the following exit options:

i) A "Certificate Course in Economics under the Framework of Honours School System" to be awarded in case the candidate exits after completing one year of the course (i.e., earning successfully 48 credits);

ii) A "Diploma in Economics under the Framework of Honours School System" to be awarded in case the candidate exits after completing two years of the course (i.e., earning successfully 96 credits);

iii) A "Bachelor of Arts (B.A.) degree in Economics under the Framework of Honours School System" to be awarded in case the candidate exits after completing three years of the course (i.e., earning successfully 144 credits); and

iv) A "Bachelor of Arts Honours degree [B.A.(Honours)] in Economics under the Framework of Honours School System" to be awarded in case the candidate completes full four years of the course (i.e., earning successfully 192 credits);

or

v) A "Bachelor of Arts Honours degree with Research [B.A.(Honours) with Research] in Economics under the Framework of Honours School System" to be awarded in case the candidate completes full four years of the course (i.e., earning successfully 192 credits) after successfully submitting and defending the dissertation of 12 credits in lieu of three papers (one paper in Semester-VII and two papers in Semester-VIII);

Notes:

i) A Candidate having B.A.(Hons.)/B.A.(Hons.) with Research degree along with a percentage of marks at least 75% shall be eligible to compete for attaining the eligibility to JRF/Assistant Professorship/Ph.D. Admission through NET examination conducted by NTA/UGC without pursuing M.A. in Economics.

ii) The syllabi have been given in 3 Lectures 1 Tutorial format. A teacher may use tutorial as lecture hour as both Lecture and Tutorial are of one hour and one credit.

B.A. (Hons.) Under the Framework of Honours School System
Semester I

Economics Core Course 1: ECO (HS) DSC 1: Introductory Microeconomics (4 credits)

Economics Core Course 2: ECO (HS) DSC 2: Mathematical Methods for Economics-I (4credits)

Skill Enhancement CourseI: ECO (HS) SEC 1: Data Collection and Handling Techniques (3credits)/Anyother course offered by the Department of Mathematics/Statistics/Computer Science of the Panjab University of same or more credits

MultidisciplinaryCourse1: MDSC1: Fundamentals of Sociology (3credits) /any other course offered by the Panjab University of same or more credits

Ability Enhancement Compulsory Course (AECC)–1(2credits) (offered by Department of English)

Value Added Course1: ECO (HS) VAC 1: Indian Economic Thought (2credits)/ anyother course offered by the Panjab University of same or more credits

Value Added Course 2: ECO (HS) VAC 2: Introduction to data Analysis Using Excel-1 (2 credits)/Any other course offered by Departments of Mathematics/Statistics/ComputerScience of the Panjab University of same or more credits

MDC1:MAT-DSC-101: Calculus (4credits)/ any other course offered by Department of Mathematics of the Panjab University of same or more credits.

B.A. (Hons.) Under the Framework of Honours School System
Semester II

EconomicsCoreCourse3: ECO (HS) DSC3: Introductory Macroeconomics(4 credits)

Economics CoreCourse4: ECO(HS) DSC 4: Mathematical Methods forEconomics-II (4credits)

Skill Enhancement Course 2: ECO (HS) SEC 2: Data Analysis for Beginners (3 credits)/ Any other course offered by Departments of Mathematics/ Statistics/ Computer Science of the Panjab University of same or more credits

MultidisciplinaryCourse2: PUB MDSC 2: Basics of Indian Administration (3credits)/ Any other course offered by the Panjab University of same or more credits

AbilityEnhancement CompulsoryCourse(AECC)–II (2credits) (offeredby Department of English)

ValueAddedCourse3:ECO (HS) VAC 3: Introduction to Environmental Economics(

2credits)/Any other course offered by the Panjab University of same or more credits

Value Added Course 4: ECO (HS) VAC 4: Introduction to Data Analysis Using Excel-II (2 credits)/ Any othercourseofferedbyDepartmentofMathematics /Statistics/ComputerScience ofthe Panjab University of same or more credits

MDC-2: MAT-DSC 151: Algebraand Ordinary DifferentialEquations(4credits)/ Anyother course offeredbythe Department of Mathematics of thePanjab University of same or more credits

B.A. (Hons.) Under the Framework of Honours School System
Semester III

Economics Core Course 5: ECO (HS) DSC 5: Intermediate Microeconomics-I (4 credits)

EconomicsCoreCourse6: ECO (HS) DSC 6: IntroductoryStatisticalMethodsforEconomic Analysis (4credits)

Economics Core Course 7: ECO (HS) DSC 7: History of Economic Thought (4 credits)

SkillEnhancement Course 3:ECO (HS) SEC 3: Working with Open-Source Software(Python and R)(3credits)/Anyothercourseoffered by Department of Mathematics/Statistics/Computer Science ofthePanjab University of same or more credits

MultidisciplinaryCourse3: MDSC 3: SocialStratification(3credits)/Anyothercourseoffered bythe Panjab University of same or more credits

AbilityEnhancementCompulsoryCourse(AECC)–III(2credits)(Punjabi/Hindi)(Offeredby Department of Punjabi/Hindi)

MDC 3: MAT-M-3: Partial Differential Equations and Numerical Analysis (4credits)/Any other course offered by Department of Mathematics of the Panjab University of same or more credits.

B.A. (Hons.) Under the Framework of Honours School System
Semester IV

EconomicsCoreCourse8:ECO (HS) DSC 8: IntermediateMicroeconomics-II(4credits)

EconomicsCoreCourse9: ECO (HS) DSC 9: IntermediateMacroeconomics(4 credits)

EconomicsCoreCourse10: ECO (HS) DSC 10: IntroductoryEconometrics(4 credits)

EconomicsCoreCourse11: ECO (HS) DSC 11: GrowthModels(4 credits)

AbilityEnhancementCompulsoryCourse(AECC)–IV(2credits)(Punjabi/Hindi)(Offeredby Punjabi/Hindi Department)

MDC 4: Statistics-STAT-M-162- Introductory Probability (4credits)/Any other course offered by Departments ofStatistics ofthe Panjab University of same or more credits

Or

Mathematics: MAT-M-4: Number Theory & Group Theory (4 credits)

Internship/CW/FP (2 Credits)

B.A. (Hons.) Under the Framework of Honours School System
Semester V

ECO (HS) DSC 12: Intermediate Econometrics (4 credits)

ECO (HS) DSC 13: International Economics (4 credits)

ECO (HS) DSC 14: Indian Economy-I (4 credits)

ECO (HS) DSC 15: Introductory Public Finance (4 credits)

MDC 5: Statistics-STAT-M APPLIED STATISTICS (4 credits**)**

Or

Mathematics: Offered by Department of Mathematics

ECO (HS) VAC (Value Added Course) 5: Financial Literacy**Or**

Introduction to Research (2 credits)

Internship/CW/FP: Internship/CW/FP (2 Credits)

B.A. (Hons.) Under the Framework of Honours School System
Semester VI

ECO (HS) DSC 16: Economics of Development & planning (4 credits)

ECO (HS) DSC 17: Indian Economy-II (4 credits)

ECO (HS) DSC 18: Introductory Agriculture Economics (4 Credits)

ECO (HS) DSC 19: Introductory Industrial Economics (4 Credits)

ECO (HS) DSC 20: Basics of Financial Markets (4 Credits)

MDC 6: Statistics-STAT-M BASICS OF STATISTICS INFERENCE (4 credits)**Or**

Mathematics: Offered by the Department of Mathematics

One of the Following Two Options in 4th Year for the Candidate Scoring At-least 75% Marks up to Three Years

B.A. (Hons.) Under the Framework of Honours School System

Semester VII

ECO (HS) DSC 21: International Trade and Trading Systems(4 credits)

ECO (HS) DSC 22: Fundamental Issues in Political Economy (4 credits)

ECO (HS) DSC 23: Optimisation Techniques and Game Theory (4 credits)

ECO (HS) RBCRM(A): Research Methodology(4 credits)

ECO (HS) DSE(#): Any Discipline Specific Elective Paper from Optional Papers of Economics (4 credits)
MDC 7: Statistics- Offered by the Department of Statistics (4 credits) **Or**
Mathematics: Offered by the Department of Mathematics (4 credits)

B.A. (Hons.) Under the Framework of Honours School System

Semester VIII

ECO (HS) DSC 24: Advanced Issues in Public Finance (4 credits)

ECO (HS) DSC 24: Environmental Economics (4 credits)

ECO (HS) DSC 26: National Income Accounting (4 credits)

ECO (HS) RBCRM(B): Research Publication and Ethics (4 credits)

ECO (HS) DSE(#): Any Discipline Specific Elective Paper from Optional Papers of Economics (4 credits)
MDC 8: Statistics- Offered by the Department of Statistics (4 credits) **Or**
Mathematics: Offered by the Department of Mathematics (4 credits)

Or

B.A. (Hons.) with Research Under the Framework of Honours School System

Semester VII

ECO (HS) DSC 21: International Trade and Investment(4 credits)

ECO (HS) DSC 22 Fundamental Issues in Political Economy (4 credits)

ECO (HS) DSC 23: Optimisation Techniques and Game Theory (4 credits)

ECO (HS) RBCRM(A): Research Methodology(4 credits)

ECO (HS)-RBCDST(A): Dissertation Paper (4 credits)

MDC 7: Statistics- Offered by the Department of Statistics (4 credits) **Or**
Mathematics: Offered by the Department of Mathematics (4 credits)

B.A. (Hons.) with Research Under the Framework of Honours School System
Semester VIII

ECO (HS) DSC 24: Advanced Issues in Public Finance (4 credits)

ECO (HS) RBCRM(B): Research Publication and Ethics (4 credits)

ECO (HS) DSE(#): Any Discipline Specific Elective Paper from Optional Papers of Economics (4 credits)

ECO (HS)-RBCDST(A): Dissertation Paper (8 credits)

MDC 8: Statistics- Offered by the Department of Statistics (4 credits) **Or**
Mathematics: Offered by the Department of Mathematics (4 credits)

Semester-I

ECO (HS) DSC1:INTRODUCTORY MICRO ECONOMICS

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**100**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Course Objective: -This course is designed to expose the students to the basic principles of microeconomic theory. The emphasis will be on thinking like an economist and the course will illustrate how microeconomic concepts can be applied toanalyze real-life situations.

Course Outcomes: To rigorouslyand comprehensively equip the students with theoretical concepts, methodology and process of reasoning involved in analyzing economic behavior of individuals, firms andmarkets using, in general, a static and partial equilibrium framework.

PedagogyoftheCourseOutcomes:80% Classroom teaching.

20%InternalAssessment.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES: -

1. Themaximummarksforthepaperwillbe 100. Thequestionpaperwillbe of 80 marks and continuous evaluation 20 marks. Time allowed will be 3hours.
2. Thepaper-settermustputanoteinthequestionpaperin thisregard.
3. Thereshallbe9questionsinall.
4. The first question is compulsory comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of two marks (10 x 2= 20). Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

UNIT-I

Exploring the subject matter of Economics. Why study economics? Scope and method of economics; the economic problem: scarcity and choice; the question of what to produce, how to produce and how to distribute output

CardinalUtilityAnalysis, Ordinal Utility Analysis: The consumption decision - budget constraint, consumption and income/price changes, demand for all other goods and price changes; description of preferences(representing preferences with indifference curves); properties of indifference curves; consumers optimum choice; income and substitution effects.

UNIT-II

Theoryofproduction: lawsgoverningshortrunandlongrun. Producers' equilibrium and expansion path. CostTheory:Shortrunandlongruncostcurves.

TheFirmandIndustryunderPerfectMarketStructure:Price and output determination; short run and long run.

UNIT-III

Determinants of individual demand/supply; demand/supply schedule and demand/supply curve; market versus individual demand/supply; shifts in the demand/supply curve, demand and supply together; how prices allocate resources. The basic competitive model. Supply and Demand How Markets Work, Markets and Welfare Markets and competition Elasticity of demand: its measurements, applications and determinants.Consumer surplus;producer surplusandtheefficiencyof the markets.

Concept ofdeadweight loss;Govt.interventionintheformofTaxes and Subsidies.

UNIT-IV

Singlesellermonopoly:Priceandoutputdetermination.

Input Markets Labour and land markets - basic concepts (derived demand, productivity of an input, marginal productivity of labour, marginalrevenueproduct);demandforlabour;inputdemandcurves;shifts in input demandcurves; competitive labour markets and firms profit maximisationconditionsin input market; and labour markets andpublicpolicy.Derivationoflaboursupplyandsavingsdecision - choicebetweenleisureandconsumption.

Essential Readings:-

1. N. Gregory Mankiw. (2007) *Economics: Principles and Applications*. (4th ed.). India edition by South Western, a part of Cengage Learning, Cengage Learning India Private Limited.
2. Karl,E.Case.,and RayC. Fair, (2007).*Principlesof Economics (*8th ed.). Pearson Education Inc.

SuggestedReadings:-

1. Joseph E. Stiglitz, and Carl E. Walsh. (2007). Economics(4th ed.). W.W.Norton & Company, Inc., New York, International Student Edition.
2. Salvotore.D(2006)*TheoryandProblemsofMicroeconomicTheory*(Sachemsseries) (3rd ed.)TataMcGraw-HillPublishingCompany Ltd.
3. Salvotore.D. (2007). *Micro economic theory and Applications* (4th ed.)Oxford UniversityPress.

ECO (HS) DSC2: MATHEMATICAL METHODS FOR ECONOMICS–I

Topologyofthe code	**Major**
LeveloftheCourse	**100**
Course Duration /LTP (Lectures, Tutorials,Practical)	**L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+Internal Assessment20marks**
Time	**3hours**

Course Objective: -This is the first part of a compulsory two-course sequence. The objective of this sequence is to transmit the body of basic mathematics that enables the study of economic theory at the undergraduate level, specifically the courses on microeconomic theory, macroeconomic theory, statistics and econometrics set out in thissyllabus.Inthiscourse,particulareconomicmodelsarenottheends,butthe means for illustrating the method of applying mathematical techniques to economic theory in general. The level of sophistication with which the material is to be taught is indicated by the contents of the prescribed textbook.

Course Outcomes: - 1. Students are able to understand the preliminaries and functions ofone realvariable and their economic applications. 2.Students are able to learn about differentiable functions and also single variable optimization using calculus and its applications. 3. Students are able to understand and apply the integration of functions and their simple economic applications. 4. Students are able to understand and apply the Difference Equations and their simple economic applications.

PedagogyoftheCourseOutcomes: -80%Classroomteaching. 20%InternalAssessment

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

Thereshallbe**9**questionsinall.

The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shallattempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT-I

Preliminaries: EssentialsofSetTheory,Logic,MathematicalProofs, Mathematical Induction, The real Number System.

Functionsofonerealvariable: Graphsoffunctions,Linear,quadratic, polynomial, power, exponential, logarithmic.

PropertiesofFunctions: ShiftingofGraphs,NewfunctionsfromOld,Inverse functions, Graphs of Equations, Distance in Plane.

Limits,continuityandseries.

UNIT-II

Differentiation: Differentiationconcepts, properties with respect to various operations and applications; second and higher order derivatives: properties and applications.

ConvexandConcaveFunctions and Optimization:

ConvexandConcavefunctions:Propertiesandapplications;LocalExtreme Points (Maxima and Minima) and applications.

UNIT-III

Integration: Indefinite Integrals, Definite Integrals, Methods of Integration by Parts, Substitution and partial fractions andsimple economic applications

UNIT-IV

Differenceequations: Introduction,solutionofdifferenceequationsup to2^{nd} order, simple economic applications.

EssentialReadings: -

1. K. Sydsaeter and P. Hammond, (2016). Mathematics for Economic Analysis (11th Edition), Pearson:NewDelhi{Chapter1:Section1.4, Chapter6:Section6.1to6.7,Chapter9:section9.1to9.6;Chapter10:section10.1to10.4;Chapter 11:Section11.1to11.4;Chapter20:section20.1to20.5}

2. K.Sydsaeter, P. Hammond,A.Strom and A.Carvajal (2021).Essential Mathematicsfor EconomicAnalysis(6thedition),Pearson.{Chapter1:Section1.1to1.4;Chapter4:section4.1to4.10;Chapter5:Section5.1to5.5;Chapter6:section6.1to6.11;Chapter8:Section8.1to8.6;chapter9:Section9.1to9.6;Chapter10:section10.1to10.7}

SuggestedReadings: -

1. Weber,JeanE(1982).MathematicalAnalysisBusinessandEconomicApplications (4thEd.)YamaneTaro(1972).MathematicsforEconomists.

ECO (HS) SEC1: DATA COLLECTION AND HANDLING TECHNIQUES

Topologyofthecode	**Skill Enhancement Course**
LeveloftheCourse	**100**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=30hours(2 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**3**
Max.Marks	**75Marks=Theory60marks+InternalAssessment 15 marks**
Time	**3hours**

Course Objective: - To provide skills to undertake data-based research so that students get the competency of executing sample surveys and knowledge about secondary data collection.

Course Outcomes:-It will enable the students to qualify forjobs relating to fieldinvestigation in any field of social sciences.

PedagogyoftheCourseWork: -80%Classroomteaching.

20%InternalAssessment.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

- Themaximummarksforthepaperwillbe75. Thequestionpaper will be of 60 marks and internal assessment 15.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe **7**questionsinall.

The first question is **compulsory** comprising 8 short answer type questions spread over the whole syllabus. The candidates are requiredto attempt 5 questions. Each question shall be of 3marks (5x3=15).

Restofthepapershallcontainthreeunits.Two questionsshallbeasked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each questionwill carry15 marks. (15x3=45).

UNIT-I

Research: Meaning, objectives, Characteristics, Types, Process and Utility, Researchmethod Vs. Methodology, ResearchDesign-Meaning, TypesandFeatures of Good Research Design.TypesofData:PrimaryvsSecondary**,**CrossSectionor MicroData,TimeSeriesData,Pooling of data, Panel data, Census vs sampled data.

UNIT-II

Sampling – Concepts of Statistical Population, Sample, Sampling Frame, Sampling and non-Sampling Errors, Sample Size. Census and Sampling survey, Characteristics of good sampling.

Sampling Techniques: Simple Random Sampling; PPS Sampling; Stratified Random Sampling; Multistage Sampling (Applications only), determination of sample size.

UNIT-III

Methods of collection of Primary data (observation method, interview method, collectionof datathroughquestionnaires- Designing ofquestionnaire, collectionof data through schedules-Designing of schedules). Measurement and scaling, validity and reliability.

CollectionofSecondaryData

Selectionofappropriatemethodofdatacollection.

**** Practical Work: Students are required to formulate a research problem and need to prepare a questionnaire for the same along with appropriate measurement and scaling techniques.

EssentialReadings: -

1. Kothari,C.R.(2009).*ResearchMethodsandTechniques*, WileyEastern.

SuggestedReadings: -

1. Bethlehem,J.(2009).*Appliedsurveymethods:Astatisticalperspective.*Wiley.
2. Cochran,W.(2008).*Samplingtechniques,3rded.*Wiley.
3. Cooper,D.,Schindler, P.,Sharma, J.(2012).*Businessresearchmethods*,*12thed.*McGraw-Hill.
4. Flick,U.(2012).*Introducingresearchmethodology:Abeginner'sguide to doing aresearch project*. Sage Publications.
5. Groves,R.,Fowler,F.,Couper,M.,Lepkowski,J.,Singer,E.,Tourangeau,R. (2009). *SurveyMethodology*.Wiley
6. Kumar,R.(2014).*Researchmethodology:Astepbystep guide forbeginners*, *4th ed.* Sage Publications.

ABILITY ENHANCEMENT COURSE 1

CourseTitle: LanguageSkills:Listening and Speaking
Course Code: ENG-AEC-1-101
TypologyofCourse: AEC
Level: 100
Course Duration: 60hoursofpracticalinonesemester (4hr/week)
Credits: 2credits
MaximumMarks: 50marks(30Theory+15Practical+5InternalAssessment)

Practical&Tutorialsinallsubjectstothecapacityof only20studentsperlectureclass.

Learning Objectives of the Course:

- Tounderstandthestructureoflanguageandhowlanguagevariesovertime,across social situations and social groups
- Tobeproficientinarticulatingthoughtsandcommunicatingideasinlanguage
- Toparticipateindebatesanddiscussionswhereideasaretestedandsharpenedand honing the skill of public speaking
- Studentsshouldbeabletoholdmeaningfulconversations
- StudentshouldbeabletoeffectivelyuseEnglishlanguagetomeetprofessionaland personal goals

Pedagogyofthe Course:

a) Thepaperrequires4hoursofpracticalsperweekduringwhichstudentswillbetaught and explained the theory of communication through practical and hands-on training in various aspects of communication.
b) Thepracticalevaluation,whichcarries15marks,willbedonebyanexternalexaminer either from the same institution/department or from outside.

Course Content:

ThePrescribedbookforalltheUnitsis:*Essence:ACourseinCommunicativeEnglish*, Cambridge University Press, 2025

UnitI: SoundsofEnglish

Unit2: LanguageUseinVarious Situations

Unit3: ArtofPublic Speaking

a. GroupDiscussio
b. Interview

SuggestedReadings:

- *FluencyinEnglishPartII*,OxfordUniversityPress.
- *ATextbookofPhoneticsforIndianStudents*,T.Balasubramian,Macmillan.
- *BusinessEnglish*, Pearson
- *Language,LiteratureandCreativity*,OrientBlackSwan.

- *LanguagethroughLiterature*,GauriMishra,RanjanaKaul,BratiBiswas(Eds), Primus Books.
- *EnglishCommunication,*MacmillanEducation.
- *EnrichYourEnglish*,S.R.InthiraandV.Saraswathi,OxfordUniversityPress.
- *CommunicationandLanguageSkills*,SanghitaSen,etal,CambridgeUniversity Press.
- *EnglishandCommunicationSkillsI*,TasneemShahnaaz,C.SavithaandCherlylRao, Cambridge University Press.
- *PracticalEnglish*,K.K.Singh,APHPublishingCorporation.

TestingScheme:

Max.Marks:50marks
Theory: 30 marks
Practical:15 marks
InternalAssessment:05marks Time: 1.5 Hours

Note-Questionno.1iscompulsory.AttemptanyfourquestionsfromQ2to7.

Q.1.Shortnotes.**Twooutoffive**tobeattemptedinnotmorethan150wordseach.

(10 marks)

Q2. to Q.7. from the prescribed syllabus of Listening and Speaking. **Four out of six** questions to be attempted in not more than 150 words each.

(20 marks)

ECO (HS) VAC-1: INDIAN ECONOMIC THOUGHT

Topologyofthecode	**ValueAddedCourse**
LeveloftheCourse	**100**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=30hours(2hoursaweekfor15weeks= 01semester), T=0, P=0**
Credits	**2**
Max.Marks	**50Marks=Theory40marks+InternalAssessment 10 marks**
Time	**1.5 hours**

CourseObjective: -Thecourseaimstoenablethestudentstoacquireand demonstrate the knowledge on economic thoughtwith its historical perspective.

Course Outcomes: -

1. Studentsthroughthis course become familiar withthe historicalevolution ofeconomic ideas. Theyare able to learn the foundational contributions of the classical economists to the economic literature.

2. Students are able to develop understanding about the philosophical foundationsof economic analysis of capitalist system. They also learn the theories of imperialism which are important to understand the exploitation process ofthird world countries by the metropolis nations.

3. Studentsareabletotheoriginofmarginalist,neoclassicalandAustrianschoolof thought and their main features. This is important to understand the differencesandsimilaritiesamongtheir ideas.4.Studentsareabletolearnabout the Keynesian revolution in economic theory. They also understand the causesofgreat depression. Moreover, they learn about the Kaleckian version ofmacroeconomics.

PedagogyoftheCourseWork:80%Classroomteaching.

20%InternalAssessment.

InstructionsforPaper-setterandcandidates:

• The maximummarksforthepaperwillbe50. Thequestionpaperwillbeof40 marks and continuous evaluation 10 marks. Time allowed will be 2 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe5questionsinall.

The first questioncompulsorycomprising 10 short answer type questions spread overthewholesyllabus.Thecandidatesarerequiredtoattempt 5 questions.Each question shall be of two marks (5x2= 10).

Rest ofthe paper shall contain two units. Two questions shall be asked fromeach unit andthecandidatesshallbegiveninternalchoice.Thecandidatesshallattempt one question from each unit. Each question will carry 15 marks (15x2=30).

UNIT-I

Pre-independenceIndianEconomicthought

Kautilya's Arthasastra:Theeconomic functionsofthestate,Taxationandpricing policy, Theland system

Dadabhai Naorojiandthedraintheory:What isthedrain?CriticismsandLong-run relevance

Ranadeandtheeconomicsofdevelopment:Povertyand industrialisation, Agrarian policy, Railway investment.

UNIT-II

IndianEconomicthoughttowardsindependence

Gokhaleand theeconomicsofeducation

Gandhian economics: Methodological issues, Consumption behaviour: the limitationofwants,Swadeshi;Trusteeshipandindustrialrelations;Charity, leisure and the sanctity of work

EssentialReadings:-

1. Dasgupta,A.K.(1993).AHistoryofIndianEconomicThought (1sted.). Routledge.

Suggested Readings:-

1. Kautilya.&Shamasastry,R.(1967).*Kautilya'sArthasastra*.Mysore,IndiaMysorePublishingandPrin tingHouse.
2. Naoroji,D.(1901).PovertyandUn-BritishRuleinIndia.United Kingdom: S.Sonnenschein.
3. Chandra,B.,Ranade,M.G.(1990).Ranade'sEconomicWritings. India: GianPublishing House.
4. Gokhale, G. K. (2018). Gopal Krishna Gokhale: Select Speeches and Writings. India: National Book Trust, India.
5. VillageSwarajbyM.K.Gandhi.(2021).(n.p.):PrabhatPrakashan.
6. Dasgupta,A.K.(1996). Gandhi'sEconomicThought.(n.p.):Taylor &Francis.

ECO (HS) VAC-2: INTRODUCTION TO DATA ANALYSIS USING EXCEL-I

Topologyofthecode	**ValueAddedCourse**
LeveloftheCourse	**100**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=15hours(1hoursaweekfor15 weeks= 01semester),T=0, P=30 (02 Practical per week)**
Credits	**2**
Max.Marks	**50Marks=Theory40marks+Internal Assessment10 marks**
Time	**1.5 hours**

Objective: The objective of the paper is to rigorously and comprehensively equip the students with applied data handling skills using spreadsheet based graphical and statistical tools of reasoninginvolved in empirical analysis of economic behaviour. The students would be evaluated at the end of each semester through subjectivetype questions/answers (bothshort and essaytype). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

PedagogyoftheCourseOutcomes:80%Classroomteaching.

20%InternalAssessment.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

The maximummarks forthepaperwillbe50. Thequestionpaperwillbeof 40 marks and continuous evaluation will be of 10 marks. Time allowed in the paper will be 2 hours.

Thepaper-settermustputanoteinthequestionpaperinthisregard.

- Thereshallbe5questions inall.
- The first question compulsory comprising 10 short answer type questions spread over the whole syllabus. The candidates are required to attempt 05 questions. Each question shall be of two marks (5x2= 10).
- Rest of the paper shall contain two units. Two questions shall be asked from each unit and thecandidates shall be given internalchoice. The candidates shallattempt one question from each unit. Each question will carry 15 marks (15x2=30).

CourseOutcomes

After successful completion, the students will be able to understand the use ofMS-Excel in data handling and presenting. The filtering techniques, lookup methods, pivot tabling, graphics presentation, matrix analysis etc. shall sharpen a student in the field of quantitative economics and econometrics.

UNIT-I

IntroductiontoSpreadsheets

Introductiontospreadsheets,readingdata,manipulatingdata.Arithmeticoperationsand basic functions.

Spreadsheet FunctionstoOrganizeData

IntroductiontosomemoreusefulfunctionssuchastheIF,nestedIF,VLOOKUPand HLOOKUP functions in Excel.

UNIT-II

IntroductiontoFiltering,PivotTables and Charts

Introduction to the Data filtering capabilities of Excel, the construction of Pivot Tables to organize data and introduction to charts in Excel. Constructing various Line, Bar and Pie charts. Using the Pivot chart features ofExcel. Understanding and constructing Histograms and Scatterplots

References:

1. Lloyd,Chris, J. (2011).DataDrivenBusinessDecisions, JohnWiley&Sons.
2. Levine,D.M.,Stephen,D.andSzabat,K.A.(2016).StatisticsforManagersUsing Microsoft Excel, Pearson Education.
3. Day,AlastairL.(2013).MasteringFinancialMathematicsinMicrosoftExcel,Prentice Hall.
4. Piascik,Chester(2001).AppliedMathematicswithMicrosoftExcel,PennsylvaniaState University.
5. Jelen,Bill(2013). Excel2013:ChartsandGraphs inMicrosoftExcel.QuePublishing

Semester-II

ECO(HS) DSC3: INTRODUCTORY MACRO ECONOMICS

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**100**
CourseDuration/LTP (Lectures, Tutorials, Practical)	**L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Course Objective

Macroeconomics or aggregative economics analysis establishes the functional relationship between the large aggregates. The aggregated analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essential for the proper comprehension of the different issues and policies. Macroeconomics now is not only a scientific method of analysis; but also a body of empirical economic knowledge.

The paper entitled "IntroductoryMacroeconomics" equips the students at the undergraduate level to understand systemic facts and latest theoretical developments for empirical analysis.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

PedagogyoftheCourseOutcomes:80%Classroomteaching.

20%InternalAssessment.

Course Outcomes

After completion of the course, a student must be capable of understanding various measures of National income and accounting. Also the paper shall inculcate the fundamental macroeconomic modelling abilities among students to handle macro-economic relationships for their empirical testing. After completion of course, the candidate shall be specifically able to identify the determinants of consumption and investment functions along with the measuresof national income;a major cause of variation in two functions.

INSTRUCTIONSFORTHEPAPER-SETTERAND CANDIDATES

1. Thesyllabusofthispaper hasbeendividedinto four units.

2. Thereshallbe9questionsinall.

3. The first question, which would be compulsory, shall be short answer type (word limit 25-30

each).It would carry15 short questions, spread over the entire syllabus. The candidate will be requiredtoattempt any10short answertypequestions.Eachshort answertypequestionwould carry 2 marks (10 × 2 = 20).

4. Restofthepapershallcontain4units. Eachunit shallhavetwoquestionsandthecandidatesshall berequired to attempt onequestion fromeach Unit— 4 inall. Eachquestionshallcarry15 marks (15 × 4 = 60).

UNIT-I

Conceptual Introduction: Aggregates of National and Personal income. Measure of disposable income. The measurement of these aggregates at market and constant prices.

Methods of National Accounting: Expenditure, Value Added, and Income Methods. Mixed methods of national accounting.

Social Accounting: Techniques of Maintaining National Accounts such as Double Entry system, Social Accounting Matrix, National Accounting Equations. The Input Output Tables as base of Social Accounting Matrix. Introduction to Supply and Use Tables.

UNIT-II

***IncomeandEmploymentDetermination*:**IntegratedClassicaland KeynesianModelsofIncomeand Employment Determination; commodity, money (including bond market of Keynes), and labourmarkets.

Wage-PriceFlexibilityandAutomaticFullEmployment:ClassicalVersus KeynesianApproach.

UNIT-III

Consumption and Consumption Function: Keynes Consumption and saving functions under psychological law of consumption, Consumption Puzzle: Absolute Income hypothesis, Relative Income hypothesis, Permanent Income hypothesis and Life Cycle Hypothesis.

UNIT-IV

Investment and Investment Function: Type of Investment, Role of investment using Investment Multiplier, Classicaland KeynesianTheories ofInvestment, Accelerator TheoryofInvestment, Neo-Classical Theory of Investment and Tobin's-q Theory of Investment. Investment in the Housing Market.

RecommendedReadings:

1. Ackley,G.(2007).MacroeconomicTheory. Macmillan.
2. Allen, R.G.D. (1967). Macro-Economic Theory: A Mathematical Treatment. Palgrave Macmillan.
3. Beckerman,W.(1980).IntroductiontoNationalIncome Analysis, LittlehamptonBookServicesLtd.
4. Blanchard, O. (2009). Macroeconomics. New Delhi: Pearson Education Inc.

5. Blanchard, O.J.&Fischer. (1996). Lectureson Macroeconomics, Delhi: PHILearningPvt.Ltd.
6. Branson, W. H. (1989). Macro Economic Theory and Policy (3rd ed.).
7. Harper Collins. Cooper, R.,Edey, Harold C., and Peacock, A.T.(2003). National Income and Social. Routledge.
8. D'Souza,E.(2008).Macroeconomics. PearsonEducation.
9. ***Dornbusch,R.,Fisher,S&Startz,R.(2002).Macroeconomics,(9thed.).NewDelhi:Tata McGraw Hills.***
10. Levacic,R.&Rebmann,A.(1982). Macroeconomics:AnIntroductionto KeynesianNeoclassical Controversies. Macmillan.
11. Miller,R.L.&Pulsinelli,R.(1986).Macroeconomics.
12. NewYork:Harper&Row. Ola Olsson (2010). Essentials of Advanced Macroeconomic Theory, Routledge.
13. Romer,D.(1996).AdvancedMacroeconomics.McGraw-Hill.
14. UmaDattaRoyChoudhury(2000).NationalIncomeAccounting,Macmillan.

FurtherReadings:

1. Abel, A. B., Bernanke, B. S. (2007). Macroeconomics. New Delhi: Pearson Education Inc.
2. Dornbusch,R.(1980).OpenEconomyMacroeconomics.NewYork:\
3. Frisch, H. (1983). Theories of Inflation. Cambridge: Cambridge University Press.
4. Froyen,R.T.(2002).MacroEconomics. NewDelhi:PearsonEducationInc.
5. Gordon, R. J. (2012). Macroeconomics. Delhi: PHI Learning Pvt. Ltd.
6. Heijdra,B.J.(2017).TheFoundationsofModernMacroeconomics.OxfordUniversityPress.
7. Hicks, J. R. (1950). A Contribution to the Theory of Trade Cycle Oxford: Clarendon Press.
8. Lucas,R.(1981).StudiesinBusinessCycleTheory,Massachusetts:MITPressCambridge.
9. Makin, A. J. (2002). International Macroeconomics. Princeton Hall.
10. Mankiw,N. G.&Romer,D.(1991).NewKeynesianEconomics.MIT,2Volumes.
11. Mathews, R. C. O. (1964). The Trade Cycle. Cambridge: Cambridge University Press.
12. Pentecost,E.(2000). Macroeconomics–AnOpenEconomyApproach. MacMillanPressLtd.
13. Raghbendra,J.(1964). ContemporaryMacroeconomics:TheoryandPolicy. NewAgeInternational Publishers.
14. Sherffrin,S.(1999).RationalExpectations.CambridgeUniversityPress.
15. Snowden,B&Vane,H.R.(2005). ModernMacroeconomics:ItsOrigin,Development andCurrent Stat., UK: Edward Press.
16. Wendy,Carlin/Soskice,D.(2007).Macroeconomics:Imperfections,InstitutionsandPolicies. Oxford University Press.

ECO(HS) DSC 4: MATHEMATICAL METHODS FOR ECONOMICS–II

Topologyofthecode	**Major**
LeveloftheCourse	**100**
CourseDuration/LTP (Lectures, Tutorials,Practical)	**L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20marks**
Time	**3hours**

Course Objective :-This is the first of a compulsory two-course sequence. The objective of this sequence is to transmit the body of basic mathematics that enables the study of economic theory at the undergraduate level, specifically the courses on microeconomic theory, macroeconomic theory, statistics and econometrics set out in this syllabus. In this course, particular economic models are not the ends, but the means for illustrating the method of applying mathematical techniques to economic theory in general. The level of sophistication atwhich the material is to be taught is indicated bythe contents of the prescribed textbook.

Course Outcomes: -1. Students are able to understand the preliminaries and functions of one real variable and their economic applications. 2. Students are able to learn about differentiable functions and also single variable optimization using calculus and its applications. 3.Studentsareabletounderstandand applytheintegrationo functionsandtheirsimple economic applications. 4. Students are able to understand and apply the DifferenceEquations and their simple economic applications.

PedagogyoftheCourseOutcomes:-80%Classroomteaching.

20%InternalAssessment.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:
Themaximum marks for the paper will be 100.The question paper will be of 80marks and continuous evaluation 20 marks. Time allowed will be 3 hours.
Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe**9**questionsinall.
- Thefirstquestion**compulsory**comprising15shortanswertypequestionsspreadover
- thewholesyllabus. Thecandidatesarerequiredto attempt 10questions. Eachquestion shall be of **two** marks (10 x 2= 20).
- Rest ofthe paper shallcontain four units. Two questions shall be asked fromeachunit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT-I

Differential Equation*:* Introduction; Solution: Variable Separable Case, Homogeneous Case, Standard Linear Differential Equation, Bernoulli's form, Exact Equation; Solution of Linear differential Equation with Constant Coefficients; Simple Application Questions.

UNIT-II

Linear algebra: Vector spaces: algebraic and geometric properties, scalar products, norms, orthogonality; linear transformations: properties, matrix representations and elementary operations; systems of linear equations: properties of their solution sets; determinants: characterization, properties and economic applications.

UNIT-III

Functions of several real variables: Geometric representations: graphs and level curves; differentiable functions: characterizations, properties with respect to various operations and applications; second order derivatives: properties and applications; the implicit function theorem, and application to comparative statics problems; homogeneous and homothetic functions: characterizations and economic applications.

UNIT-IV

Multi-variable optimization: Convex sets; geometric properties of functions: convex functions, their characterizations, properties and applications; further geometric properties of functions: quasi convex functions, their characterizations, properties and applications; unconstrained optimization: geometric characterizations,characterizationsusingcalculusandapplications;constrainedoptimizationwithequality constraints: geometric characterizations, lagrange characterization using calculus and applications; properties of value function: envelope theoremand economic applications.

EssentialReadings:-

1. K.SydsaeterandP.Hammond(2002).MathematicsforEconomicAnalysis,Pearson Educational Asia Delhi.

2. K. Sydsaeter and P. Hammond, (2016). MathematicsforEconomic Analysis (11thEdition), Pearson: NewDelhi{Chapter5,Chapter12,Chapter14:Section14.1-14.3,Chapter15:Section15.1-15.3,Chapter16,Chapter17,Chapter18:Section18.1-18.7,Chapter21}

SuggestedReadings:-

1. Weber,JeanE.(1982).MathematicalAnalysisBusinessandEconomicApplications(4thEd.)

2. Yamane,Taro (1972). Mathematics for Economists.

ECO(HS) SEC2: DATA ANALYSIS FOR BEGINNERS

Topologyofthecode	**SkillEnhancementCourse**
LeveloftheCourse	**100**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=30hours(2 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**3**
Max.Marks	**75Marks=Theory60marks+InternalAssessment15 marks**
Time	**3hours**

CourseObjective:-Toprovideskillsto undertakedatabasedresearchsothatstudentsgetthe competencyof executing sample surveys and knowledge about secondarydata collection.

CourseOutcomes: -It willenablethestudentstoqualified for jobsrelatingto field investigation in any field of social sciences.

PedagogyoftheCourseWork:-80%Classroomteaching.

20%InternalAssessment.

InstructionsforPaper-setterandcandidates:

Themaximummarksforthepaperwillbe75. Thequestionpaperwillbeof 60marks and internal assessment 15.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe**7**questionsinall.
- Thefirst questionis **compulsory**comprising 8 short answertypequestions spreadover the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of **3** marks (5x3=15).
- Rest of the papershall contain three units. Two questions shall be asked from each unit andthecandidatesshallbegiveninternalchoice.Thecandidatesshallattemptonequestionfromeachunit.Eachquestionwillcarry15marks. (15x3=45).

UNIT-I

Collectionofprimarydatathroughquestionnaire(online(google form)/ faceto face) Errors in surveys: Misunderstanding of question and answers, problem of non- response.

Processing of Data: Questionnaire checking, Editing, Coding, classification, Tabulation, Graphicalrepresentation (Bar charts, Pie Charts, line graphs) Data cleaning, Data Adjusting.

UNIT-II

Analysis of Data and Interpretation: Measures of central tendency (Arithmetic mean median and mode),Dispersion,simplecorrelation(KarlPearsonandSpearmanRankCorrelation)and simpleregression.

UNIT-III

PreparationofaReport:TypesofReport, ResearchReportFormat,PrinciplesofWritingReports, Documentation, Footnotes and References & Bibliography.
Useofcomputersin research.
Ethicsandscientificintegrity:standardsofconduct,privacyindata,Plagiarism.

****PracticalWork:Project Report(collectiondatathroughquestionnaireon aresearch problemand preparation, analysis and interpretation of data).

EssentialReadings:-

1. Bethlehem,J. (2009).*Appliedsurveymethods:Astatisticalperspective.*Wiley.
2. Cochran,W.(2008).*Samplingtechniques,3rded.*Wiley.

SuggestedReadings:-

1. Cooper,D.,Schindler, P.,Sharma, J.(2012).Businessresearchmethods,12thed.McGraw-Hill.
2. Flick,U.(2012).Introducingresearchmethodology:Abeginner'sguidetodoinga research project. Sage Publications.
3. Groves,R.,Fowler, F.,Couper,M.,Lepkowski,J.,Singer,E.,Tourangeau, R.(2009).SurveyMethodology.Wiley.
4. Kothari,C.R.(2009).ResearchMethodsandTechniques,WileyEastern.
5. Kumar,R.(2014).Research methodology:Astepbystepguideforbeginners,4th ed.SagePublications.

ABILITY ENHANCEMENT COURSE 2

Course Title: **LanguageSkills: Reading and Writing**
Course Code: ENG-AEC-2-201
TypologyofCourse: AEC
Level: 100
Course Duration: 60hoursofpracticalinonesemester (4hr/week)
Credits: 2credits
MaximumMarks: 50marks(30Theory+15Practical+5InternalAssessment)

Practical & Tutorials in all subjects to the capacity of only 20 students per lecture class.

Learning Objectives of the Course:

- Learntocomposecomprehensiveandwell-structuredparagraphs
- Learntoutilizeliterarytermsandvocabulary,criticalmethods,andvariouslensesof interpretation in their writing
- Toapplyrulesofgrammar
- Tolearntheformattinganddocumentingconventionsofthediscipline

Pedagogyofthe Course:

a) Thepaperrequires4hoursofpracticalperweekduringwhichstudentswillbetaughtand explained the aspects of reading and writing through practical and hands-on training.
b) Forthepracticalexam,studentwillbeaskedtoreadatext,paraphraseagiventext, answer questions and interpret graphs and charts.
c) Thepracticalevaluation,whichcarries15marks,willbedonebyanexternalexaminer either from the same institution/department or from outside.

Course Contents:

TheprescribedbookforalltheUnitsis:*Essence:ACourseinCommunicativeEnglish*, Cambridge University Press, 2025

UnitI:Reading

1. Close Reading
2. Comprehension
3. Summary
4. Paraphrasing
5. InterpretingGraphsandCharts

UnitII.Writing

1. ParagraphWriting
2. Note-Making
3. Letter/E-mailWriting
4. ResumeWriting
5. Report Writing

SuggestedReadings:

- *OnTrack:EnglishSkillsforSuccess*,OrientBlackSwan.
- *Embark:EnglishforUndergraduates*,SteveHart,ArvindR.NairandVeenaBharti, Cambridge University Press.
- *Connect:CourseinCommunicativeEnglish*,DebashisBandyopadhyayandMalathi

Krishnan, Cambridge University Press.

- *EnrichYourEnglish*,S.R.InthiraandV.Saraswathi,OxfordUniversityPress.
- *CommunicationandLanguageSkills*,SanghitaSen,etal,CambridgeUniversityPress.
- *PracticalEnglish*,K.K.Singh,APHPublishingCorporation.
- *Language,LiteratureandCreativity*,OrientBlackSwan.
- *TheElementsofStyle,*WilliamStrunkJr.andE.B.White,Pearson.
- *OnTheArtofWriting*,ArthurQuiller-Couch,NotionPress.
- *FluencyinEnglishPartII*,OxfordUniversityPress.

TestingScheme:

Max.Marks:50 marks
Theory: 30 marks
Practical:15 marks
InternalAssessment:05 marks
Time: 3 hours

Note-Question no.1 is compulsory. Attempt two questions from Section B and two questions from Section C.

Section-A

Q.1.ShortNotes.**Twooutoffive**tobeattemptedinnotmorethan150wordseach. (10 marks)

Section-B

Attempt two questions out of given four: (10marks) Q2.
Reading Comprehension
Q.3. Summariseagiven passage.
Q.4. Writeaparaphraseofthegivenpassage.
Q.5. InterpretingGraphsandCharts

Section-C

Attemptanytwoquestionsoutoffive: (10 marks)
Q.6. ParagraphWriting
Q.7. Note-Making
Q.8. Letter/E-mailWriting
Q.9. ResumeWriting
Q.10. Report Writing

ECO(HS) VAC-3: INTRODUCTION TO ENVIRONMENTAL ECONOMICS

Topologyofthecode	**ValueAddedCourse**
LeveloftheCourse	**100**
CourseDuration/LTP (Lectures,Tutorials,Practical)	**L=30hours(3hoursaweekfor15weeks=01 semester), T=0, P=0**
Credits	**2**
Max.Marks	**50Marks=Theory40marks+InternalAssessment 10 marks**
Time	**1.5 hours**

CourseObjective:-Thispaperaimstomakeawarestudentsabouttheimportanceofenvironment in economics and vice-versa. It helps them to know the ways of sustaining our natural resources by optimally allocating them for future use.

CourseOutcomes:-Thispapercoverstopicsrelatedtoecosystems,naturalresources,biodiversity, conservation environmental pollution and policies.

PedagogyoftheCourseWork:80%Classroomteaching.

20% InternalAssessment.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

Themaximummarks forthepaperwillbe50.Thequestionpaperwillbeof40 marks and continuous evaluation 10 marks. Time allowed will be 2 hours.

Thepaper-settermustputanoteinthequestionpaperinthisregard.

- Thereshallbe5questionsinall.
- The first questioncompulsorycomprising 10 short answer type questions spread over the whole syllabus.Thecandidatesarerequiredtoattempt5questions.Eachquestionshallbeoftwomarks (5x2= 10).
- Restofthepaper shallcontaintwo units.Twoquestionsshallbeasked fromeachunit andthe candidatesshallbegiveninternalchoice.Thecandidatesshallattemptonequestionfromeach unit. Each question will carry 15 marks (15x2=30).

UNIT-I

Introduction:Environmentaleconomics:Nature,Scope,andSignificance.Relationbetween Environment

and Economy (inter linkages); Environment Kuznets Curve.

Commonpropertyresourcesandtheeconomicproblem;EnvironmentalExternalities inProduction and consumption. Managing externalities with taxes and subsidies and Property Rights; Coase Theorem.

UNIT-II

GlobalEnvironmentalPollution:Airandwaterpollution,Acidrain,OzonedepletionandGlobal warming (causes, consequences, and solutions).

Indian Law pertaining to Environmental protection: Environment Protection Act 1986; Air (PreventionandControlofPollution)Act;Water(Preventionandcontrolof Pollution) Act;Wildlife Protection Act; Forest Conservation Act.

ConsumerismandWastemanagement.

EssentialReadings:-

1. Kolstad.C.D.(2011).IntermediateEnvironmentalEconomics.OxfordUniversityPress.
2. Goodwin,N.,Harris,J.,Nelson,J.,Rajkarnikar,P.,Roach,B.,&Torras,M.(2019).Principlesof economics in context. Routledge, Taylor & Francis Group, New York & London.
3. Bharucha,E.(2021).TextbookforEnvironmentalStudies(Thirdedition).OrientBlackswanPvtLtd.
4. Eugine,T.(2005).EnvironmentalEconomics.VarindaPublications,Delhi.

FurtherReadings:-

1. Hussen,Ahmed(2004).PrinciplesofEnvironmentalEconomic(secondedition),Routledge,Tay lor & Francis Group, London & New York.
2. Goodstein,E.S.(2002).EconomicsandtheEnvironment(thirdedition)JohnWiley&Sons., Inc.
3. Maddala,G.S.(2004).Microeconomics;Theoryand Practice.McGraw-HillEducation.

ECO (HS) VAC-4: INTRODUCTION TO DATA ANALYSIS USING EXCEL-II

Topologyofthecode	**ValueAddedCourse**
LeveloftheCourse	**100**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=15hours(1 hours aweekfor15weeks=01semester), T=0, P=30(2 hours aweekfor15weeks=01semester)**
Credits	**2**
Max.Marks	**50Marks=Theory40marks+InternalAssessment10 marks**
Time	**1.5 hours**

Objective: The objective of the paper is to rigorously and comprehensively equip the students with applied data handling skills using spreadsheet based graphical and statistical tools of reasoning involved in empirical analysis of economic behaviour. The students would be evaluated at the end of each semester through subjective type questions/ answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy ofthe Course Work: Thecourserelies on a combination of lectures, solving problems and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporarythemes and issues fromthesyllabi. Specialtutorials/contact hour for one-to-onestudent-teacherinteractions.

InstructionsforPaper-setterandcandidates:

Themaximum marks for the paper will be 50. The question paper will be of 40 marks and continuous evaluation 10 marks. Time allowed will be2 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe5questions inall.
- The first question compulsorycomprising 10 short answer type questions spread over thewholesyllabus. Thecandidatesarerequiredtoattempt 5questions. Eachquestionshallbeof two marks (5x2= 10).
- Restofthepapershallcontaintwo units.Twoquestionsshallbeasked fromeachunit andthe candidates shall be given internal choice. The candidates shall attempt one question from eachunit.Each question will carry 15 marks (15x2=30).

UNIT-I

Macro RecordingsandMathematicalanalysis

Introduction to Excel Macros. Using Excel-Macro recording for data analyses (without using

VB),Solution of matrices.

UNIT-II

StatisticalAnalysis

CentralTendencyandDispersionmeasures,Regressionanalysis, Correlationanalysis.

References:

1. Lloyd,Chris, J. (2011).DataDrivenBusinessDecisions, JohnWiley&Sons.
2. Levine, D.M., Stephen, D. and Szabat, K.A. (2016). Statistics for Managers Using Microsoft Excel, PearsonEducation.
3. Day, Alastair L. (2013). Mastering Financial Mathematics in Microsoft Excel, Prentice Hall.
4. Piascik,Chester(2001).AppliedMathematicswithMicrosoftExcel,PennsylvaniaStateUniversity. Jelen, Bill (2013). Excel 2013: Charts and Grap Microsoft Excel. Que Publishing.

Semester-III

ECO(HS) DSC 5: INTERMEDIATEMICROECONOMICS–I

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**200**
CourseDuration/LTP (Lectures,Tutorials,Practical)	**L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

CourseDescription

The course is designed to provide a sound training in microeconomic theoryto formally analyze the behaviour of individual agents. Since students are already familiar with the quantitative techniques in the previous semesters, mathematical tools are used to facilitate understanding of the basic concepts. This course looks atthe behaviourofthe consumer and the producerand also covers the behaviour of a competitive firm.

PedagogyoftheCourseOutcomes:80%Classroomteaching.

20%InternalAssessment.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

The maximum marks for the paper will be 100. The questionpaper will be of80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

The paper-setter must put a note in the question paper in this regard.

- Thereshallbe**9**questions inall.
- The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).
- Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT-I

Budgetconstraint;propertiesofbudgetset,howbudgetlinechanges. Preference; Indifference curves, Marginal rate of substitution.

Utility;perfectsubstitutes,perfectcomplements,

Choice;optimalchoice;neutralsandbads,discretegoods, concave preferences.

Demand;incomeoffercurvesandEngelcurves,priceoffercurveanddemandcurve.Revealed preference

UNIT- II

Slutskyequation;the substitutioneffect,thetotalchangeindemand,ratesofchange.

BuyingandSelling;netgrossdemand, offercurvesanddemandcurvesIntertemporalchoice.

UNIT- III

Uncertaintyandrisk aversion

Technology:ProductionFunctions; Homogenousandhomothetic,Cobb DouglasandCES: Properties,elasticity of substitution; Eulers theorem.

UNIT-IV

Perfect competition: Assumptions: theory of a firm under perfect competitiondemand and revenue; equilibrium of the firm in the short run and long run; long run industry supply curve: Increasing, decreasing and constant cost industries.

Price controls: Price Ceiling; price floors; production import quotas Welfare: allocative efficiency under perfect competition.

Essential Readings:

ScopeasinVarian,H.R(2010).*IntermediateMicroeconomics,a ModernApproach*(8thed.).W.W.NortonandCompany/AffiliatedEast- West Press (India) given as follows;
Unit1 chapters2to 7
Unit2chapters 8,9,10
Unit3 chapters12to 18
Unit4 chapters19,22,23

SuggestedReadings:

1. Varian.H.R (2010). Intermediate Microeconomics, a ModernApproach (8th ed.) W.W. Norton and Company/Affiliated East-West Press (India).The workbook by Varian and Bergstrom could be used for problems.
2. C. Snyder and W. Nicholson.(2010).Fundamentals ofMicroeconomics, Cengage Learning(India).
3. Maddala,G.S.andMiller,R.L.(2006).Intermediate microeconomics:theory,issues, applications.McGraw-HillCompanies.
4. Koutsoyiannis, A.(2003).Modernmicroeconomics.(2nded.).Paperback
5. B. Douglas Bernheim and Michael D. Whinston. (2009).Microeconomics, Tata McGraw-Hill (India).
6. Bergstrom, T.C., &Varian, H.R.(1990).Workouts inintermediatemicroeconomics. WWNorton.
7. Pindyck, Robert S., and Daniel L. Rubinfeld. Microeconomics. Upper Saddle River, N.J.:Pearson/Prentice Hall, 2009.
8. Mankiw, G. (2000). Microeconomics.In *Principles of Microeconomics*(Vol. 1,pp. 3–6,270–272).

ECO (HS)DSC6: INTRODUCTORY STATISTICAL METHODS FOR ECONOMIC ANALYSIS

Topologyofthecode	Major/Minor
LeveloftheCourse	200
CourseDuration/LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Course Objective: The Objective of this paper is to make students familiar with the use of statistical tools in analyzing economic problems.The course content contains simple tools and techniques, considerednecessary for data collection, presentation and analysis with basic idea of probability concepts which are essential for understanding advance topics related to statistical inference.

Course Outcome: On completion of this course students will be able to: (i) Get familiar with various statistical methods used in summarization and analysis of data. (ii) Understand and able to calculate the measures of central tendency and variability. (iii) Understand about the types of correlations and linear regression for bivariate data and their uses in real life problems. (iv) Understand the basic cocepts of probability to be used in statistical inference.

PedagogyoftheCourseWork:

80% Classroom teaching.
20% Internal Assessment

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

The maximum marks forthepaperwillbe100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

The paper-setter must put a note in the question paper in this regard.

1. There shall be **9** questions in all.

2. The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10x2= 20).

3. Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT–I

Statistics, Data and Statistical Thinking

ScienceofStatistics,Typesofstatisticalapplications,FundamentalelementsofStatistics,Typesof Data, Collecting data. Classification and Tabulation of data, Graphic and Diagrammatic representationof data.The role of statisticsin critical thinking and ethics.

UNIT–II

Numerical Descriptive Measures

Measures of Central Tendency: Mean, Median, Mode, GeometricMean andHarmonicMean; *MeasuresofVariability:*Range,QuartileDeviation,Mean DeviationandStandardDeviation: Interpreting the Standard Deviation.

UNIT–III

Techniques to quantify linear relationship

Correlation and Analysis: Simple Correlation (Karl Pearson's and Rank Correlation);Partial and Multiple Correlation Analyses;

Regression Analysis: EstimationofaRegressionlineinaBivariateDistribution, Least Squares Method: Interpretation of Correlation Coefficients with help of Regression Coefficients, Coefficient of Determination

UNIT–IV

Basics of Probability

Sample spaces and events, Definitions of probability (Classical, Empirical and Axiomatic), Unions and intersections, complementary events.

Theadditiveruleandmutuallyexclusiveevents,conditionalprobability,themultiplicative rule and independent events, random sampling and counting rules.

Essential Readings:

1. Devore, J. (2012). Probabilityand statistics for engineers, 8th ed. Cengage Learning. (Chapters 1, 2and 12- 12.1, 12.2 and 12.5-508 to 511 covering unit III)

2. JamesMcClave,TerrySincich.(2013).StatisticsforBusinessandEconomics,PearsonPublication.

(Chapter 1-1.1to 1.6, Chapter 2-2.1to 2.6, 2.9and 2.10and chapter 3-3.1to 3.8 covering units I, II and IV)

Suggested Readings

1 Larsen,R.,Marx,M.(2018),AnIntroductiontoMathematicalStatisticsand itsApplications,Pearson. Gupta,SC.(8th edition).FundamentalsofStatistics,HimalayaPublishingHouse

2 Miller,Irwinand Miller,Marylees(2006):JohnE.Freund'sMathematicalStatisticswithApplications,(7thEdn.),PearsonEducation,Asia.

ECO (HS)DSC 7:HISTORY OF ECONOMIC THOUGHT

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**200**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Objective:

- Evolution of economic ideas is both a response to contemporary economic problems and a self-conscious attempt to refine earlier ideas to integrate them as a part of current social thought.
- Theobjectiveofthiscourse isto familiarizethestudentswiththe historicalevolutionofeconomic ideas into the contemporary economics.
- The time-testedmethod ofimparting verbalinstructionsthought lectureswould be used. Examples,so far as possible, would be selected from everyday life-experience. The students would be evaluated at the end of semester through subjective type questions/answers (both short and essay type).

PedagogyoftheCourseWork:

80% Classroom teaching.

20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

1. Themaximummarksforthepaperwillbe100.Thequestionpaperwillbeof 80marksandcontinuous evaluation 20 marks. Time allowed will be 3 hours.

2. Thepaper-setter mustputanoteinthequestionpaper inthis regard.

3. Thereshallbe9questionsinall.

4. The first question compulsory comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of two marks (10x2= 20).

5. Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internalchoice. The candidates shall attempt one question fromeach unit. Each question will carry 15 marks (15x4=60).

UNIT–I

The Economic Revolution; Scope and Significance of the Study of History EconomicThought;

Mercantilism: Factors Responsible for the Rise of Mercantilism; The MercantilistConception of foreign trade, State and Money.

Physiocracy: Rise of Physiocracy; Concept of Net Product; Natural Order; Circulation of Wealth

UNIT – II

Classical School: Adam Smith: Theories of Value, Distribution and Growth.

Ricardo Economics: Theories of Value, Distribution and Views on Machinery.

Malthus: Theory of Glut and Glut Controversy.

Karl Marx: Labour Theory Value

UNIT – III

The Marginalist Revolution: Jevons, Walras and Menger

Neo Classical Economics - Main features and contribution.

Keynes: Postulates ofclassical economics and Principle of Effective Demand.

Friedman: Introduction of monetary school.

UNIT – IV

Evolution of Economic thought post world war -II. 1960s: The decade of Development

1970s: Inequality debate (Kuznets etal.)

Phases of Neo-Liberalism: 1980s and 1990s;

Rise of Neo-Nationalism in 2000s

RECOMMENDED READINGS

1. Jevons,W.S.(1888):TheTheoryofPoliticalEconomy,MacmillanandCo.,NewYorkandLondon. Chapter 1 and 6.
2. Marshall,(1890):Principles of Economics 8th Edition,Macmillan andCo., NewYorkandLondon Book III, Chapter 6 and Appendix C51.
3. Pigou,A.C.(1920):TheEconomicofWelfare,MacmillanandCo., NewYorkandLondon,Chapter20.
4. KeynesJ.M.(1936):TheGeneralTheoryofEmployment,InterestandMoney,ReprintedbyAtlantic Publication, New Delhi. Chapter 2, 3, 8,9,10 and12.
5. Kalecki M. (1971): Selected Essays on the Dynamics of Capitalism, University Press, Cambridge.
6. Friedman,M.(1962):CapitalismandFreedom,TheUniversityofChicago Press,Chicago andLondon. Foster, J. B. and: The Endless Crisis, Monthly Review Press. McChesney (2012)
7. Harvey, D.(2005): ABriefHistoryofNeoliberalism,Oxford UniversityPress,NewYork.
8. Roy, R.K. (2010)UniversityPress, New York.

FurtherReadings:

1. Screpanti(2006): OutlineofHistoryofEconomicThought.Oxford UniversityPress, Calcutta...
2. Blaug, M (1983): Economic Theory in Retrospect. Vikas Publishing House, New Delhi.
3. Dasgupta,AK(1986):EpochsofEconomicTheory.OxfordUniversityPress,NewDelhi
4. Collander, D C (1984): Neo-Classical Political Economy, Ballinger, Cambridge, M A.
5. Mandel, E (1971): Marxist Economic Theory. Upra & Co.
6. Schumpeter,JA(1954):HistoryofEconomicAnalysis,OxfordUniversityPress,Oxford

ECO (HS) SEC-3: WORKING WITH OPEN-SOURCE SOFTWARE (PYTHON AND R)

Topology of the code	Skill Enhancement Course
Level of the Course	200
Credits	3
Course Duration/LTP (Lecture, Tutorials, Practical)	L=30hours(2 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Total Marks	75 (Including Internal Assessment =15)
Time Allowed for Examination	3 Hrs.

Course Objective: -

- To provide students with basic knowledge of Open-Source Software i.e. Python and R to enhance their competency in using these tools for data analysis.
- Course Outcome: It will introduce students to foundational skills in Python and R, enabling them to pursue basic mathematic and statistical operations and analyze the datasets.

Pedagogy of the Course work: -

80% Classroom Teaching

20% Internal Assessment

UNIT-I

Introduction to R: Overview of R Programming language, Needs and Applications, Significance in Economics, Installing R and R Studio, Understanding R Studio Interface, Understanding Basic Syntax, Installing and loading Packages.
Variables Types and Objects: Numeric, Integers, Characters, Dates, Vectors, Matrix, Array, List, Data Frames.

Operator and Comments: Assignment, Relational, Arithmetic, Logical and Comments. Flow Control: If, If-else, For loop, Break.

UNIT-II

Basic Inbuilt Functions: Mathematical Functions, Statistical Functions and Plotting Functions.
Working with Data: Reading and writing data from tables, CSV files, and Excel worksheets. Creating and removing variables in a data frame, replacing values of variables, and replacing missing values. **Basic Statistics-** Mean, Median, Standard Deviation and Range.

UNIT-III

Introduction to Python: Overview of Python, Uses and its Applications, Installation Process, Python Syntax, Python Scripts, Modules and Packages.
Objects in Python: Variables, Objects in Numpy and Pandas

UNIT-IV

Import and Export of Data Files: Working with data frames, Importing files from Excel, .csv files, and other sources.

Machine Learning: Brief Introduction of Machine Learning, Regression Analysis, Scikit-learn.

Essential/ Recommended Readings

1. Using R for Introductory Econometrics (2020) by *Florian Heiss and Daniel Brunner* (2nd Edition)
2. *R Programming Fundamentals* (2018) by Kaelen Medeiros
3. R Cookbook, James J.D Long and Paul Teetor, 2nd Edition, O'Reilly, 2019
4. *Learning R Programming* (2016) by Kun Ren
5. *Hands-On Programming with R* (2014)Garett Gloremund
6. Using Python for Introductory Econometrics (2020) by *Florian Heiss and Daniel Brunner (1st Edition)*
7. Python for Data Analytics; Data Analysis and Science Using Pandas, Matplotlib and the Python by *Fabio Nelli* (3rd Edition)
8. Python Programming: An Introduction to Computer Science by *John Zelle*
9. Introduction to Machine Learning with Python: A Guide for Data Scientists by *Andreas Muller* (2016).

Semester III

COMPULSORY PUNJABI

FYUGP B.A /B.A. (Hons.) in Economics

ਲਾਜ਼ਮੀ ਪੰਜਾਬੀ

ਨਾਟਕ, ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਅਤੇ ਗੁਰਮੁਖੀ ਲਿਪੀ

Natak, Punjabi Bhasha Ate Gurmukhi Lipi

B.A. (ECO) PUN (COMPL-01)-301

Semester- III

ਕ੍ਰੈਡਿਟ-2

Credit distribution, Eligibility and the Pre- Requisites of the Course

Course title & Code	Credits	Credit distribution of the Course			Eligibility criteria	Pre- requisite of the course (if any)
		Lecture	Tutorial	Practical/ Practice		
Natak, Punjabi Bhasha Ate Gurmukhi Lipi B.A. (ECO) PUN (COMPL-01)-301	2	3	1	0	Class 10 & 12th pass in any stream	Studied Punjabi up to 10th Standard or working knowledge of Punjabi Language

Credit: 2

Learning objective

- ਇਸ ਕੋਰਸ ਰਾਹੀਂ ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਨਾਟਕ ਸੰਬੰਧੀ ਮੁੱਢਲਾ ਗਿਆਨ ਪ੍ਰਾਪਤ ਕਰਨਗੇ।
- ਇਸ ਕੋਰਸ ਰਾਹੀਂ ਵਿਦਿਆਰਥੀ ਨਾਟਕਕਾਰ ਈਸ਼ਵਰ ਚੰਦਰ ਨੰਦਾ ਦੇ ਨਾਟਕ ਰਾਹੀਂ ਨਾਟ-ਕਲਾ ਤੇ ਰੰਗਮੰਚ ਬਾਰੇ ਗਿਆਨ ਹਾਸਿਲ ਕਰਨਗੇ।
- ਇਸ ਕੋਰਸ ਰਾਹੀਂ ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਦੀਆਂ ਉਪ-ਭਾਸ਼ਾਵਾਂ ਦੀ ਵਿਲੱਖਣ ਪਛਾਣ ਤੇ ਸ਼ਬਦਾਵਲੀ ਸੰਬੰਧੀ ਗਿਆਨ ਹਾਸਿਲ ਕਰਨਗੇ।
- ਵਿਦਿਆਰਥੀਆਂ ਵਿੱਚ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਅਤੇ ਗੁਰਮੁਖੀ ਲਿੱਪੀ ਦੇ ਸਬੰਧ ਨੂੰ ਸਮਝਣ ਦੀ ਸੂਝ ਪੈਦਾ ਹੋਵੇਗੀ।

Learning outcomes

- ਇਸ ਨਾਲ ਵਿਦਿਆਰਥੀਆਂ ਅੰਦਰ ਪੰਜਾਬੀ ਨਾਟਕ ਦੀ ਵਿਧਾ ਸੰਬੰਧੀ ਮੁੱਢਲੀ ਸੂਝ ਪੈਦਾ ਹੋਵੇਗੀ।
- ਇਸ ਨਾਲ ਵਿਦਿਆਰਥੀ ਨਾਟਕ ਦੇ ਹਵਾਲੇ ਨਾਲ ਪੰਜਾਬੀ ਨਾਟ-ਕਲਾ ਤੇ ਰੰਗਮੰਚ ਦੀ ਬਣਤਰ ਸੰਬੰਧੀ ਵਿਵੇਕ ਹਾਸਿਲ ਕਰ ਪਾਉਣਗੇ।
- ਇਸ ਕੋਰਸ ਰਾਹੀਂ ਵਿਦਿਆਰਥੀਆਂ ਵਿਚ ਪੰਜਾਬੀ ਦੀਆਂ ਉਪ-ਭਾਸ਼ਾਵਾਂ ਦੀ ਵਿਲੱਖਣ ਪਛਾਣ ਸੰਬੰਧੀ ਸਮਝ ਪੈਦਾ ਹੋਵੇਗੀ।
- ਇਸ ਕੋਰਸ ਰਾਹੀਂ ਵਿਦਿਆਰਥੀਆਂ ਵਿਚ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਅਤੇ ਗੁਰਮੁਖੀ ਲਿੱਪੀ ਦੇ ਸਬੰਧ ਨੂੰ ਸਮਝਣ ਦੀ ਸੂਝ ਪੈਦਾ ਹੋਵੇਗੀ।

ਕ੍ਰੈਡਿਟ : 2

ਨਿਰਧਾਰਿਤ ਕੋਰਸ ਦਾ ਸਮਾਂ: 60 ਘੰਟੇ

ਕੁੱਲ ਅੰਕ = 50

L-100

ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ ਅੰਕ = 10

ਲਿਖਤੀ ਅੰਕ = 40

(ਨੋਟ: ਇਸ ਕੋਰਸ ਵਿਚ ਦੋ ਯੂਨਿਟ ਹੋਣਗੇ। ਯੂਨਿਟ ਪਹਿਲਾ ਵਿਚ ਨਾਟਕ ਨਾਲ ਸੰਬੰਧਿਤ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣਗੇ।ਇਸ ਯੂਨਿਟ ਦੇ ਕੁੱਲ 20 ਅੰਕ ਹੋਣਗੇ। ਯੂਨਿਟ ਦੂਜਾ ਵਿਚ ਗੁਰਮੁਖੀ ਲਿਪੀ ਦੇ ਮਹੱਤਵ ਤੇ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦੀਆਂ ਉਪ ਭਾਸ਼ਾਵਾਂ ਨਾਲ ਸੰਬੰਧਿਤ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣਗੇ ਅਤੇ ਕੁੱਲ 20 ਅੰਕ ਹੋਣਗੇ।)

ਯੂਨਿਟ ਪਹਿਲਾ : ਈਸ਼ਵਰ ਚੰਦਰ ਨੰਦਾ, ਵਰ-ਘਰ, ਲਾਹੌਰ ਬੁੱਕ ਸ਼ਾਪ, ਲੁਧਿਆਣਾ, 1930

30 ਘੰਟੇ

(ੳ) ਨਾਟਕ ਦਾ ਵਿਸ਼ਾ-ਵਸਤੂ ਤੇ ਕਲਾ ਪੱਖ

(ਅ) ਕਿਸੇ ਪਾਤਰ ਦਾ ਪਾਤਰ-ਚਿਤਰਣ

(ੲ) ਨਾਟਕ ਦੀ ਰੰਗਮੰਚੀ ਸਾਰਥਕਤਾ

ਯੂਨਿਟ ਦੂਜਾ : ਗੁਰਮੁਖੀ ਲਿਪੀ ਅਤੇ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦੀਆਂ ਉਪ-ਭਾਸ਼ਾਵਾਂ 30 ਘੰਟੇ

(ੳ) ਗੁਰਮੁਖੀ ਲਿੱਪੀ ਦਾ ਮਹੱਤਵ ਅਤੇ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਲਈ ਇਸਦੀ ਸਾਰਥਕਤਾ

(ਅ) ਭਾਸ਼ਾ ਅਤੇ ਉਪ-ਭਾਸ਼ਾ: ਅੰਤਰ ਸੰਬੰਧ ਤੇ ਨਿਖੇੜਾ

(ੲ) ਪੰਜਾਬੀ ਦੀਆਂ ਉਪ-ਭਾਸ਼ਾਵਾਂ

ਨੋਟ: ਲਾਜ਼ਮੀ ਪੰਜਾਬੀ ਵਿਸ਼ੇ ਦੇ ਕ੍ਰੈਡਿਟ 2 ਹੋਣਗੇ ਅਤੇ ਪੇਪਰ 50 ਅੰਕਾਂ ਦਾ ਹੋਵੇਗਾ। ਵਿਦਿਆਰਥੀ ਹਫ਼ਤੇ ਵਿਚ ਘੱਟੇ-ਘੱਟ ਪੰਜ ਦਿਨ ਪੜ੍ਹਨਗੇ।

ਨਿਰਧਾਰਿਤ ਪੁਸਤਕਾਂ

1. ਈਸ਼ਵਰ ਚੰਦਰ ਨੰਦਾ , ਵਰ-ਘਰ, ਲਾਹੌਰ ਬੁੱਕ ਸ਼ਾਪ, ਲੁਧਿਆਣਾ, 1930
2. ਸਤੀਸ਼ ਕੁਮਾਰ ਵਰਮਾ, ਪੰਜਾਬੀ ਨਾਟਕ ਦਾ ਇਤਿਹਾਸ, ਪੰਜਾਬੀ ਸਾਹਿਤ ਅਕਾਦਮੀ, ਦਿਲੀ, 2005
3. ਕਮਲੇਸ਼ ਉੱਪਲ, ਪੰਜਾਬੀ ਨਾਟਕ ਅਤੇ ਰੰਗਮੰਚ, ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ, ਪਟਿਆਲਾ, 2004
4. ਸ. ਸ. ਖਹਿਰਾ, ਪੰਜਾਬੀ ਭਾਸ਼ਾ : ਵਿਆਕਰਣ ਅਤੇ ਬਣਤਰ, ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ, ਪਟਿਆਲਾ।
5. ਜੋਗਿੰਦਰ ਸਿੰਘ ਪੁਆਰ ਅਤੇ ਹੋਰ, ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦਾ ਵਿਆਕਰਨ, ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਅਕਾਦਮੀ, ਜਲੰਧਰ 1999
6. ਬੂਟਾ ਸਿੰਘ ਬਰਾੜ (ਡਾ.), ਪੰਜਾਬੀ ਵਿਆਕਰਨ: ਸਿਧਾਂਤ ਅਤੇ ਵਿਹਾਰ, ਚੇਤਨਾ ਪ੍ਰਕਾਸ਼ਨ, ਲੁਧਿਆਣਾ

ਪੇਪਰ ਸੈਟਰ/ ਅਧਿਆਪਕ ਲਈ ਹਦਾਇਤਾਂ ਅਤੇ ਪ੍ਰਸ਼ਨ-ਪੱਤਰ ਦੀ ਰੂਪ-ਰੇਖਾ :

1. ਪ੍ਰਸ਼ਨ-ਪੱਤਰ ਦੇ ਕੁੱਲ 50 ਅੰਕ ਹਨ ਜੋ ਕਿ 2 ਕ੍ਰੈਡਿਟ ਦੇ ਬਰਾਬਰ ਹੈ ਅਤੇ ਕੋਰਸ ਦਾ ਨਿਰਧਾਰਤ ਸਮਾਂ 60 ਘੰਟੇ ਹੋਵੇਗਾ।
2. ਲਿਖਤੀ ਪ੍ਰਸ਼ਨ-ਪੱਤਰ 40 ਅੰਕਾਂ ਦਾ ਹੋਵੇਗਾ।
3. ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ ਦੇ 10 ਅੰਕ ਹੋਣਗੇ।
4. ਪੇਪਰ ਦਾ ਸਮਾਂ ਤਿੰਨ ਘੰਟੇ ਹੋਵੇਗਾ।

ਯੂਨਿਟ ਅਤੇ ਅੰਕਾਂ ਦੀ ਵੰਡ

1. ਪਹਿਲਾ ਪ੍ਰਸ਼ਨ ਲਾਜ਼ਮੀ ਹੋਵੇਗਾ, ਜੋ ਸਮੁੱਚੇ ਸਿਲੇਬਸ 'ਤੇ ਆਧਾਰਤ ਹੋਵੇਗਾ। ਇਸ ਵਿੱਚ 7 ਪ੍ਰਸ਼ਨ ਪੁੱਛੇ ਜਾਣਗੇ ਜਿਨ੍ਹਾਂ ਵਿੱਚੋਂ ਕਈ 5 ਪ੍ਰਸ਼ਨਾਂ ਦੇ ਲਘੂ ਉੱਤਰ ਦੇਣੇ ਹੋਣਗੇ। ਇਹ ਪ੍ਰਸ਼ਨ 5X2=10 ਅੰਕਾਂ ਦੇ ਹੋਣਗੇ।
2. ਯੂਨਿਟ ਇਕ ਵਿਚੋਂ 'ਵਰ-ਘਰ' ਨਾਟਕ ਵਿਚੋਂ ਵਿਚ ਨਾਟਕ ਦੇ ਵਿਸ਼ਾ-ਵਸਤੂ, ਕਲਾ ਪੱਖ, ਨਾਟਕ ਦੇ ਕਿਸੇ ਪਾਤਰ ਦਾ ਪਾਤਰ-ਚਿਤਰਣ ਅਤੇ ਨਾਟਕ ਦੀ ਰੰਗਮੰਚੀ ਸਾਰਥਕਤਾ ਬਾਰੇ ਤਿੰਨ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣਗੇ।ਵਿਦਿਆਰਥੀ ਨੇ ਇਹਨਾਂ ਵਿੱਚੋਂ ਦੋ ਸੁਆਲ ਕਰਨੇ ਹੋਣਗੇ। ਇਸ ਭਾਗ ਦੇ 8X2=16 ਅੰਕ ਹਨ।

.3 ਯੂਨਿਟ ਦੋ ਵਿੱਚੋਂ ਗੁਰਮੁਖੀ ਲਿਪੀ ਦੇ ਮਹੱਤਵ ਅਤੇ ਪੰਜਾਬੀ ਦੀਆਂ ਉਪ-ਭਾਸ਼ਾਵਾਂ ਬਾਰੇ ਤਿੰਨ ਪ੍ਰਸ਼ਨ ਪੁੱਛੇ ਜਾਣਗੇ, ਇਹਨਾਂ ਵਿੱਚੋਂ ਵਿਦਿਆਰਥੀ ਨੇ ਦੋ ਸੁਆਲ ਕਰਨੇ ਹੋਣਗੇ। ਇਸ ਭਾਗ ਦੇ 7X2=14 ਅੰਕ ਹਨ।

हिन्दी

समसटर–III

MIL-HIN-01-301

Course Title : हिन्दी **औपचारिकलखन**
HINDI AUPCHARIKLEKHAN
Course Code : MIL-HIN-01-301
Typology of Course: Modern Indian Language
Course Duration : 30 hours in One semester/2 hrs per week
Credit : 2
Maximum Marks : 50 (45 Theory+5 Internal Assessment)

पाठयकरम का उ□□शय–(Course Objectives)

- विद्यार्थीकीभाषायी□क्षताऔरलेखन-कौशलकोबढ़ावा□ना।
- कार्यालयीवव्यावसायिकहिन्दीकीसमझविकसितकरना।
- हिन्दीभाषा-□क्षताऔरतकनीककेअन्तःसंबंधसेपरिचितकरना।
- कार्यालयोंमेंव्यवहारिककार्यकेविभिन्नपक्षोंसेअवगतकराना।
- हिन्दीप्रयोगसेजुड़ेफील्डवर्कआधारितविश्लेषणऔरलेखनपरबल□ना।

पाठयकरम का परिणाम–(Course Learning Outcomes)

- विद्यार्थीकार्यालयीऔरव्यावसायिकहिन्दीकीविशेषताओंसेपरिचितहोंगे।
- सूचनाकेअधिकारकेलिएलेखनकरसकेंगे।
- कार्यालयोंमेंहोनेवालेव्यवहारिककार्यकाज्ञानप्राप्तहोगा।
- मार्केटसर्वेक्षणहेतुप्रश्नावलीकानिर्माणतथाउसकाविश्लेषणकरनाजानेंगे।
- विद्यार्थीटिप्पण,प्रारूपण,प्रतिवे□न,प्रेस-विज्ञाप्तितैयारकरनासीखसकेंगे।

इकाई-I
लेखन□कषताकाविकास

Max.Marks:50
Theory : 45
Internal Assessment : 05
Time : 03 Hours

1. निमनलिखितविषयोमेंसे**सपरशनपछजाएग।**

क. कार्यालयीहिन्दी

ख. व्यावसायिकहिन्दी

ग. टिप्पणऔरप्रारूपणकासामान्यपरिचय

घ. प्रतिवेदनऔरप्रेसविज्ञाप्ति

उपरोक्तविषयोंसे**चार**प्रश्नपूछेजाएंगे।किन्हीं□ोंकेउत्तरदेनेहोंगे। (**शब्द**सीमा 150) **अंक-10**

2. कार्यालयीटिप्पणियाँ(सूचीसंलग्न)

प□रहटिप्पणियाँ दी जाएंगी। किन्हीं□**स**का हिन्दी रूप लिखनाहोग **अंक-10**

इकाई-II
औपचारिक लेखन क परकार

1. क□र्य□लय□ी(केवलश□सक□ीय)पत्र-**व्य**
□ोपत्रदिएजाएंगे□**एक**लिखनाहोगा। **अंक-10**

2. निमंत्रणपत्रएवंविज्ञापनकाप्रारूपतैयारकरना।
(□ोमेंसे**एक**) **अंक-8**

3. स्ववृतलेखन(Bio-data) **अंक-7**

सहायकपुस्तक

1. प्रयोजनमूलकऔरकार्यालयीहिन्दी,कृष्णाकुमारगोस्वामी
2. प्रयोजनमूलकहिन्दीकीनईभूमिका,कैलाशचन्द्रपाण्डेय
3. प्रयोजनमूलकहिन्दी:सिद्धांतऔरप्रयोग,दंगलझाल्टे
4. प्रशासनिकहिन्दी,हरिमोहन,तक्षशिलाप्रकाशन
5. प्रशासनिक हिन्दी, □ शोक और □ शोक, □ रुण पब्लिकेशन, Chandigarh.

मलयाकनप□धति(Assessment method)

कुल□ंक–50
लिखितपरीक्षा–45□ंक
आंतरिकमूल्यांकन–5

टिपपणिया-

1. Abriefnoteisplaced below	संक्षिप्तटिप्पणियाँनीचेप्रस्तुतहै।
2. Acknowledgereceiptof this	□सकीपावतीभेजिए।
3. Actionasproposedmaybetaken	यथाप्रस्तावितकार्यवाहीकीजाए।
4. Agendaofthemeetingisputup	बैठककीकार्यसूचीप्रस्तुतहै।
5. Applicationmayberejected	आवेदन□ स्वीकारकरदियाजाए।
6. Approvedasproposed	प्रस्तावके□ नुसार□ नुमोदित।
7. Administrativeapprovalmaybeobtained	प्रशासनिक□ नुमोदनप्राप्तकियाजाए।

8. Awaitreply	उत्तरकीप्रतीक्षाकरें।
9. Awaitfurtherreport	औरविवरणकीप्रतीक्षाकरें।
10. Ascertainthispositionplease	कृपयास्थितिकापतालगायें।
11. Amendeddraftissubmittedforapproval	संशोधितप्रारूप□ वलोकनार्थप्रस्तुतहै।
12. Briefresumeofthecaseisgivenbelow	मामलेकासंक्षिप्तसारनीचेप्रस्तुतहै।
13. Callforexplanation	स्पष्टीकरणमाँगे।
14. Callforreport	रिपोर्टमँगवाए।
15. Complywiththeorders	आदेशोंकापालनकरें।
16. Clarifythepositionplease	कृपयास्थितिकोस्पष्टकरें।
17.Copyenclosedforreadyreference	सुलभसंदर्भकेलिएप्रतिलिपिसंलग्नहै।
18. Copyisenclosed	प्रतिलिपिसंलग्नहै।
19. Copyforwardedforinformationand Necessary action	प्रतिलिपिसूचनार्थएवंआवश्यककार्यवाही केलिए प्रेषित है।
20. Casemaybekeptpending	मामलेको□ भी□ निर्णीतरखाजाए।
21. Delayshouldbeavoided	विलम्बनकियाजाए।
22. Disciplinaryproceedingsmaybeinitiated	□ नुशासनिककार्यवाहीशुरूकीजाए।
23. Draftreplyisputup	उत्तरकामसौदाप्रस्तुतहै।
24. Enquirymaybeconducted	जाँचकीजाए।
25. Expediteaction	शीघ्रकार्यवाहीकरें।
26. Explanationmaybecalledfor	स्पष्टीकरणमाँगाजाए।
27. Factsforthecasemaybeputup	मामलोंकेतथ्यप्रस्तुतकरें।
28. Forperusalandreturn	देखकरलौटानेकेलिए।
29. Forcommentsplease	कृपयाटिप्पणीदें।
30. Forsympatheticconsideration	सहानुभूतिपूर्णविचारकेलिए।
31. Forwardedandrecommended	प्रेषितऔरसंस्तुत।
32. Iconcurwiththeproposal	प्रस्तुतप्रस्तावसेमेरीसहमतिहै।
33. Ihavenoremarkstooffer	मुझेकोईटिप्पणीनहींकरनीहै।
34. Informallconcerned	सभीसंबंधितव्यक्तियोंकोसूचितकरें।
35. Issueacircular	परिपत्रजारीकरें।
36. Keeppending	निर्णयकेलिएरोकरखें।
37. Kindlyaccordsanction	कृपयास्वीकृतिदीजिए।
38. Kindlyconfirm	कृपयापुष्टिकीजिए।
39. Matterisunderconsideration	मामलाविचाराधीनहै।
40. Noactionisnecessary	कोईकार्यवाहीउपेक्षितनहींहै।
41. Needfulhasbeendone	आवश्यककार्यवाहीकीजाए।
42. Ordermaybeissued	आदेशजारीकियाजाए।
43. Pleaseseetheproceedingnotes	कृपयापिछलीटिप्पणियाँदेखलें।
44. Putuptherelevantpapers	संबंधितकागज़प्रस्तुतकरें।

45. Remindermaybesent	□ नुस्मारकभेजाजाए।
46. Reportcomplianceimmediately	□ नुपालनकरकेतुरंतसूचितकरें।
47. Specificreasonmaybegiven	विशिष्टकारणदिएजाए।
48. Submittedforinformation	सूचनाकेलिएप्रस्तुतहै।
49. Thismaybetreatedasconfident	□सेगोपनीयसमझाजाए।
50. Verifiedandfoundcorrect	जाँचाऔरसहीपाया।

Semester-IV

ECO(HS) DSC8: INTER MEDIATE MICRO ECONOMICS– II

Topologyofthecode	Major/Minor
LeveloftheCourse	200
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+Internal Assessment20 marks
Time	3 hours

CourseDescription

This course is a sequel to Intermediate Microeconomics I. The emphasis will be on giving conceptual clarity to the student coupled with the use of mathematical tools and reasoning.It covers general equilibrium and welfare, imperfect markets and topics underinformation economics.

PedagogyoftheCourseWork:

80%Classroomteaching.

20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

InstructionsforPaper-setterandcandidates:

The maximum marks for the paper will be 100.The question paperwillbe of 80marks andcontinuousevaluation20marks. Time allowedwillbe 3 hours.
Thepaper-settermustputanoteinthequestionpaperinthisregard.

- Thereshallbe**9**questions inall.
- The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

- Rest of the paper shall contain four units. Two questions shall be asked from eachunit andthe candidatesshallbegiveninternalchoice. Thecandidatesshall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT-I

ImperfectCompetition

Monopoly;pricing;measurementofmarketpower; Multiplant monopoly Price discrimination; Inter temporal; peak-load pricing; two-part tariff Regulation of Monopoly
Monopolisticcompetition:priceand output determination Economic efficiency and resource allocation

UNIT-II

ImperfectCompetition

Oligopoly;Priceleadership;Cournot,StackelbergandBertrandModels

Gametheoryandcompetitivestrategies:DominantStrategy;Mixedstrategies;Prisoners Dilemma;Repeated andSequentialgames

UNIT-III

General Equilibrium, Efficiency and Welfare Equilibriumand efficiency under pure exchange and production;overallefficiencyWelfareeconomics;Social welfare, welfare maximisation, envy and equity,

UNIT-IV

MarketFailure

Externalities; Productionand Consumption, CoaseTheorem, Market Signals, Public Goods: Provision of Public Goods,Free Riding,DifferentLevels of PublicGoods, Asymmetric Information: Market for Lemons, Moral Hazard, Adverse Selection, Signaling.

Essential Readings:

ScopeasinVarian.H.R(2010).*IntermediateMicroeconomics,a ModernApproach*(8thed.)W.W.NortonandCompany/AffiliatedEast-West Press (India) given as follows:

Unit1chapters24,25

Unit2chapters27,28

Unit3 chapters31,32,33

Unit4 chapters34,36,37

SuggestedReadings:

1. Varian.H.R (2010). *Intermediate Microeconomics, a Modern Approach* (8th ed.)W.W. Norton and Company/Affiliated East-West Press (India).The workbook by Varian and Bergstrom could be used for problems.

2. C. Snyder and W. Nicholson.(2010).*Fundamentals of Microeconomics,*Cengage Learning (India).

3. Maddala, G.S. and Miller, R. L. (2006). *Intermediate microeconomics: theory, issues, applications*. McGraw-Hill Companies.

4. Koutsoyiannis,A.(2003).*Modernmicroeconomics.(2nd ed.).*Paperback

5. B. Douglas Bernheim and Michael D. Whinston. (2009). Microeconomics Tata McGraw-Hill(India).

6. Bergstrom,T.C.,& Varian,H.R. (1990).Workouts in intermediate microeconomics.WWNorton.

7. Pindyck,RobertS.,andDanielL.Rubinfeld.Microeconomics.Upper SaddleRiver,N.J.:Pearson/Prentice Hall, 2009.

8. Mankiw,G.(2000). Microeconomics.In*Principlesof Microeconomics*(Vol. 1,pp.3–6,270–272).

ECO(HS) DSC9-INTERMEDIATE MACRO ECONOMICS

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**200**
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

CourseObjective

Macroeconomics or aggregative economics analysis establishes the functional relationship between the large aggregates. The aggregated analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essentialfor the propercomprehensionofthe different issuesand policies. Macroeconomicsnow is not only a scientific method of analysis; but also a bodyof empiricaleconomic knowledge.

The paper entitled "Intermediate Macroeconomics" equips the students at the undergraduatelevel to understand systemic facts and latest theoretical developments for empirical analysis. The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course Outcomes

After completion of the course, a student must be capable of understanding the determinants of moneydemandandsupply. Also thepapershallinculcatetheabilitiesamong studentsto develop open economy IS-LM based models to work out the key macroeconomic targets. An understanding shall be developed on behaviour of general price level and its use as a key policy instrument to target real variables such as unemployment and output. The theories of explaining trade cycles shall make a student capable of understanding nature and causes of fluctuations in economic activity.

PedagogyoftheCourseWork: 80% Classroom Teaching

20% Internal Assessment

INSTRUCTIONSFORTHEPAPER-SETTERAND CANDIDATES

1. Thesyllabusofthispaper hasbeendividedinto four units.

2. Thereshallbe9questionsinall.

3. The first question, which would be compulsory, shall be short answer type (word limit 25-30 each).It would carry15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answertype question would carry 2 marks ($10 \times 2 = 20$).

4. Rest of the paper shall contain 4 units. Each unit shall have two questions and thecandidates shall be required to attempt one question fromeach Unit— 4 in all. Each question shall carry15 marks ($15 \times 4 = 60$).

UNIT-I

Supply of Money: Theoretical Debate and Empirical Attempts to define money; Components of Supply of Money, Credit Creation by Commercial Banks, Money Multiplier.

Demand for Money: Classical Quantity Theory, Keynesian Theory, Baumol and Tobin's Contributions. Friedman's Restatement of Quantity Theory of Money.

UNIT-II

IS and LM Framework: Derivation, Properties, Shifts and Rotations of IS and LM Curves under closed and open economy systems. Derivation, Properties, Shifts, and Rotations of BP Curve. Simultaneous Equilibrium in Money and Product Markets. Impact of Opening-up on simultaneous equilibrium (i.e., IS-LM-BP simultaneous equilibrium).

MonetaryandFiscalPolicies: Objectives,ConflictsamongObjectives.RelativeEffectivenessof Monetary and Fiscal Policies under Different Situations in IS-LM-BP Framework. International Trinity and Quadrilemma choices under IS-LM-BP framework.

UNIT-III

Inflation: Effects of Inflation. Theories of Inflation: Quantity Theory, Keynesian Theory, Monetarist views on Inflation, Modern theory of Inflation, Structural Theory.

The menu of policy choices: Philips Curve Analysis –Short Run and Long Run views. The Monetarist-Keynesian Debate and the Phillips Curve.

UNIT-IV

Trade Cycle: Hansen–Samuelson Accelerator–Multiplier Interaction Model, Hicks Model, Kaldor Model, Goodwin model of endogenous cycles.

RecommendedReadings:

1. Ackley,G.(2007).MacroeconomicTheory. Macmillan.
2. Allen,R.G.D.(1967). Macro-EconomicTheory:AMathematicalTreatment.Palgrave Macmillan.
3. Beckerman,W.(1980).IntroductiontoNationalIncomeAnalysis, LittlehamptonBookServices Ltd.
4. Blanchard,O.(2009).Macroeconomics.NewDelhi:PearsonEducationInc.
5. Blanchard, O.J. & Fischer. (1996). Lectures on Macroeconomics, Delhi: PHI Learning Pvt. Ltd.
6. Branson, W. H. (1989). Macro Economic Theoryand Policy (3rd ed.). Harper Collins. Cooper,R.,Edey,HaroldC.,andPeacock,A.T.(2003). NationalIncomeandSocialAccounting. Routledge.
7. D'Souza,E.(2008).Macroeconomics. PearsonEducation.
8. Dornbusch,R.,Fisher,S&Startz, R.(2002).Macroeconomics, (9thed.). NewDelhi: TataMcGrawHills.
9. Levacic,R.&Rebmann,A.(1982). Macroeconomics:AnIntroductionto Keynesian Neoclassical Controversies. Macmillan.
10. Miller,R.L.&Pulsinelli,R.(1986).Macroeconomics.NewYork:Harper&Row.
11. Ola Olsson (2010). Essentials of Advanced Macroeconomic, Routledge.
12. Romer,D.(1996).AdvancedMacroeconomics.McGraw-Hill.

13. UmaDattaRoyChoudhury(2000). NationalIncomeAccounting,Macmillan.

FurtherReadings:

1. Abel,A.B.,Bernanke,B.S.(2007). Macroeconomics.NewDelhi:PearsonEducationInc.

2. Dornbusch, R. (1980). Open Economy Macroeconomics. New York: WeltwirtschaftlichesArchive.

3. Frisch,H.(1983).TheoriesofInflation.Cambridge:CambridgeUniversityPress. Froyen, R. T. (2002). Macro Economics. New Delhi: Pearson Education Inc.

4. Gordon,R.J.(2012).Macroeconomics. Delhi:PHILearningPvt.Ltd.

5. Heijdra,B.J.(2017).TheFoundationsofModernMacroeconomics.OxfordUniversityPress. Hicks, J. R. (1950). A Contribution to the Theory of Trade Cycle Oxford: Clarendon Press.

6. Lucas,R.(1981).StudiesinBusinessCycleTheory,Massachusetts:MITPressCambridge. Makin, A. J. (2002). International Macroeconomics. Princeton Hall.

7. Mankiw,N. G.&Romer,D.(1991).NewKeynesianEconomics.MIT,2Volumes.

8. Mathews, R. C. O. (1964). The Trade Cycle. Cambridge: Cambridge University Press.

9. Pentecost,E.(2000). Macroeconomics–AnOpenEconomyApproach. MacMillanPressLtd.

10. Raghbendra, J. (1964). Contemporary Macroeconomics: Theory and Policy. New Age International Publishers.

11. Sherffrin,S.(1999).RationalExpectations.CambridgeUniversityPress.

12. Snowden,B&Vane,H.R.(2005). ModernMacroeconomics:ItsOrigin,Development and Current Stat., UK: Edward Press.

13. Wendy,Carlin/Soskice,D.(2007).Macroeconomics:Imperfections,InstitutionsandPolicies. Oxford University Press.

ECO (HS) DSC10–INTRODUCTORY ECONOMETRICS

Topologyofthecode	Major/Minor
LeveloftheCourse	200
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

CourseObjective: Application of economic theory need a reasonable understanding of economic relationship and relevant statistical methods. The econometric theory thus becomes a very powerful tool for understanding of applied economic relationships and for meaningful research in economics. This paper accordingly is devoted to equip the students with basic theory of econometrics and relevant applications of the methods. The topics covered in the course include various problems faced in estimation of both single equations and simultaneous equations models.

Besides the time-tested method of imparting verbal instructions through lectures, description and derivation of econometric models would be demonstrated by extensive useofblackboard. Examples, in so far as possible, would be selected fromeveryday life/experience. The feedback would be sought through the students writing assignments and written examinations.

PedagogyoftheCourseWork:

80%Classroomteaching.

20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

- The maximum marks for the paper will be 100. The question paper will be of 80 marks andcontinuous evaluation 20 marks. Time allowed will be 3 hours. The paper-setter must put a note in the question paper in this regard.

- There shall be **9** questions in all. The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each questionshall beof**two** marks(10 x2=20) Rest of thepaper shallcontain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT-I

Introduction: NatureandScopeofEconometrics,MethodologyofEconometrics. Specification of an econometric model. Nature of Data used in estimation.

Simple Regression Models: Estimation of model by method of Ordinary Least Squares; PropertiesoftheLeast Squaresestimators, theGauss-Markovtheorem. Goodness offit -R Square & Adjusted R Square. Testing the significance of Regression Coefficients.

UNIT-II

Multiple Regression Models: Estimation of model by method of Ordinary Least Squares; Properties of the Least Squares estimators, the Gauss-Markov theorem.

Specification Errors: Types of Specification errors. Consequences and Remedial measures for Omission of a relevant variable; and Inclusion of irrelevant variable.

UNIT-III

ErrorsofMeasurement:ErrorsofMeasurementinDependentandIndependent variables.

EncompassingandNon-EncompassingModels: Introduction.ModelSelectionCriteria.

UNIT-IV

Econometrics Problems: Nature, Consequences, Detection and Remedial measures of the problems of Multicollinearity, Heteroscedasticity and Autocorrelation.

EssentialReading:

GujaratiDamodarN.(2004):BasicEconometrics(4thed.),Singapore.McGrawHillsCo.

FurtherReadings:

1. Green,William(2000):EconometricAnalysis(4thEd.),PearsonEducation.
2. Johnston,J&Dinardo,J. (2007):Econometric(4thEd.), McGrawHillsCo.
3. Maddala,GS(2007):IntroductiontoEconometrics,3rdEdition,JohnWiley&Sons,Ltd., England.
4. Pindyck, R S & Rubinfeld, D L (1998): Econometric Models & Economic, Forecasts, McGrawHills Co.
5. Intriligator, M J & Bodkin, H (1996):Econometric Models Techniques and Application. Prentice Hall.

ECO(HS) DSC11- GROWTH MODELS

Topologyofthecode	**Major/Minor**
LeveloftheCourse	200
CourseDuration/LTP(Lectures,Tutorials, Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssess ment 20 marks
Time	3hours

Course Objective

A continuous economic growth is the primary condition for economic development. The determinantsofgrowthdiffer onthe basis ofthe stageofeconomic development acountryfaces. Thus, a studyofthe macroeconomic GrowthModels is necessaryto understand the determinants of economic growth. The idea and analysis of growth has therefore, assumed such a great significance in recent times that a prior understanding of growth theories is considered essential for the proper comprehension of the different macroeconomic issues and policies. Growthmodels now are not only a scientific method of economic analysis; but also, a body of empirical economic knowledge.The paper entitled "GrowthModels" equipsthe studentsatthe undergraduatelevel to understand systemic facts and latest theoretical developments for theoretical and empirical analysis of growth dynamics.The students will be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts will be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course Outcomes

The knowledge of theories and growth models would help the students better understand the entire economic system and its growth over time. The basic models mainly focus on the contribution of inputs to production while assuming the technology is constant. The course also covers measuring the level of technology and its role in changing economic structure. The importance of main fundamentals such as public infrastructure, institutions, and efficientresourceallocationhasalso beencoveredto understandcurrent development issues. Overall, this course would lay the foundation for the students to understand the real issues in economic development.

PedagogyoftheCourseWork: 80% Classroom Teaching

20% Internal Assessment

INSTRUCTIONSFORTHEPAPER-SETTERAND CANDIDATES

1. Thesyllabusofthispaper hasbeendividedinto four units.
2. Thereshallbe9questionsinall.
3. The first question, which would be compulsory, shall be short answer type word limit (25-30 each).It would carry15 short questions, spread over the entire syllabus. The candidate

will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10 × 2 = 20).

4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question fromeach Unit— 4 in all. Each question shall carry15 marks (15 × 4 = 60).

UNIT-I

Initial Discussion on Growth Theory:

Constant capital output ratio-based growth models: Domar model of growth, Harrod Model, Similarities between two models, Stability property of Harrod and types of growth, Deviations from knife edge and possible mismatches among three types of growth rates.

UNIT-II

Production Function and Growth Theories:

Neo-Classical Model including Solow Model; conditions of steady state, deviation from steady state. Solow residualas technological progress, the Golden rule. Overlapping Generations and Dynamic Efficiency

UNIT-III

Economic Growth and Convergence:

The Convergence Hypothesis; Beta and *sigma* convergences, idea and empirical evidences.Some advance measures of convergence; Club and Gamma convergences, some empirical evidences.

UNIT-IV

Endogenous Growth:

Limitations of neoclassical growth theories and need of endogenising growth. The AK model as initial endogenous growth model. Neoclassical endogenous growth; capital accumulation, externalities and human capital. Directed Technical Change: Endogenous Skill-Bias and Endogenous Labor-Augmenting Technological Change. Competition and technological change.

Recommended Readings:

1. Acemoglu, Daron(2009).IntroductiontoModernEconomicGrowth. PrincetonUniversityPress, 2009. ISBN: 9780691132921.

2. Freitas, M.L. (2022). Economic Growth Models: A Primer / Student's Guide, Nova School of BusinessandEconomics,

3. Jones,Hywel G.(1976).An Introductionto ModernTheories ofEconomicGrowth,McGraw- Hill, Republished in 1976.

4. Solow,Robert(1970).GrowthTheory:AnExposition,ClarendonPress,Oxford,UK.

Semester IV

COMPULSORY PUNJABI

FYUGP B.A./B.A. (Hons.) in Economics

ਲਾਜ਼ਮੀ ਪੰਜਾਬੀ

ਪੰਜਾਬੀ ਕਹਾਣੀ ਅਤੇ ਵਿਹਾਰਕ ਪੰਜਾਬੀ

Punjabi Kahani Ate Viharak Punjabi
B.A. (ECO) PUN (COMPL-02)-401

Semester- IV

ਕ੍ਰੈਡਿਟ-2

Credit distribution, Eligibility and the Pre- Requisites of the Course

Course title & Code	Credits	Credit distribution of the Course			Eligibility criteria	Pre- requisite of the course (if any)
		Lecture	Tutorial	Practical/ Practice		
Punjabi Kahani Ate Viharak Punjabi B.A. (ECO) PUN (COMPL-02)-401	2	3	1	0		

Credit: 2

Learning objective

- ਇਸ ਕੋਰਸ ਰਾਹੀਂ ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਕਹਾਣੀ ਸੰਬੰਧੀ ਮੁੱਢਲਾ ਗਿਆਨ ਹਾਸਿਲ ਕਰਨਗੇ।
- ਇਸ ਕੋਰਸ ਰਾਹੀਂ ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਕਹਾਣੀ ਦੇ ਵਿਸ਼ੇ ਤੇ ਕਲਾ ਪੱਖ ਸੰਬੰਧੀ ਮੁੱਢਲਾ ਗਿਆਨ ਹਾਸਿਲ ਕਰਨਗੇ।
- ਇਸ ਕੋਰਸ ਰਾਹੀਂ ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ ਨਾਲ ਸੰਬੰਧਿਤ ਨਿਬੰਧ ਲੇਖਣ ਦੇ ਹੁਨਰ ਨੂੰ ਗ੍ਰਹਿਣ ਕਰਨਗੇ।
- ਇਸ ਕੋਰਸ ਰਾਹੀਂ ਵਿਦਿਆਰਥੀ ਅਖਾਣਾਂ ਤੇ ਮੁਹਾਵਰਿਆਂ ਦੀ ਵਰਤੋਂ ਸੰਬੰਧੀ ਜਾਣਕਾਰੀ ਪ੍ਰਾਪਤ ਕਰਨਗੇ।

Learning outcomes

- ਇਸ ਨਾਲ ਵਿਦਿਆਰਥੀ ਅੰਦਰ ਪੰਜਾਬੀ ਕਹਾਣੀ ਲੇਖਣ ਸੰਬੰਧੀ ਮੁੱਢਲੀ ਸੂਝ ਪੈਦਾ ਹੋਵੇਗੀ।
- ਇਸ ਨਾਲ ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਕਹਾਣੀ ਦੇ ਵਿਸ਼ਿਆਂ ਨੂੰ ਸਮਝਣ ਦੇ ਯੋਗ ਬਣਨਗੇ।
- ਇਸ ਨਾਲ ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ ਨਾਲ ਸੰਬੰਧਿਤ ਨਿਬੰਧ ਲੇਖਣ ਦੀ ਬਣਤਰ ਨੂੰ ਸਮਝਣ ਦੇ ਯੋਗ ਬਣਨਗੇ।
- ਇਸ ਨਾਲ ਵਿਦਿਆਰਥੀ ਅਖਾਣਾਂ ਤੇ ਮੁਹਾਵਰਿਆਂ ਦੀ ਵਰਤੋਂ ਸੰਬੰਧੀ ਸੂਝ ਪ੍ਰਾਪਤ ਕਰ ਪਾਉਣਗੇ।

ਕ੍ਰੈਡਿਟ : 2

ਨਿਰਧਾਰਿਤ ਕੋਰਸ ਦਾ ਸਮਾਂ: 60 ਘੰਟੇ

ਕੁੱਲ ਅੰਕ = 50

L-100

ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ ਅੰਕ = 10

(ਇਸ ਕੋਰਸ ਵਿਚ ਦੋ ਯੂਨਿਟ ਹੋਣਗੇ। ਯੂਨਿਟ ਪਹਿਲਾ ਵਿਚ ਨਿਰਧਾਰਿਤ ਕਹਾਣੀ ਸੰਗ੍ਰਹਿ 'ਕਥਾ-ਕਹਾਣੀ' ਵਿੱਚੋਂ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣਗੇ। ਇਸ ਯੂਨਿਟ ਦੇ ਕੁੱਲ 20 ਅੰਕ ਹੋਣਗੇ। ਯੂਨਿਟ ਦੋ ਵਿਚ ਵਿਹਾਰਕ ਪੰਜਾਬੀ ਨਾਲ ਸੰਬੰਧਿਤ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣਗੇ, ਜੋ 20 ਅੰਕਾਂ ਦਾ ਹੋਵੇਗਾ।)

ਯੂਨਿਟ ਪਹਿਲਾ : ਡਾ. ਧਨਵੰਤ ਕੌਰ (ਸੰਪਾ.), ਕਥਾ-ਕਹਾਣੀ, ਪਬਲੀਕੇਸ਼ਨ ਬਿਊਰੋ, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ।

30 ਘੰਟੇ

(ੳ) ਕਹਾਣੀ ਦਾ ਵਿਸ਼ਾ-ਵਸਤੂ, ਮੰਤਵ ਤੇ ਸਾਰ

(ਅ) ਕਹਾਣੀ ਦੇ ਕਿਸੇ ਪਾਤਰ ਦਾ ਪਾਤਰ-ਚਿਤ੍ਰਣ

(ੲ) ਕਹਾਣੀਕਾਰ ਦਾ ਜੀਵਨ ਅਤੇ ਸਾਹਿਤਕ ਯੋਗਦਾਨ

ਯੂਨਿਟ ਦੂਜਾ : ਵਿਹਾਰਕ ਪੰਜਾਬੀ

30 ਘੰਟੇ

(ੳ) ਨਿਬੰਧ ਲੇਖਣ: ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ ਨਾਲ ਸੰਬੰਧਿਤ (ਦੋ ਵਿਚੋਂ ਇੱਕ)

(ਅ) ਮੁਹਾਵਰਿਆਂ ਦੀ ਵਾਕਾਂ ਵਿਚ ਵਰਤੋਂ (ਸੱਤ ਵਿਚੋਂ ਪੰਜ)

(ੲ) ਅਖਾਣਾਂ ਦੀ ਵਾਕਾਂ ਵਿਚ ਵਰਤੋਂ (ਸੱਤ ਵਿਚੋਂ ਪੰਜ)

ਨੋਟ: ਲਾਜ਼ਮੀ ਪੰਜਾਬੀ ਵਿਸ਼ੇ ਦੇ ਕ੍ਰੈਡਿਟ 2 ਹੋਣਗੇ ਅਤੇ ਪੇਪਰ 50 ਅੰਕਾਂ ਦਾ ਹੋਵੇਗਾ।ਵਿਦਿਆਰਥੀ ਹਫ਼ਤੇ ਵਿਚ ਘੱਟੇ-ਘੱਟ ਪੰਜ ਦਿਨ ਪੜ੍ਹਨਗੇ।

ਨਿਰਧਾਰਿਤ ਪੁਸਤਕਾਂ

1. ਡਾ. ਧਨਵੰਤ ਕੌਰ (ਸੰਪਾ.), ਕਥਾ-ਕਹਾਣੀ, ਪਬਲੀਕੇਸ਼ਨ ਬਿਊਰੋ, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ,2012
2. ਸੰਤ ਸਿੰਘ ਸੇਖੋਂ, ਕਹਾਣੀ ਸ਼ਾਸਤਰ, ਲੋਕ ਗੀਤ ਪ੍ਰਕਾਸ਼ਨ, ਸਰਹਿੰਦ, 1991
3. ਟੀ.ਆਰ.ਵਿਨੋਦ, ਪੰਜਾਬੀ ਕਹਾਣੀ ਅਧਿਐਨ, ਰਵੀ ਸਾਹਿਤ ਪ੍ਰਕਾਸ਼ਨ, ਅੰਮ੍ਰਿਤਸਰ, 1988
4. ਡਾ.ਹਰਕੀਰਤ ਸਿੰਘ ਅਤੇ ਗਿਆਨੀ ਲਾਲ ਸਿੰਘ, ਕਾਲਜ ਪੰਜਾਬੀ ਵਿਆਕਰਨ, ਪੰਜਾਬ ਸਟੇਟ ਯੂਨੀਵਰਸਿਟੀ ਟੈਕਸਟ ਬੁੱਕ ਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ।
5. ਵਣਜਾਰਾ ਬੇਦੀ, ਲੋਕ ਆਖਦੇ ਹਨ, ਆਰਸੀ ਪਬਲੀਕੇਸ਼ਨ, 2006
6. ਸੁਖਦੇਵ ਮਾਦਪੁਰੀ, ਲੋਕ ਸਿਆਣਪਾਂ: ਅਖਾਣ ਤੇ ਮੁਹਾਵਰੇ, ਲਾਹੌਰ ਬੁੱਕ ਸ਼ਾਪ, ਲੁਧਿਆਣਾ
7. ਸ. ਸ. ਖਹਿਰਾ, ਪੰਜਾਬੀ ਭਾਸ਼ਾ : ਵਿਆਕਰਣ ਅਤੇ ਬਣਤਰ, ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ,

ਪਟਿਆਲਾ।

ਪੇਪਰ ਸੈਟਰ/ਅਧਿਆਪਕ ਲਈ ਹਦਾਇਤਾਂ ਅਤੇ ਪ੍ਰਸ਼ਨ-ਪੱਤਰ ਦੀ ਰੂਪ-ਰੇਖਾ :

1. ਪ੍ਰਸ਼ਨ-ਪੱਤਰ ਦੇ ਕੁੱਲ 50 ਅੰਕ ਹਨ ਜੋ ਕਿ 2 ਕ੍ਰੈਡਿਟ ਦੇ ਬਰਾਬਰ ਹੈ ਅਤੇ ਕੋਰਸ ਦਾ ਨਿਰਧਾਰਿਤ ਸਮਾਂ 60 ਘੰਟੇ ਹੋਵੇਗਾ।
2. ਲਿਖਤੀ ਪ੍ਰਸ਼ਨ-ਪੱਤਰ 40 ਅੰਕਾਂ ਦਾ ਹੋਵੇਗਾ।
3. ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ ਦੇ 10 ਅੰਕ ਹੋਣਗੇ।
4. ਪੇਪਰ ਦਾ ਸਮਾਂ ਤਿੰਨ ਘੰਟੇ ਹੋਵੇਗਾ।

ਯੂਨਿਟ ਅਤੇ ਅੰਕਾਂ ਦੀ ਵੰਡ

1. ਯੂਨਿਟ ਇਕ ਵਿਚ 'ਕਥਾ-ਕਹਾਣੀ' (ਕਹਾਣੀ ਸੰਗ੍ਰਹਿ) ਪੁਸਤਕ ਵਿਚੋਂ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣਗੇ।ਇਹ ਯੂਨਿਟ 20 ਅੰਕਾਂ ਦਾ ਹੋਵੇਗਾ।
    - ੳ ਇਸ ਭਾਗ ਵਿਚ ਕਹਾਣੀਆਂ ਅਤੇ ਕਹਾਣੀ ਲੇਖਕਾਂ ਸੰਬੰਧੀ 7 ਲਘੂ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣਗੇ, ਵਿਦਿਆਰਥੀ ਨੇ ਇਹਨਾਂ ਵਿੱਚੋਂ ਪੰਜ ਸੁਆਲ ਕਰਨੇ ਹੋਣਗੇ। ਇਸ ਭਾਗ ਦੇ 5x2=10 ਅੰਕ ਹੋਣਗੇ।
    - ਅ ਇਸ ਭਾਗ ਵਿਚ ਕਹਾਣੀ ਦੇ ਵਿਸ਼ਾ-ਵਸਤੂ, ਮੰਤਵ, ਸਾਰ, ਪਾਤਰ ਦਾ ਪਾਤਰ-ਚਿਤਰਣ ਬਾਰੇ ਦੋ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣਗੇ। ਵਿਦਿਆਰਥੀ ਨੇ ਇਹਨਾਂ ਕੋਈ ਇੱਕ ਸੁਆਲ ਕਰਨਾ ਹੋਵੇਗਾ। ਇਸ ਸੁਆਲ ਦੇ 5x1=5 ਅੰਕ ਹਨ।
    - ੲ ਇਸ ਭਾਗ ਵਿਚ ਕਹਾਣੀਕਾਰਾਂ ਦੇ ਜੀਵਨ ਅਤੇ ਸਾਹਿਤਕ ਯੋਗਦਾਨ ਬਾਰੇ ਦੋ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣਗੇ। ਵਿਦਿਆਰਥੀ ਨੇ ਇਹਨਾਂ ਕੋਈ ਇੱਕ ਸੁਆਲ ਕਰਨਾ ਹੋਵੇਗਾ। ਇਸ ਸੁਆਲ ਦੇ 5x1=5 ਅੰਕ ਹਨ।
2. ਯੂਨਿਟ ਦੋ ਵਿਚ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ ਨਾਲ ਸੰਬੰਧਤ ਨਿਬੰਧ ਲੇਖਣ, ਮੁਹਾਵਰਿਆਂ ਅਤੇ ਅਖਾਣਾਂ ਬਾਰੇ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣਗੇ।ਇਸ ਯੂਨਿਟ ਦੇ ਕੁੱਲ 20 ਅੰਕ ਹੋਣਗੇ।
    - ੳ ਇਸ ਭਾਗ ਵਿਚ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ ਨਾਲ ਸੰਬੰਧਤ ਨਿਬੰਧ ਲਿਖਣ ਬਾਰੇ ਦੋ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣਗੇ। ਵਿਦਿਆਰਥੀ ਨੇ ਇਹਨਾਂ ਵਿੱਚੋਂ ਇਕ ਸੁਆਲ ਕਰਨਾ ਹੋਵੇਗਾ। ਇਸ ਸੁਆਲ ਦੇ 10 ਅੰਕ ਹਨ।
    - ਅ ਇਸ ਭਾਗ ਵਿਚ ਮੁਹਾਵਰਿਆਂ ਦੀ ਵਾਕਾਂ ਵਿਚ ਵਰਤੋਂ ਸੰਬੰਧੀ (ਸੱਤ ਵਿਚੋਂ ਪੰਜ) ਬਾਰੇ ਸਵਾਲ ਪੁੱਛਿਆ ਜਾਵੇਗਾ। ਇਸ ਭਾਗ ਦੇ 5x1=5 ਅੰਕ ਹੋਣਗੇ।
    - ੲ ਇਸ ਭਾਗ ਵਿਚ ਅਖਾਣਾਂ ਦੀ ਵਾਕਾਂ ਵਿਚ ਵਰਤੋਂ ਸੰਬੰਧੀ (ਸੱਤ ਵਿਚੋਂ ਪੰਜ) ਬਾਰੇ ਸਵਾਲ ਪੁੱਛਿਆ ਜਾਵੇਗਾ। ਇਸ ਭਾਗ ਦੇ 5x1=5 ਅੰਕ ਹੋਣਗੇ।

हिन्दी
समसटर–IV

Course Title	:	**भाषा एव वयाकरण** **BHASHA EVAM VYAKARAN**
Course Code	:	MIL-HIN-02-401
Course Duration	:	30 hours in One
semester/2 hrs per week Credit	:	2
Maximum Marks	:	50(45Theory+5Internal Assessment)

पाठयकरमकाउ□□शय–(CourseObjectives)

- भाषाकेव्यवहारिकपपक्षकाबोधकराना।
- विद्यार्थीकीभाषायीदक्षताऔरलेखन-कौशलकोबढ़ावादेना।
- हिन्दीभाषा-दक्षताऔरतकनीकके□ न्तःसंबंधोंसेपरिचितकराना।

पाठयकरमकापरिणाम - (CourseLearning0utcomes)

- व्यवहारिकहिन्दीकीविशेषताओंएवंरूपोंसेपरिचितहोंगे।
- व्याकरणभागसेभाषा-व्यवहारमेंकुशलताकासंचारहोगा।
- शब्द-भण्डारमें□ भिवृद्धिहोगी।

इकाई-I

Max. Marks:50
Theory : 45
InternalAssessment:05
Time : 03 Hours

1. भाषा:व्यवहारिकपक्ष

(क) भाषाएवंबोलीसे□ भिप्राय, भाषाकेविविधरूप मूलभाषा, परिनिष्ठितभाषा, मातृभाषा, राजभाषा, राष्ट्रीयभाषा, □ न्तरराष्ट्रीयभाषा, विभाषा, □ पभाषा, विशिष्टभाषा, कूटभाषा, कृत्रिमभाषा।

(**छह**मेंसे**चार**केउत्तरदेनेहोंगे) □ंक-12

(ख) फाइल-प□्धति-नियम

सामान्यपरीचय, □ंकन, छेदनवक्रमसंख्या, सन्दर्भदेना, नईफा□लखोलना, हवाला देनातथानीशान्कयोग, फा□लोंपर□ ग्रत□ंकन।

(**चार**मेंसे□ोकेउत्तरदेनेहोंगे।) □ंक - 8

इकाई–II

2. वयाकरण

(क)	मुहावरेऔरलोकोक्तियाँ (**आठ**मेंसे**पाँच)**	ंक-10
(ख)	**आठ**शब्दोंमेंसेकिसी**पाँच**केविलोमशब्दलिखनेहोंगे।	ंक-5
(ग)	**आठ**शब्दोंमेंसेकिसी**पाँच**केपर्यायवाचीशब्दलिखनेहोंगे।	ंक-5
(घ)	**आठ**शब्दोंमेंसेकिन्हींपाँचकेशुद्धरूपलिखनेहोंगे।	ंक-5

आतरिकमलयाकन अंक

सहायकपसतक:

1. प्रयोजनमूलकऔरकार्यालयीहिन्दी, कृष्णाकुमारगोस्वामी
2. प्रयोजनमूलकहिन्दीकीनईभूमिका, कैलाशचन्द्रपाण्डेय
3. प्रयोजनमूलकहिन्दी:सिद्धांतऔरप्रयोग, दंगलझाल्टे
4. प्रशासनिकहिन्दी, हरिमोहन, तक्षशिलाप्रकाशन
5. प्रशासनिकहिन्दी, शोकऔर शोक, रुणपब्लिकेशन, चण्डीगढ़।
6. भाषा-विज्ञानएवंभाषा-शास्त्र, आचार्यकपिलदेवद्विवेदी, विश्वविद्यालयप्रकाशन, वाराणसी।

Semester- V

ECO (HS) DSC 12 – INTERMEDIATE ECONOMETRICS

Topologyofthecode	Major/Minor
Levelofthe Course	300
Course Duration /LTP(Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max. Marks	100 Marks= Theory 80 marks + Internal Assessment 20 marks
Time	3hours

CourseObjective:

Applicationofeconomictheoryneedareasonableunderstandingofeconomicrelationship and relevant statistical methods. The econometric theory thus becomes a very powerfultool for understanding of applied economic relationships and for meaningful research in economics.This paper accordingly is devoted to equip the students with basic theory of econometrics andrelevant applications of the methods. The topics covered in the course include various problemsfacedinestimation ofboth singleequationsand simultaneous equationsmodels.Besides the time-tested method of imparting verbal instructions through lectures, description andderivationofeconometricmodelswouldbedemonstratedbyextensiveuseofblackboard.Examples, in so far as possible, would be selected from everyday life/experience. The feedbackwouldbesoughtthroughthestudents writingassignments andwrittenexaminations.

PedagogyoftheCourseWork:

80%Classroomteaching.

20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

InstructionsforPaper-setterandcandidates:

- The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours. The paper-setter must put anotein thequestion paperin this regard.
- There shall be **9** questions in all. The first question **compulsory** comprising 15 short answertype questions spread over the whole syllabus. The candidates are required to attempt 10questions.Each questionshall beof**two** marks (10 x2=20) Rest of the paper shall contain four units. Two questions shall be asked from each unit and thecandidates shall be given internal choice. The candidates shall attempt one question from each unit.Eachquestion will carry15

marks (15x4=60).

UNIT-I

The General Linear Model: Review of Assumptions, Estimation and Properties of Estimators: BLUE

UNIT -II

Simultaneous Equation Models –Structural v/s Reduced Form Models, Model Identification Conditions, simultaneous equation bias. ModelsEstimation Techniques: ILS, 2SLS,3SLS, SUR, LIML, and FIML.

UNIT- III

Dummy Variables in Regression: Models with Dummy Independent variables. Types of dummy variable.

Dichotomous Dependent Variable Models: LPM, Probit and Logit estimations. Odd Ratios and Marginal Effects. Types of marginal effects.

UNIT-IV

Censored Dependent Variable Models: Censoring v/s Truncation. Tobit Regression; estimation and interpretation.

Polychotomous Dependent Variable Models: Multinomial, Ordered and Sequential Models: Estimation and Interpretation.

Note: Applications of all above models using computer software must be delivered.

Essential Reading:

1. Gujarati Damodar N. (2004): Basic Econometrics (4th ed.), Singapore. McGraw Hills Co.
2. Hill, R. C., Griffiths, W. E., & Lim, G. C. (2018). *Principles of econometrics*. John Wiley & Sons.
3. G.S. Madala (1983): Limited Dependent and Qualitative Variables in Econometrics.Cambridge University Press
4. Stock J.H. & Watson, M.W. (2020). Introduction to Econometrics (4th edition). Pearson

Further Readings:

1. Green, William (2000): Econometric Analysis (4th Ed.), Pearson Education.
2. Johnston, J &Dinardo, J. (2007): Econometric (4th Ed.), McGraw Hills Co.
3. Maddala, G S (2007): Introduction to Econometrics, 3rd Edition, John Wiley & Sons, Ltd., England.
4. Pindyck, R S & Rubinfeld, D L (1998): Econometric Models & Economic, Forecasts, McGraw Hills Co.
5. Intriligator, M J & Bodkin, H (1996): Econometric Models Techniques and Application. Prentice Hall

ECO (HS) DSC 13: INTERNATIONAL ECONOMICS

Topologyofthecode	Major/Minor
LeveloftheCourse	300
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

CourseDescription:

Thiscoursedevelopsasystematicexposition ofmodelsthattry to explain thecomposition, direction, and consequencesofinternationaltrade,andthedeterminantsandeffectsoftradepolicy.Itthenbuildsonthe models of open economy macroeconomics, focusing on national policies as well as international monetary systems. It concludes with an analytical account of the causes and consequences of the rapid expansion
ofinternationalfinancialflowsinrecentyears.Althoughthecourseisbasedonabstracttheoreticalmodels, students will also be exposed to real-world examples and case studies.

PedagogyoftheCourseWork:

80%Classroomteaching.

20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

Themaximummarksforthepaperwillbe100.Thequestionpaperwillbeof80marksandcontinuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperinthis regard.

- Thereshallbe**9**questionsinall.
- The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).
- Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

UNIT-I

Whatisinternationaleconomicsabout?TheoriesofAbsoluteAdvantage,ComparativeAdvantageand Opportunity Costs.

Heckscher-OhlinTheoryofTrade–ItsmainFeatures,Assumptionsand Limitations.

NewApproachestoTradeTheory:TheProductCycleTheory,TechnologyGapTheory,Intra-industrytrade.

UNIT-II

ConceptandtypesofTermsoftrade.
EffectsofTariffunderPartialEquilibriumAnalysis.NontariffBarriers.PoliticaleconomyofNon-tariff Barriers.
EconomicIntegration:ConceptandFormsofRegionalIntegration;StaticandDynamicEffectsofCustom Union.

UNIT-III

ForeignExchangeMarket:Fixedversusflexibleexchangerates,Concept,RoleandTypesofForeign Exchange Transactions: Speculation, Arbitrage and Hedging.

TheProcessofAdjustmentintheBalanceofPaymentsunderFixedandFlexibleExchangeRateSystems.

UNIT-IV

Internationalmonetarysystems:InternationalMonetaryFund;EurocurrencyMarket.Financial Globalization and Financial crises : Causes and case studies.

Readings:

1. Krugman, P., Obstfeld, M., & Melitz, M. (2012). International economics: Theory and policy (9th ed.). Addison-Wesley (Pearson Education Indian Edition).
2. Salvatore, D. (2011). International economics: Trade and finance (10th ed.). John Wiley & Sons (International Student Edition).

ECO (HS) DSC 14:INDIAN ECONOMY –I

Topologyofthecode	Major/Minor
LeveloftheCourse	300
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Objective: Theobjectiveofthispaperistoacquaintthestudentswiththestrategyof economic development of the Indian Economy since Independence. The paper also aims at creatingawareness amongthe students regardingthe contemporaryissues pertaining to various sectors of the economy.

PedagogyoftheCourseWork:

80%Classroomteaching.
20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

InstructionsforPaper-setterandcandidates:

Themaximummarksforthepaperwillbe100.Thequestionpaperwillbeof80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperinthis regard.

- Thereshallbe**9**questionsinall.
- Thefirstquestion**compulsory**comprising15shortanswertypequestionsspreadover the whole syllabus. The candidates are required to attempt 10 questions. Eachquestion shall be of **two** marks (10 x 2= 20).
- RestofthepapershallcontainfourunitsTwoquestionsshallbeaskedfromeachunit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT–I

EconomicPolicy: EconomicPolicy–Definition.Types-QualitativeandQuantitative. Need for Economic Policy - Economic Institutions and Policies in India (only introduction). Problems in Policy making in India.

UNIT–II

EconomicDevelopmentStrategyinIndia: AdoptionofPlanninginIndia:

Objectives,StrategyandAssessment(Planwisedetailstobeexcluded).NeedforEconomicReforms -State of the Economy in 1991.NitiAyog- its Objectives and Functions.

UNIT-III

DemographyandEconomicDevelopment:TheoryofDemographicTransition.

Demographic Features of India's Population. Population Growthin India – Promotingor Retarding EconomicGrowth.UrbanisationinIndia-Extent,CausesandConsequencesofRapidUrbanisation.

UNIT-IV

Structural Changes in India: Changes in Sector Income Distribution and OccupationalStructure.Inter-StateDisparitiesinthePatternofDevelopment- Causes, Indicators and Policy Measures to Remove Regional Imbalances.

EssentialReadings:

1. Nayak,P.(Ed.)(2015):*EconomicDevelopmentofIndia*,RoutledgeLondon.
2. Jalan,Bimal(1992):*TheIndianEconomics–ProblemsandProspects*,Viking,NewDelhi.
3. Lucas, R. E., & Papanek, G. F. (1988). *The Indian economy: recent development and future prospects*.
4. Brahmananda, P. R., &Panchamukhi, V. R. (Eds.). (1987). *The development process of the Indian economy*. Mumbai: Himalaya Publishing House.
5. Tinbergen, J. (1952).*On the Theory of Economic Policy.* North Holland, Amsterdam.
6. Ramaswamy,VS&Namakumari,S(1996):*StrategicPlanningandFormulationofCorporateStrategy– Text and Cases*, MacMillan, New Delhi.

FurtherReadings:

1. Byres,TJ(Ed.)(1998):*TheIndianEconomy:MajorDebateSinceIndependence Economic Survey* : Government of India: Various Issues.
2. EconomicandPoliticalWeekly:Various Issue

ECO (HS) DSC15: INTRODUCTORY PUBLIC FINANCE

Topologyofthecode	Major/Minor
LeveloftheCourse	300
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

CourseDescription

Public economics is the study of government policy from the points of view of economic efficiency and equity. The paper deals with the nature of government intervention and its implications for allocation, distribution and stabilization.

Inherently, this study involves a formal analysis of government taxation and expenditures. The subject encompasses a host of topics including public goods, market failures and externalities. The paper is divided into two sections, one dealing with the theory of public economics and the other with the Indian public finances.

PedagogyoftheCourseWork:

80%Classroomteaching.

20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

InstructionsforPaper-setterandcandidates:

Themaximummarksforthepaperwillbe100.Thequestionpaperwillbeof80marksandcontinuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperinthis regard.

- Thereshallbe**9**questionsinall.
- The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).
- Rest of the paper shall contain four units. Two questions shall be asked from each unit and the

candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT–I

TheoryofMaximumSocialAdvantage.PrivateGoods,PublicGoods,mixedgoodsandMeritGoods. Optimal Provision of Public Goods: partial equilibrium approach;freeridingproblem.Externalities:PigouvianandCoarsesolutions.OverviewofDifferentRoleof Government: Allocation, Stabilization and Distribution.

UNIT–II

Regressive,ProgressiveandProportionalTaxes.DirectandIndirectTaxes.IncidenceofIndirectTaxes. Effects of Taxation: concepts of Deadweight losses and Excess Burden.

UNIT–III

TypesofGovernmentBudgets.DifferentConceptsofBudgetaryDeficitsintheIndianContext.Economic and Functional Classification of Central Govt. Budgets.

UNIT-IV

Changing Trends and Pattern of Direct and Indirect Taxes in India. Changing Trends and Pattern of GovernmentExpenditurein India.RecentTaxandExpenditureReformsin India. IssuesNeedandobjectives of fiscal federalism in India.

Readings:

1. Cullis, J., & Jones, P. (1998). *Public finance and public choice* (1st ed.). Oxford University Press.
2. Basu, K., & Maertens, A. (Eds.). (2013). *The new Oxford companion to economics in India*. Oxford University Press.
3. M.M.Sury,(1990).*GovernmentBudgetinginIndia*.
4. Ulbrich,H.(2003),*PubicFinanceinTheoryandPractice*.Thomson.
5. Parikh,S.(1999).*IndianDevelopmentReport1999-2000*.NewDelhi.OxfordUniversityPress.
6. Datt.R&Sundaram,K.P.M.(2003).*IndianEconomy*.NewDelhi.S.Chand&Company.
7. Aronson,J.R. (1985).*PublicFinance*.NewYork:McGraw-HillInternational.
8. Houghton,R.W.(1973).Publicfinance.London:PenguinEducation.
9. Musgrave,R.A.,&Musgrave,P.B.(1989).*PublicFinanceinTheoryandPractice*.NewYork.McGraw- Hill International.

ECO (HS) VAC 5 (a): FINANCIAL LITERACY

Topologyofthecode	ValueAddedCourse
LeveloftheCourse	300
Course Duration /LTP (Lectures,Tutorials,Practical)	L=30hours(2hoursaweekfor15 weeks= 01 semester),T=0, P=0
Credits	2
Max.Marks	50Marks=Theory45marks+Internal Assessment 05 marks
Time	3 hours

Course Objectives: To create awareness in students about the need for possessing financial literacy education.

CourseOutcomes:

1. The students will be able to understand the importance of financial literacy and prepare financial plans and budgets.

2. The student will be able to describe the importance of insurance services as social security measures.

Pedagogy of the Course Work: Classroom Lecture, Interactive sessions, PPT, Tutorials, Group Discussions, Case Studies, etc.

Instructionsforpapersettersandcandidates:

Note: The question paper of each subject covering the entire course shall be divided into three sections.

1. **Section A (10 marks)**: This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions. Each question will carry 2.5 marks; total weightage being 10 marks.

2. **SectionB(15marks)**:Thissectionwillconsistofessaytype/numericalquestions fromUnit I of the syllabus. The candidate will be required to attempt two questions out of four. Each question will carry 7.5 marks; total weightage being 15 marks.

3. **Section C (15 marks)**: This section will consist of essay type/numerical questions from UnitII of the syllabus. The candidate will be required to attempt two questions out of four. Each question will carry 7.5 marks; total weightage being 15 marks.

UNIT–I

IntroductiontoFinancialPlanningandBudgeting:Meaning,importance,andneedfor financialplanning; PersonalBudget,FamilyBudget,BusinessBudget;Procedurefor financialplanningandpreparingabudget;

avenuesforsavingsfromsurplus.FinancialLiteracy - Components of Financial Literacy - Financial Institutions: Meaning, Banking and Non-Banking Financial Institutions.

UNIT–II

BankingandInsurance:MeaningandTypesofBanks,Variousservicesofferedbybanks, types of bank deposit accounts, Formalities to open various types of bank accounts, KYCnorms.Various typesof Loans:Short-term,Medium-term,andLong-term loans. Insurance – its meaning and types: Life Insurance, Health Insurance Plan, Property Insurance, General Insurance – Types.

SuggestedReadings:

1. Avadhani, V.A. (2019), *Investment Management*, Mumbai: Himalaya Publishing House Pvt. Ltd.

2. Chandra, P. (2012), *Investment Game: How to Win*, New Delhi: Tata McGraw Hill Education.

3. Kothari, R. (2010), *Financial Services in India: Concept and Application*, New Delhi: Sage Publication India Pvt. Ltd.

4. Milling, B. E. (2003), *The Basics of Finance: Financial Tools for Non-Financial Managers*, Indiana: Universe Company.

5. Zokaityte,A.(2017),*FinancialLiteracyEducation*,London:PalgraveMacmillan.

OR

ECO (HS) VAC 5(b): INTRODUCTION TO RESEARCH

COURSE TITLE	**INTRODUCTION TO RESEARCH**
Course Code	**VAC 101**
Topology of Course	**Value Added Courses (VAC)**
Course Duration/ LP (Lectures, Practical)	**30 hours of lectures for theory (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 Semester = 15 weeks)**
Credits	**2 credits**
Max. Marks and Time	**50 Marks (40 TH +10 IA)** **1.5 Hours.**

Course Objectives: To acquaint the students in understanding the relevance of research, and train them to do research in a scientific manner

Course Outcomes: On successful completion of the course, the students will be able to:

1. Understand the importance of research methodology.
2. Gain knowledge of different research designs.
3. Develop an understanding of different sampling techniques.
4. Explore date collection methods.
5. Develop effective research writing and communication skills.

Pedagogy of the Course Work: Class Room Lecture, Interactive sessions, PPT, Group Discussions, Case Studies etc.

Instructions forpaper setters and candidates. The syllabus has been divided into TWO units. The question paper of each subject covering the entire course shall be divided into three sections

1. Section A (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks.

2. Section B (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.

3. Section C (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.

UNIT-I

Research - Meaning, Characteristics, Types, Process and Utility, Research Design - Meaning, Types and Features of good research design. Qualitative and Quantitative research. Importance of review of literature. The research proposal.

Primary and Secondary data. Methods of data collection. Measurement and scaling, Designing of Questionnaire and schedule.

UNIT-II

Sampling Design and Sampling procedure. Sampling and non-sampling errors. Data analysis and interpretation.

Business Reports and Proposals: Introduction, Meaning, Steps in Writing a Routine Business Report, Format of Proposals, Four Key Elements in Winning Business Proposals; Case Studies.

Suggested Readings:

1.Kothari, C. R. Research Methodology: Methods and Techniques. New Age International Publishers.

2.Ranjit Kumar. Research Methodology: A Step-by-Step Guide for Beginners. SAGE Publications India.

3.Panneerselvam, R. Research Methodology. PHI Learning Private Limited.

4.Creswell, J. W., & Creswell, J. D. (2017). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches. SAGE Publications.

5.Neuman, W. L. Social Research Methods: Qualitative and Quantitative Approaches. Pearson.

6.Sekaran, U., & Bougie, R. Research Methods for Business: A Skill-Building Approach. Wiley

Semester-VI

ECO (HS) DSC16: ECONOMICS OF DEVELOPMENT & PLANNING

Typologyofthecode	Major/Minor
LeveloftheCourse	300
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3 hours

CourseDescription: The course begins with elaboration of the difference between growth and development. It further builds on the evolution of concepts of economic development like measurement, poverty, inequality. It introduces students to the characteristics of developing nations followed by theories of Economic Growth and Development. In addition, issues related to planning, globalization and sustainable development have been incorporated.

PedagogyoftheCourseWork:

80%Classroomteaching.
20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

Themaximummarksforthepaperwillbe100.Thequestionpaperwillbeof80marksandcontinuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperinthis regard.

Thereshallbe**9**questionsinall.

The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT-I

Understanding Growth and Development:

Meaning of Economic Development.Difference between Economic Growth and Economic Development;Indicators of economic development: Income based measure, PQLI and HDI (Concept only); Structural changes in the composition of GDP and Occupational Structure.
Poverty: Concept and Measures. Inequality: Concept and Measures

UNIT-II

Characteristics of Developing Nations:
Relationship between income distribution and economic development (Kuznets' inverted U curve); Relationship between Population Growth and Economic Development (Theory of Demographic Transition);

Vicious Circle of Poverty (Nurkse) and Low Level Equilibrium Trap (Nelson); Critical Minimum effort: (Lebenstein), Dualism: Meaning and forms (concepts only)

UNIT-III

Theory of Economic Growth and Development: Theory of Unlimited Supplies of Labour (Lewis); The Theory of Big Push (Rosenstein Rodan); Rostow's Stages of Economic Growth; Concept of Balanced Vs. Unbalanced Growth.

UNIT-IV

Planning, Globalization and Sustainable Development: Need, Objectives, Strategies and Problems of planning; Globalization: Meaning and Role of FDI and MNCs Sustainable development: Meaning, concerns, indicators and remedial measures.

RecommendedReadings:

1. Thirlwall, A.P. (Latest edition). *Economics of development: Theory and evidence*. New York: Palgrave Macmillan.

2. Todaro, M. P., & Stephen, C. S. (Latest edition). *Economic Development*. New York: Addison-Wasley Logman.

3. Agarwala, A. N., &Singh, S. P. (Eds.). (1963).*Economics of underdevelopment:Aseriesof articles and papers*. London : Oxford University Press.

4. Cypher,J.(2014).*Theprocessofeconomicdevelopment*.London:Routledge.

5. Lucas Jr, R. E. (1988). On the mechanics of economic development.*Journal of Monetary Economics*, *22*(1),3-42 Meier, G. M., & Rauch, J. E. (Eds.). (1995).*Leading issues in economic development(Vol. 6*).New York: Oxford University Press.

6. Ray,D.(1998).*Developmenteconomics*.NewJersey:PrincetonUniversityPress.

7. Romer, P. M. (1986). Increasing returns and long-run growth.*Journal of Political Economy*,*94*(5), 1002-1037.

8. Romer,P.M.(1990).Endogenoustechnologicalchange.*JournalofPoliticalEconomy*,*98*(5,Part2), S71-S102.

9. Schultz,T. W. (1982).*Investingin people:Theeconomicsof populationquality*.USA: Universityof California Press.

10. Sen,A.(1999).*Developmentasfreedom*.NewDelhi:OxfordUniversityPress.

ECO (HS) DSC 17: INDIAN ECONOMY- II

Topologyofthecode	Major/Minor
LeveloftheCourse	300
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Objective: Theobjectiveofthispaperistoacquaintthestudentswiththestrategyof economic development of the Indian economy since Independence. The paper also aims at creatinganawareness amongthe students regardingthe contemporaryissues pertaining to various sectors of the economy.

PedagogyoftheCourseWork:

80%Classroomteaching.

20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

InstructionsforPaper-setterandcandidates:

Themaximummarksforthepaperwillbe100.Thequestionpaperwillbeof80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperinthis regard.

- Thereshallbe**9**questionsinall.
- Thefirstquestion**compulsory**comprising15shortanswertypequestionsspreadover the whole syllabus. The candidates are required to attempt 10 questions. Eachquestion shall be of **two** marks (10 x 2= 20).
- RestofthepapershallcontainfourunitsTwoquestionsshallbeaskedfromeachunit and the

candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT–I

Agricultural Development: Pattern of Growth of Indian Agriculture since 1950 – RoleofGreenRevolution,Decelerationinthe1990s–CausesandFutureChallenges.

UNIT–II

Industrial Development Strategy: Trends in Growth and Structure of Indian

Industry.CritiqueofRegulatoryIndustrialPolicyRegime(1947-1991).Deregulation and Liberalization of the Private Sector. Privatization – Nature and Extent in India.

UNIT–III

ExternalSector:TradePolicyduringPre-ReformPeriod.Globalisation –Meaningand Trade Policy during Post-Reform Period. Liberalized Exchange Rate Management System.

UNIT–IV

SomeContemporaryIssues: ServiceSectorGrowth–SustainabilityofServicesLed

GrowthinIndia.ImpactofNewEconomicPolicyonIndianEconomy.Consumer Protection Act. Competition Policy of India.

EssentialReadings:

1. Nayak,P.(Ed.)(2015)*EconomicDevelopmentofIndia*,Routledge,London.
2. Nagaraj,R(2006)*:AspectsofIndia'sEconomicGrowthandReforms*,AcademicFoundation,New Delhi.
3. Krueger,AO(Ed.)(2003):*EconomicPolicyReformsandtheIndianEconomy*,OxfordUniversityPress, New Delhi.
4. Ahluwalia, I. J., & Little, I. M. D. (1998). *India" s Economic Reforms and Development: Essays for Manmohan Singh*-Oxford University Press.

FurtherReadings:

1. Rosen,George,(1988):IndustrialChangeinIndia1970-2000AlliedPublishers,NewDelhi.
2. EconomicSurvey:GovernmentofIndia:VariousIssues.
3. EconomicandPoliticalWeekly:VariousIssues.

ECO (HS) DSC18–INTRODUCTORY AGRICULTURE ECONOMICS

Topologyofthecode	Major/Minor
LeveloftheCourse	300
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Objective: Thefocus ofthis courseistointroducestudentstotheimportance ofagricultural sector andtotheissuesassociatedwithagricultural transformation,agriculturalproductionanddemandof andsupplyforagricultural commodities.Theobjectiveistoequipstudentstoanalyzeandcritically assessissues,policiesandprogrammesinthese areaswithparticular emphasisonIndianAgriculture.

PedagogyoftheCourseWork:80%Classroomteaching.

20%Internal AssessmentComprisingUnitTest,Assignment and Class room participation.

InstructionsforPaper-setterandcandidates:

The maximum marks for the paperwill be 100.The question paperwill be of 80marks and continuous evaluation20 marks.Timeallowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperinthisregard.

- Thereshallbe**9**questionsinall.
- Thefirstquestion**compulsory**comprising15shortanswertypequestionsspreadoverthewhole syllabus.The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10x2= 20).
- Rest of thepaper shall containfour units.Twoquestionsshall beasked fromeachunit andthe candidates shallbegiveninternal choice.Thecandidates shallattempt onequestion from each unit.Each question will carry15 marks (15x4=60).

UNIT–I

Introduction: Nature and Scope of Economics of Agriculture; Specificities of Farm Organization, Agricultural ProductionandMarkets; Inter-sector Linkages of Agriculture.

Role of Agriculture in Economic Development: Contribution of Agriculture to Economic Development; Historical Perspective(ExperienceofEngland, UnitedStates, JapanandIndia).

UNIT–II

AgriculturalProduction: UniquenessofAgriculturalProduction;VariousTypesofFactor-Products, Factor-FactorandProduct-ProductRelations;Cobb-Douglas andCES ProductionFunctions.

FarmSize–ProductivityandTenancyEfficiencyDebate.

UNIT–III

Agricultural Transformation: Schultz Theory of Transformation of Traditional Agriculture; Hayami-RuttanInducedInnovationModel (HRIM) ofAgricultural Development.

LandReformsinIndianAgriculture.

UNIT–IV

Risk and UncertaintyinAgriculture: NatureandTypes ofRisks and Uncertainties in Agriculture, PublicPolicies and FarmLevel Measures to ReduceRisk and Uncertainties.

TermsofTradebetweenAgricultureandIndustry;AgriculturalPricePolicysinceIndependence.

EssentialReadings:

1. Gardner, B. L., &Rausser, G. C. (Eds.). (2001). *Handbook of agricultural economics: Vol. 1A. Agricultural production*. Elsevier.
2. Gardner, B. L., &Rausser, G. C. (Eds.). (2001). *Handbook of agricultural economics: Vol. 1B. Marketing, distribution and consumers*. Elsevier.
3. Gardner, B. L., &Rausser, G. C. (Eds.). (2002). *Handbook of agricultural economics: Vol. 2A. Agriculture and its external linkages.* Elsevier Science.
4. Gardner, B. L., &Rausser, G. C. (Eds.). (2002). *Handbook of agricultural economics: Vol. 2B. Agricultural and food policy.* Elsevier Science.
5. Eicher, C. K., & Witt, L. W. (Eds.). (1964). *Agriculture in economic development*. McGraw-Hill.

6. Ghatak, S., &Ingersent, K. A. (1984). *Agriculture and economic development.* Wheatsheaf Books.

7. Southworth, H. M., & Johnston, B. F. (Eds.). (1967). *Agricultural development and economic growth.* Cornell University Press.

8. Tsakok, I. (2011*). Success in agricultural transformation: what it means and what makes it happen*. Cambridge University Press.

9. Soni, R. N. (2005). *Leading issues in agricultural economics.* Vishal Publishing Co.

FurtherReadings:

1. Heady, E. O. (1964). *Economics of agricultural production and resource use*. Prentice-Hall of India Pvt. Ltd.

2. Metcalf, D. (1969). The economics of agriculture. Penguin Books.

ECO (HS) DSC19: INTRODUCTORY INDUSTRIAL ECONOMICS

Topologyofthecode	Major/Minor
LeveloftheCourse	300
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Objective: IntheContemporaryworldwithglobalizationandliberalizationmoreandmoreattentionisbeing giventoindustry.Sinceindustryperformancecriticallydependsonfirms'behaviorallowingequilibrium outcome, the course intends to provide a rigorous knowledge of different long-run equilibrium outcome offirms under different conditions from the point of view of public policy. The students are also equipped todeal with debates involved in the industrial development in a cogent and analytical manner, particularly inthe Indian context. However, it should be noted that Game Theoretic approach to any topic/problem is outside the scopeof thepresentsyllabus. The time-tested method of in parting verbal instructions throughlectureswouldbeused.Examples,inso faraspossible,would beselected from

everyday life/experience. The feedback would be sought through the students writing assignments, tutorials and seminars. Thestudentswould be evaluatedattheendofeachsemesterthrough subjectivetype questions/answers(both short and essay type).

PedagogyoftheCourseWork:

80%Classroomteaching.

20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

InstructionsforPaper-setterandcandidates:

Themaximummarksforthepaperwillbe100.Thequestionpaperwillbeof80marksandcontinuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperinthis regard.

- Thereshallbe**9**questionsinall.
- Thefirstquestion**compulsory**comprising15shortanswertypequestionsspreadoverthewholesyllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).
- Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shallbegiveninternalchoice.Thecandidatesshallattemptonequestionfromeachunit.Eachquestionwill carry 15 marks (15x4=60).

UNIT–I

BasicConceptsofFirm,IndustryandMarket;OrganizationalForms.SeparationofOwnershipfrom Management and Control. Alternative Goals of Firm: Contributions of R Marris and Williamson.

UNIT–II

ElementsofMarketStructure:'SellersandBuyers'Concentration,ProductDifferentiation,Conditionsof Entry. Measurement of Sellers's Concentration.

Determinants:EconomiesofScaleandBarrierstoEntry.

GrowthofFirms:VerticalIntegration,Diversification;ConceptsofMergersand Acquisitions.

UNIT–III

Oligopolistic Conduct: Evolution of Market Structure –Conduct – Performance Paradigm, Economics of Advertisement:Concepts,MarketStructureandAdvertisement:Dorfman–SteinerconditionEconomicsofR & D: Concepts, Measurement and Market Structure and Innovation: incentive and ability issues.

UNIT–IV

MarketStructureandProfitability.IssuesofAllocativeandProductiveEfficiency:DegreeofSubOptimal Capacity.

EssentialReadings:

1. Martin,Stephen(1989):*IndustrialEconomics:EconomicAnalysisand Public Policy*. McMillan Publishing Co., New Delhi
2. Clark, R. (1985). *Industrial economics*. Blackwell Publishers.
3. Shepherd, W. G. (1997). *The economics of industrial organization: Analysis, markets, policies* (4th ed.). Prentice Hall International.
4. Scherer, F. M., & Ross, D. (1990). Industrial market structure and economic performance (3rd ed.). Houghton Mifflin Company.
5. Koutsoyiannis(1979):*ModernMicro-economics*.MacMillan,London.
6. Bains, J S (1959) :*Industrial Organization.* University of California.
7. Waldman, D. E., & Jensen, E. J. (2001). *Industrial organization: Theory and practice* (2nd ed.). Addison-Wesley.
8. Hay, D. A., & Morris, D. (1991). Industrial economics and organization: theory and evidence.

ECO (HS) DSC20 –BASICS OF FINANCIAL MARKETS

Topologyofthecode	Major/Minor
LeveloftheCourse	300
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Course Objective: Theoperationoffinancial marketsandtheirregulationsaretobestudiedtoappreciate their key-role in an economy, especially after far reaching banking and financial sector reforms in India and elsewhere.The present course is designedto acquaint thestudents fullywith theoryand practiceofdifferent financial markets andinstitutions tounderstandandanalyzetheirdevelopment roleandlimitationsinshapingandinfluencingthemonetaryandrelatedpoliciesbothatthenational and international levels.

Course Outcome: Upon completion, students will be able to classify financial markets and instruments. They can evaluate risk-return and apply valuation models. They will analyze the Indian financial system's components and reforms, and explain the role of major financial institutions and regulators.

PedagogyoftheCourseWork:80%Classroomteaching.

20%Internal AssessmentComprisingUnitTest,Assignment and Class room participation.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

Themaximummarksforthepaperwillbe100.Thequestionpaperwillbeof80marksand continuous evaluation20 marks. Time allowed will be3 hours.

Thepaper-settermustputanoteinthequestionpaperinthisregard.

- Thereshallbe**9**questionsinall.
- Thefirstquestion**compulsory**comprising15shortanswertypequestionsspreadoverthewhole syllabus.The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10x2= 20).
- Rest of thepaper shall containfour units.Twoquestionsshall beasked fromeachunit andthe candidates shallbegiveninternal choice.Thecandidates shallattempt onequestion from each

unit.Each question will carry15 marks (15x4=60).

UNIT–I

Financial Markets: Role, Functions and Structure, Types and Distinctive Features of Financial Instruments, Functions of Financial Institutions.

UNIT–II

Risk and Return: Sources and Types ofRisk and Return – Historical and Expected, Measurement of Risk, Efficient Portfolio, Minimum Variance Portfolio, Capital Asset Pricing Model, Efficient Market Hypothesis, Speculation, Arbitrage andHedging– Concept andSignificance.

UNIT–III

IndianFinancialMarkets:
IndianMoneyMarket:Participants,StructureandInstruments,BankingSectorReformssince1991
IndianCapitalMarket:PrimaryandSecondaryMarket,Growth,ProblemsandReformssince1991.
Market for Derivatives – Futures and Options.

UNIT–IV

RoleandStructureofNon-BankingFinancialInstitutions. RBI – Role and Functions, MonetaryPolicy of RBI. SEBI – Role and Functions.

InternationalFinancialMarkets–StructureandSignificance.

EssentialReadings:

1. Bhole, L. M. (2003). *Financial institutions and markets* (4th ed.). New Delhi: Tata McGraw-Hill.
2. Bhole, L. M. (2000). *Indian financial system*. Allahabad: Chugh Publications.
3. Houthakker, H. S., & Williamson, P. J. (1996). *Economics of financial markets*. Oxford: Oxford University Press.
4. Fabozzi, F. J., & Modigliani, F. (2002). *Foundations of financial markets and institutions* (3rd ed.). Pearson Education.

FurtherReadings:

Khan, M. Y. (1996). *Indian financial system.* New York: Tata McGraw-Hill.

Prasad, K. N. (2001). *Development of India's financial system*. New Delhi: Sarup & Sons.

Reserve Bank of India. Various reports.

Government of India. Economic survey (Various issues).

Semester-VII

ECO (HS) DSC21 - INTERNATIONAL TRADE AND TRADING SYSTEMS

Topologyofthecode	Major/Minor
LeveloftheCourse	400
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Course Objective:- This course aims to build a comprehensive understanding of international trade theories, from classical models like Ricardian and Heckscher-Ohlin to modern frameworks involving intra-industry trade and global value chains. It seeks to analyze the effects of various trade policies and the role of international agreements. Finally, the course will explore the interconnections between international factor mobility, trade, and macroeconomic balances.

PedagogyoftheCourseOutcomes:80%Classroomteaching.

20%InternalAssessment.

InstructionsforPaper-setterandcandidates:-

- Themaximummarksforthepaperwillbe 100. Thequestionpaperwillbe of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe**9**questionsinall.

The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20). Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

UNIT – I
International Trade Theory

Trade Based on Absolute Advantage: (Adam Smith), Comparative Advantage (David Ricardo) Advantage and Opportunity Costs (Haberler's theory, Gains from trade under constant cost as well as

increasing costs).

Resources and Trade: Heckscher-Ohlin Model, Leontief Paradox.

Imperfect Competition and International Trade (Intra-industry trade), Trade Based on Dynamic Technological Differences (Technological Gap and Product Cycle Models), Gravity Model of Trade.

The Standard Trade Model: Offer Curves, Terms of Trade, Trade and Economic Growth: Immiserating Growth

UNIT – II
International Trade Policy

Theory of Tariffs: Partial Equilibrium analysis of Tariff (both small country and large country case), General Equilibrium analysis of a Tariff (both small country and large country case). Optimum tariff.

Non-Tariff Barriers and Neo-protectionism.

Economic Integration: Theory of Customs Unions. Static effects (Trade creation and trade diversion). Dynamic effects of custom unions.

International Trade Agreements: Multilateralism: The World Trade Organization (WTO), Regional Trade Agreements (RTAs), Trade, Labour, and Environmental Standards

UNIT – III
The Balance of Payments

Concept and Components of Balance of Payment.

The Price Adjustment Mechanism with Flexible and Fixed Exchange Rates, Marshall-Learner conditions, J-curve effect, Gold Standard (Price-Species Flow Mechanism).

The Income Adjustment Mechanism, Foreign Trade Multiplier.

Open-Economy Macroeconomics and Adjustment Policies: Equilibrium in the Goods Market, in the Money Market and in the Balance of Payments (Mundell-Fleming Model).

UNIT – IV
Foreign Exchange Markets and International Monetary System:

Foreign Exchange Rates, Arbitrage, Spot and Forward Rates, Currency Swaps, Futures and Options,

Foreign Exchange Risks, Hedging and Speculation. Euro currency Markets.

The International Monetary System: Past, Present and Future.

IMF and World Bank: Evolution and current relevance

Recommended Readings:

Bhagwati, J. (1987). International trade: Selected readings. (Vol. 1). MIT Press.

Clement, M.O., Pfister, R. L., & Rothwell, K. J. (1967). Theoretical issues in international economics. Constolete and Company Ltd.

Dunn, R.M., & Mutti, J. H. (Latest Edition). International economics. New Jersey: John Wiley & Sons. Inc.

Jones, R. W., & Kenen, P.B. (1984). Handbook of international economics. Vol. I, II and III. New York: Elsevier.

Krugman, P.R., & Obstfeld, M.(Latest Edition). International economics: Theory and policy. United States: Pearson Publication.

Salvatore, D. (Latest Edition). International economics. New Jersey: John Wiley& Sons Inc. Souderton, B., & Geoffrey R. (1994). International economics. London : McMillan.

Further Readings:

Ellsworth, P T & Clark L. (1975). The International economy. London: McMillan Press.

Mikic, M. (1998). International trade. London: McMillan Press.

Peter H L. (1969). International economics. New Jersey : Princeton University Press. Shenkar, O., Luo, Y., & Chi, T. (2014). International business. London: Routledge.

ECO (HS) DSC 22 - FUNDAMENTAL ISSUES IN POLITICAL ECONOMY

Topologyofthecode	Major/Minor
LeveloftheCourse	400
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Course Objective:-This course provides a critical understanding of political economy, from classical capitalism to modern globalization. It examines Marxian critiques, the state's economic role, and theories of regulation. The curriculum also analyzes the structures and impacts of the neoliberal global order.

PedagogyoftheCourseOutcomes:80%Classroomteaching.

20%InternalAssessment.

InstructionsforPaper-setterandcandidates:-

Themaximummarksforthepaperwillbe 100. Thequestionpaperwillbe of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

1. Thereshallbe**9**questionsinall.

2. The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20). Rest of the paper shall contain four units.

3. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

Unit I

Foundations of Political Economy

Scope and subject matter of political economy; Mode of production and the superstructure; Pre-capitalist modes of production: Primitive communism, Slavery and Feudalism.

Unit II

Capitalist Mode of Production

Core features of the capitalist mode of production**;** Capital accumulation and the reserve army of labour**;** The falling tendency of the rate of profit**;** The nature and dynamics of capitalist crises

Unit III

State, Monopoly Capitalism, and Accumulation

The state and its role in capital accumulation under capitalism; Development of monopoly capitalism: Concentration and centralisation of capital, Corporations, cartels, trusts, and mergers, Banks and financial institutions in monopoly capitalism

Unit IV

Agrarian Questions and Imperialism in 21st Century

The agrarian question in classical and contemporary debates; Capitalism in agriculture Imperialism: Classical theories (Hobson, Lenin, Luxemburg); Contemporary imperialism; Contemporary imperialism and its relationship to the agrarian question

Recommended Readings

Unit I

Lange, O. (1963). *Political Economy*, Vol. I. Chapters 1–2.

Nikitin, P. I. (1983). *The Fundamentals of Political Economy*. Chapter 1.

Unit II

Sweezy, P. M. (1942). *The Theory of Capitalist Development*. Chapters 4, 5, 6, and 8.

Foley, D. K. (1986). Understanding Capital. Cambridge, MA: Harvard University Press. Chpater 4.

Upadhyay, V., & Singh, P. (Eds.). (2021). *Global Political Economy: A Critique of Contemporary Capitalism*. Routledge. Chapters 2 and 4.

Howard, M. E., Howard, M. C., & King, J. E. (1988). *The Political Economy of Marx*. NYU Press. Part IV

Unit III

Sweezy, P. M. (1942). *The Theory of Capitalist Development*. Chapter 13.

Baran, P. A., & Sweezy, P. M. (1966). *Monopoly Capital*. NYU Press. Introduction and Chapter 8.

Upadhyay, V., & Singh, P. (Eds.). (2021). *Global Political Economy: A Critique of Contemporary Capitalism*. Routledge. Chapters 6 and 11.

Fine, B. and Saad-Filho, A. (2016) *Marx's Capital* . London: Pluto Press. Chapter 12.

Unit IV

Kautsky, K. (1988). *The Agrarian Question*, Vol. I. Introduction to the English Edition by Hamza Alavi & Teodor Shanin and Part I.

Patnaik, U. (Ed.). (2007). *The Agrarian Question in Marx and His Successors, Vol. I: The Development of Capitalist Relations in Agriculture*.

Hunt, E. K., & Lautzenheiser, M. (2011). *History of Economic Thought*. Chapter 13.

Amin, S. (2012). "Contemporary Imperialism and the Agrarian Question." *Agrarian South: Journal of Political Economy*, 1(1), 11–26.

Additional Readings

Bernstein, H. (2016). Agrarian questions then and now. In *Agrarian Questions* (pp. 22-59). Routledge.

Dobb, M. (1975). *Theories of Value and Distribution Since Adam Smith*.

Harvey, D. (2010). *A Companion to Marx's 'Capital'*. London: Verso

Heilbroner, R. L. (2011). *The Worldly Philosophers: The Lives, Times, and Ideas of the Great Economic Thinkers*. Simon & Schuster.

Hiferding, R. (2019). *Finance capital: A study in the latest phase of capitalist development*. Routledge.

Lenin, V. I. (1917). *Imperialism, the Highest Stage of Capitalism*.

Marx, K. (1992). *Capital*, Vol. I. Penguin Books.

Patnaik, P. (2014). "Imperialism and the Agrarian Question." *Agrarian South: Journal of Political Economy*, 3(1), 1–15.

Shaikh, A. (2016). *Capitalism: Competition, Conflict, Crises*. Oxford University Press.

ECO (HS) DSC 23–OPTIMISATION TECHNIQUES AND GAME THEORY

Topologyofthecode	Major/Minor
LeveloftheCourse	400
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Objective: The paper aims to make the students familiar with some of the mathematical techniques which have been developed in the last few decades to solve the basic and one of the fundamental problems faced by economic agents, i.e. economizing at a particular point of time or over a period of time.

PedagogyoftheCourseWork:

80%Classroomteaching.
20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

Instructions for Paper-setter and candidates:
• The maximum marks for the paper will be 100.
• The theory question paper will be of 80 marks and computer practical of 20 marks.
• Time allowed in theory paper will be 3 hours.
• The computer practical will be conducted by the teacher concerned in the computer labof the department and marks in practical paper will be awarded in followingcategories:
a) 10 marks to successful execution of the program/assignment on computer usingMatLab. Each student will be given 1 hour to run the given computer program inMatLab;
b) 5 marks to File work on MatLab programming; and
c) 5 marks to viva-voce on practical application of OTGT.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

1. Thereshallbe**9**questionsinall.

2. The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20). Rest of the paper shall contain four units.

3. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

UNIT-I

Topics in optimization: Nonlinear programming and Kuhn-Tucker condition. Theconstraint qualification. Economic Applications: Peak Load pricing and war time rationing.

Topics in linear Programming: Dual Simplex Method, Sensitivity and ParametricAnalysis.

UNIT-II

Dynamic Optimization: A Brief introduction to the calculus of variations (Eulerequation),end point (transversality) conditions. The control problem, Statement and types of control.

Pontryagin's Maximum principle, finite and infinite time horizon, interpretation ofhamiltomian, costate variables, equation of motion; end point conditions. Constructing aphase diagram and its analysis. Economic Applications of Calculus of variations andOptimum Control theory.

UNIT-III

Game Theory*:* Two persons Zero Sum Game, Max-Min Strategy. Non-Zero Sum Game,Static Games in Complete Information, Strongly and Weakly Dominant Strategies, DominantStrategy Equilibrium; Pure and Mixed Strategies;

Concept of Nash Equilibrium-Examplessuch as Prisoner's Dilemma, Battle of Sex, Matching Pennies, Co-ordination Problem andMultiple Equilibria, etc., Focal Points, Dynamic Games in Complete Information, Sub-Gamesand Sub-Game Perfect Nash Equilibrium.

UNIT-IV

Optimization using MATLAB*:* Basic Commands of MATLAB, Solving Matrices using MATLAB, Solving Linear Programming, Non-Linear Programming and Dynamic Optimization problems Using MATLAB. Solution of Games in MATLAB through Linear Programming.

Essential Readings

Chiang, A. C. (1984). *Fundamental methods of mathematical economics*. McGraw-Hill.

Chiang, A. C. (1999). *Elements of dynamic optimization*. Waveland Press.

Dixit, A. (1976). *Optimization in economic theory*. Oxford University Press.

Gibbons, R. (1992). *A primer in game theory*. Pearson Education.

Gilat, A. (2007). *MATLAB: An introduction with applications*. Wiley India.

Gravelle, H., & Rees, R. (1992). *Microeconomics*. Pearson Education.

Kirani, S. Y. (2007). *MATLAB programming*. Prentice Hall of India.

Osborne, M. J. (2004). *An introduction to game theory*. Oxford University Press.

Sundaram, R. K. (1996). *A first course in optimization theory*. Cambridge University Press.

Yang, W. Y., Cao, W., Chung, T.-S., & Morris, J. (2007). *Applied numerical methods using MATLAB*. Wiley India.

Further Readings

Dixit, A., & Skeath, S. (2004). *Games of strategy* (2nd ed.). W. W. Norton & Company.

Intriligator, M. D. (1971). *Mathematical optimization and economic theory*. Prentice Hall.

Koopmans, T. C. (1957). *Three essays on the state of economic science*. McGraw-Hill.

Lancaster, K. (1965). *Mathematical economics*. Macmillan.

Osborne, M. J., & Rubinstein, A. (1994). *A course in game theory*. MIT Press.

Rasmusen, E. (2000). *Games and information: An introduction to game theory* (3rd ed.). Blackwell Publishers.

ECO(HS) RBCRM(A) - RESEARCH METHODOLOGY

Topologyofthecode	Major/Minor
LeveloftheCourse	400
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Course Objective:- This course aims to introduce the philosophy and key approaches of social science research, including quantitative, qualitative, and interdisciplinary methods. It covers the essential steps of the research process, from problem formulation and hypothesis testing to sampling, data collection, and analysis. The course also seeks to familiarize students with major data sources for the Indian economy.

PedagogyoftheCourseWork:

80%Classroomteaching.
20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus. The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

UNIT – I

Introduction to Research Methodology: Meaning and Objectives of Research, Types of Research, Defining the Research Problem.
Inter-disciplinary Research in Social Sciences: Rationale and Approaches.
Social Science Research Process. Selection and Formulation of Research Problem under Quantitative and Qualitative research approach.

UNIT – II

Reviewing the Literature, Using Mendeley in Citing the Literature.

Triangulation: Concept, Rationale and Process.

Meta Analysis: Concept, Rationale and Process.

UNIT – III

Data Collection Techniques: Questionnaire Method: Meaning, Types, Format. Interview Method: Meaning, Types, Process and Rules of Interviews. Observation Method: Meaning and Types.
Measurement and Scaling Techniques: Measurement Scales: Types of Measurement Scales, Types and Components of Measurement.

UNIT – IV

Research Synopsis: Concept, Format and Writing.
Data Sources of Indian Economy: Nature and Sources of Micro and Macro statistical data.

Suggested Readings:

Goon, P: An Outline of Statistical Theory, World Press Pvt. Ltd., Calcutta.

Kumar Pranesh, Singh Daroga, Singh Padam: Handbook of Sampling Methods, Indian and Agricultural Statistics Research Institute, ICAR.

Anderson, T. W.(1952): An Introduction to Multivariate Statistical.

Goode, W J and Hatt, Paul K (1952): Methods in Social Research. McGraw Hill Book Company, New York.

Sukhatme and Sukwatme (1970): Sampling Theory of Surveys with Applications. John Wiley and Sons.

Tripathi, P C. (2008): A Textbook of Research Methodology, Sultan Chand.

Kumar, Ranjit (2009): Research Methodology: Methods and Techniques, New Age International, India.

Creswell, J.W. (1994). Research Design: Qualitative and Quantitative Approaches, London, Sage Publications.

Miah, Abdul Quader. (2016). Applied Statistics for Social and Management Sciences, (1st ed.) Springer, Singapore.

Moser. C.A. and Kalton G. (1979). Survey Methods in Social Investigation, (2nd ed.) The English Language Book Society and Heinemann Educational Books. London.

Delbert C. Miller. Handbook of Research Design and Social Measurement, (5th ed.) Sage Publications, London.

David Nachmias and Chara Nachmias. Research Methods in the Social Sciences. (2nd ed.) St. Martin's Press Inc, New York.

Clarie S.L. Research Methods in Social Relations, Holt Rinehart and Winston. New York.

Pauline V. Young. (2007) Scientific Social Research and Surveys, India. (4th ed.) Asia Publishing House.

William J. Goode and Paul Hatt. (2006) Methods in Social Research, (Ist Indian Reprint). USA, Mc Graw K. Hill Book Company.

Norman K. Denzin, Yvonna Lincoln. (2003). The Landscape of Qualitative Research: Theories and Issues. Sage Publications, New Delhi.

Clive Seale, Giampietro Gobo, Jaber I. Gubrium& David Silverman.(2004) Qualitative Research Practice. Sage Publications, New Delhi.

Dawn Durton. (2000). Research Training for Social Scientists, Sage Publications, New Delhi.

Royle A. Singleton, Jr. and Bruce C. Straits. (1999). Approaches to Social Science Research, Oxford University Press.

Mason, Jennifer. (2002). Qualitative Researching. Sage.

Chandrasekhar C.P.& Jandhyala B.G. Tilak (2001) India's Social-Economic Database – surveys of selected areas, Tulika, New Delhi.

ECO (HS) DSE (A): ECONOMICS OF POPULATION

Topology of the code	**Elective**
Course Duration/LTP (Lectures, Tutorials, Practicals)	**L= 60 hours (4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**
Level	**400**

Course Objective: This course aims to equip students with a clear understanding of population dynamics and their impact on economic development. It introduces key population theories, demographic measures, and India's population structure. Students will analyse fertility, mortality, migration, and urbanisation trends, and assess their socio-economic implications. The course also enables learners to critically evaluate India's population policies and demographic challenges.

PedagogyoftheCourseWork:

80%Classroomteaching.
20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

Instructions for Paper-setter and candidates:
The maximum mark for the paper will be 100. The question paper will be of 90 marks and continuous evaluation 10 marks. The time allowed will be 3 hours.

The paper-setter must put a note in the question paper in this regard.
There shall be **9** questions in all
The first question is compulsory comprising 12 short answer-type questions spread over the whole syllabus. The candidates are required to attempt 9 questions. Each question shall be of two marks (9 x 2= 18).
Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 18 marks (18x4=72).

UNIT – I

Introduction to Population Economics
Theories of Population: Malthus, Optimum; Demographic transition theory.
Population and Economic Development: inter-linkages

UNIT – II

Population Structure and Characteristics: Impact of Population growth on the age and gender structure

Economic Composition of Indian Population: Workforce Participation & Occupational Structure

Fertility Mortality: Concepts and Measurement

Infant Mortality in India: Cause and Trends

Facility: Current Status

UNIT – III

Migration: Meaning, Causes and Consequences. Migration in India (Rular and Urban, Male-Female): Causes and Trends

Urbanisation: Meaning, Determinants and Consequences, Trends of Urbanisations in India.

UNIT – IV

Population Policy in India: Features of the Population Policy in the post-independence era.

Suggested Readings:

1. Aggarwal, A. S. N. (1972): *India's Population Problem* Tata McGraw Hills Co., Bombay.
2. Becker, G. (1960): *An Economic Analysis of Fertility in Demographic Change and Economic Change in Developed Countries* (edited by National Bureau of Economic Research), Princeton University Press, Princeton.
3. Boserup, E. (1981): *Population and Technological Change* University of Chicago Press, Chicago.
4. Bouge, D. J. (1971): *Principles of Demography*, John Wiley, New York.
5. Bose, Ashish (1978): *India's Urbanization*, Tata McGraw Hills Publishing Co. Ltd., New Delhi.
6. Coale, A. and E. Hoover (1958): *Population Growth and Economic Development in Low Income Countries: A Case Study of India's Prospects.* Princeton University, Press, Princeton.
7. Chenery, H. and T.N. Srinivasan (eds.) (1989): *Handbook of Development Economics*, Vol. 1 & 2, Elsevier, Amsterdam.
8. Chandna, R. C. (2000): *A Geography of Population: Concepts, Determinants and Patterns*, Kalyani Publishers, New Delhi.
9. Easterlin, R. A. (1968): *Population, Labour Force and Long Swings in Economic Growth*: National Bureau of Economic Research, New York.
10. Ghosh, B. N. (ed.) (1987): *Studies in Population and Economic Development*, Deep and Deep Publications, New Delhi
11. John R. Weeks (2002): *Population: An Introduction to Concepts and Issues* (7th edition), Wadsworth Publishing Company, New York.
12. Leibenstein, H. (1957): *Economic Backwardness and Economic Growth*, New York, John Wiley and Sons
13. Mishra, S.K. &Puri, V.K: *Indian Economy,* Himalaya Publishing House, Mumbai (Latest Edition).

OR

ECO (HS) DSE (B): LABOUR ECONOMICS

Topologyofthecode	Discipline Specific Elective Course
LeveloftheCourse	400
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Objective: This course aims to provide a comprehensive understanding of labour market theories, from classical to contemporary perspectives. It examines the functioning of labour institutions, wage determination, and the impact of legislation and industrial relations. The course also evaluates contemporary challenges in the Indian labour market, including informalisation, gender disparities, and the evolving nature of work.

Outcomes: This course shall enable students to comprehend the theoretical and empirical foundations of labour markets and employment dynamics. It shall provide insights into the functioning of wage determination mechanisms, labour market segmentation, and institutional frameworks governing labour relations. The course shall also equip students to critically analyse contemporary issues in the Indian Labour Market such as informalisation, wage inequality, gender disparities, and structural transformations, thereby preparing them for advanced research and policy analysis in labour and employment economics.

Pedagogy of the Course Work: The students would be evaluated at the end of each semester through subjective-type questions/answers (both short and essay type). The scripts would be evaluated by the teacher/instructor teaching the paper and the grade shall be submitted accordingly.
The course relies on a combination of lectures, solving problems and discussing academic articles or real-life situations. The teacher will assign topics for assignments on contemporary themes and issues from the syllabi.

Instructions For the Paper-Setter and Candidates:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question, which would be compulsory, shall be a short-answer type (word limit 2530 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short-answer type questions. Each short-answer-type question would carry 2 marks (10× 2 = 20).
4. The rest of the paper shall contain 4 units. Each unit shall have two questions, and the candidates

shall be required to attempt one question from each Unit. Each question shall carry 15 marks (15 × 4 = 60).

UNIT-I

Labour Markets and Employment: Nature and characteristics of labour markets in developing countries
like India; Paradigms of labour market analysis – Classical, neo-classical and dualistic economy; Demandfor labour; Supply of labour in relation to growth of labour force; Labour market policies.
Employment and Development Relationship: Poverty and unemployment in developing countries; Unemployment – Concept, Types, and Measurement, particularly in India; Impact of rationalization, technological change and modernization on employment.

UNIT-II

Wage Determination: Classical, Neo-classical and bargaining theories of wage determination; Concepts of minimum wage, living wage and fair wage in theory and practice; Discrimination in labour markets; Wage determination in various sectors – rural, urban, organized, unorganized and informal sectors; Non-wage component of labour remuneration; Inflation-wage relationship at micro and macro levels; Productivity and wage relationship; Analysis of rigidity in labour markets; Asymmetric information and efficiency of labour markets in wage determination; National wage policy.

UNIT-III

Industrial Relations: Theories of Labour Movement – Growth, pattern and structure of labour unions in India, Achievements of labour unions; Causes of industrial disputes, their settlement and prevention mechanism; Role of tripartism; Current trends in collective bargaining.

UNIT-IV

State and Labour: State and Social Security of Labour – Concept of Social Security and its evolution; Social assistance and social insurance; Review and appraisal of State policies with respect to social security and labour welfare in India; Special problems of labour; Receding State and its effect on working of labour markets; Labour market reforms – Exit policy, need for safety nets, measures imparting flexibility in labour markets; Second National Commission on Labour; Globalization and Labour Markets.

Recommended Readings:

Binswanger, H. P. & M. R. Rosenzweig, Contractual Arrangements, Employment, and Wages (eds.) (1984) in Rural Labour Markets in Asia, Yale University Press, New Haven.

Datt, G. (1996), Bargaining Power, Wages and Employment: An Analysis of Agricultural LabourMarkets in India, Sage Publications, New Delhi.

Deshpande, L. K., P. R. Brahmananda, Employment Policy in a Developing Economy, and EAG Robinson (eds.)(1983) Vols. I & II, Macmillan, London.

Deshpande, L. K. and J. C. Sandesara, Wage Policy and Wages Determination in (eds.) (1970) India, Bombay University Press, Bombay.

Deshpande, S. G. Standing and L. K., Labour Flexibility in a Third World Metropolis, Deshpande (1998) Commonwealth Publishers, New Delhi.

Foster, A. D. & M. R. Rosenberg (1993), Information Flows and Discrimination in Labour Markets in Rural Areas in Developing Countries, Annual Conference on Development Economics, World Bank, Washington D C.

Hauseman, S. (1991), Industrial Restructuring with Job Security, Harvard University Press, Cambridge, Mass.

Hicks, J. R. (1932), The Theory of Wages, Clarendon Press, Oxford.

Lester, R. A. (1964), Economics of Labour (2nd edition), Macmillan, New York.

McConnell, C. R. & S. L. Brue (1986), Contemporary Labour Economics, McGraw-Hill, New York.

McCormick, B. & Smith (eds.) (1968), The Labour Market, Penguin, Harmondsworth.

Semester-VIII

ECO (HS) DSC 24: ADVANCED PUBLIC FINANCE

Topologyofthecode	Major/Minor
LeveloftheCourse	400
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Course Objective:This course analyzes government intervention for market failure, optimal taxation, and public expenditure. It explores fiscal sustainability, political economy, and fiscal federalism with a focus on Indian reforms and institutions.

PedagogyoftheCourseWork:

80%Classroomteaching.
20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question, which would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks ($10 \times 2 = 20$).
4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question from each Unit— 4 in all. Each question shall carry 15 marks ($15 \times 4 = 60$).

UNIT – I

Scope of Public Finance, Provision of Private Goods and Public Goods in Partial and General Equilibrium (Pareto's Optimality criteria).

Equity in Distribution. Various approaches to distributive Justice. Public Choice and Fiscal Politics. Alternate Voting rules.

UNIT – II

Various Approaches of Equity in Taxation: Benefit Principle including Lindahl Theory. Ability to Pay Approach.

Incidence analysis of various taxes in product and factor markets.

Effects of Taxation on Work Effort, Savings and Investment.

UNIT - III

Deficit Financing: Concept and its relation with Inflation, Deficit Financing in India.

Issues relating to Public Debt: Debt Burden Analysis and Management of Public Debt, Domar's concept of Debt Sustainability Public Debt in India.

Need for rule based fiscal consolidation. Fiscal Responsibility and Budget management (FRBM) act, 2003. Recent amendments to FRBM act.

UNIT – IV

Theories of Public Expenditure: Wagner's Law and Peacock - Wiseman Hypothesis. Structure and Classification of Public expenditure in India.

Principles of Multiunit Finance. Financial aspects of 73rd and 74th Constitutional Amendments.

Centre – State Financial Relations in India: Assessment of Horizontal and vertical imbalances. Constitutional provisions relating to fiscal and financial powers of the States, FinanceCommissions and their formulae for sharing taxes, Financial aspect of Sarkaria Commission Report.

Essential Readings:

Atkinson, A.B., & Stiglitz, J.E.(2007). *Lectures on public economics*. New Jersey: Princeton University Press.

Bhatia, H. L. (2006). *Public finance*. Vikas Publishing House Private Ltd.

Cullis, J., Jones, P., & Jones, P.R. (2009). *Public finance and public choice: Analytical perspectives*. New York: Oxford University Press.

Musgrave, R.A., & Musgrave, P.B. (1989). *Public finance in theory and practice*. New York: McGraw Hill Book Company.

Rao, M.G., & Sen, T. K. (1996). *Fiscal federalism in India: Theory and practice*. Macmillan India.

Recommended Readings:

Aronson, J.R. (1985). *Public finance*. New York: McGraw-Hill International.

Basu, K., & Maertens, A. (2012). *The new Oxford companion to economics in India*. New York: Oxford University Press.

Brown, C.V., & Jackson, P. M. (1990). *Public sector economics*. United Kingdom: Wiley-Blackwell.

Buchanan, J. M. (1958). *Public principles of public debt*. Homewood, Illinois: Richard D. Irwin.

Buchanan, J. M., & Musgrave, R. A. (1999). *Public finance and public choice: Two contrasting visions of the state*. Cambridge, Massachusetts: MIT Press.

Chelliah, R.J. (1996). *Towards sustainable growth: Essays in fiscal and financial sector reforms in India*. New York: Oxford University Press.

Datt, R., & Sundaram, K.P.M. (2003). *Indian economy*. New Delhi: S. Chand & Company.

Houghton, R. W. (1970). *Public finance: Selected readings*. Penguin.

Ulbrich, H.H. (2013). *Public finance in theory and practice Second edition*. London: Routledge.

Kurian, N. J. (1999). State government finances: A survey of recent trends. *Economic and Political Weekly*, *34* (19), 1115-1134.

ECO (HS) DSC25 / ECO (HS) DSE (E) - ENVIRONMENTAL ECONOMICS

Topologyofthecode	Major/Minor
LeveloftheCourse	450
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Course Objective: This paper aims to make aware students about the importance of environment in economics and vice-versa. It helps them to know the ways of sustaining our natural resources by optimally allocating them for future use.

PedagogyoftheCourseWork:

80%Classroomteaching.
20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.

2. There shall be 9 questions in all.

3. The first question, which would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks ($10 \times 2 = 20$).

4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question from each Unit— 4 in all. Each question shall carry 15 marks ($15 \times 4 = 60$).

UNIT – I

Introduction: Basic concepts of Environment, Economics and Ecology. Relation betweenEnvironment and Economy (inter linkages), the natural environment and the human economy-theneoclassical economics (introductory) and first two laws of thermodynamic under ecologicalperspective.

Pollution and resource degradation as externalities: the open access and the public good problem; common property resources and economic problem.

UNIT- II

Economics of pollution: Damage and abatement cost functions; Marginal decision; Optimum levelof pollution; Property Rights or Coasian Theorem; role of state in environment preservation-reviewof environment legislation in India.

UNIT- III

Cost-Benefit Analysis and Environment: measuring the costs and benefits of environmentprotection. Environment regulations and legislations- Air, Water, Land, chemicals and pesticides;
Understanding Environment Impact Assessment (EIA), Green National Accounts andEnvironmental Accounting.

UNIT- IV

Global Environmental Pollution: acid rain, ozone depletion and global warming (causes,consequences and solutions or international responses); Earth Summits 1992 and 2002.

Economic growth and development; the environmental Kuznets curve; Economics of sustainable development; Trade, development and environment under WTO regime.

Essential Readings

Eugine, T. (2005). *Environmental Economics*. Delhi: Varinda Publications.

Goodstein, E.S. (2002). *Economics and the Environment* (3rd ed.). John Wiley & Sons. Inc.

Hanley, N., Shorgen, J. F. & Ben, W. (2001). *Introduction to Environmental Economics*. Oxford University Press.

Hussen, A. (2004). *Principles of Environmental Economics* (2nd ed.). London & New York: Routledge, Taylore & Francis Group.

Kolstad. C.D. (2011). *Intermediate Environmental Economics(second edition).*Oxford University Press Inc.

Nandimath, O.V. (2009). *Handbook of Environmental Decision Making in India*. Oxford University Press.

Further Readings

Bhattachrya, R. N. (2001). *Environmental Economics-An Indian Perspective*. Oxford University Press.

Biswas, M. (2007). *Environmental Economics*. New Delhi: Mittal Publications.

Karpagam, M. (1991). *Environmental Economics*. New Delhi: Sterling Publishers.

Sankar, U. (2001). *Environmental Economics*. Oxford University Press.

Singh, K. & Shishodia, A. (2007). *Environmental Economics: Theory and Applications*. Sage Publication.

Tietenberg, T. (2003). *Environmental and Natural Resource Economics,*(6th ed.). Pearson Education.

ECO (HS) DSC26 / ECO (HS) DSE (F) - NATIONAL INCOME ACCOUNTING

Topologyofthecode	Major/Minor
LeveloftheCourse	400
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Course Objective:-This course aims to establish a foundational understanding of national income concepts, accounting frameworks, and measurement methods. It seeks to explain the circular flow of income and the principles of social accounting, including double-entry systems and sectoral accounts. The curriculum is designed to develop practical skills in computing national income and constructing basic Input-Output Tables and Social Accounting Matrices.

PedagogyoftheCourse:80%Classroomteaching.

20%InternalAssessment.

InstructionsforPaper-setterandcandidates:-

- Themaximummarksforthepaperwillbe 100. Thequestionpaperwillbe of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe**9**questionsinall.

The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20). Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

UNIT I

Basic Concepts and Definitions: GDP, GNP, NNP, National Income at factor cost, Personal Income, Disposable Income, Per Capita Income.

Circular Flow of Income: Two-sector, three-sector, and four-sector models.

Measurement Methods:

Income Method – factor payments, mixed income, transfer payments.

Expenditure Method – private consumption, government expenditure, investment, net exports.

Value Added Method – sectoral contribution, intermediate consumption, problem of double counting.

Difficulties in Measurement: Conceptual, statistical, and coverage issues.

UNIT II

Concept of Social Accounting: Need for comprehensive income and expenditure recording.

Double-Entry System: Principles of double-entry book-keeping applied to macroeconomic accounts.

Sectoral Accounts: Households, Government, Corporates, Rest of the World.

Social Accounting Matrices (Introduction): Basic structure, flows between production, income, and expenditure accounts.

UNIT III

Practical Exercises:

Computation of National Income using all three methods (Income, Expenditure, and Value Added).

Numerical problems involving depreciation, indirect taxes, subsidies, mixed incomes, imputed rent, stock changes, and factor payments.

Numerical exercises on sectoral balances, disposable income, and saving–investment identity.

Practice on Social Accounts: Preparation of simple transaction tables from given data.

UNIT IV

Input–Output Tables:

Structure and components of I–O table (intermediate consumption, primary inputs, final demand).

Leontief's model and assumptions, open and closed models.

Construction of I–O Tables: Steps in building technology matrix, use coefficients, and transaction matrix.

Supply–Use Tables vs Input–Output Tables: Conceptual differences, purpose, and conversion from supply–use to symmetric I–O tables.

Social Accounting Matrix (SAM): Construction methodology, integration with I–O tables, policy applications.

Suggested Readings

1. CSO/NSSO Reports: National Accounts Statistics (latest)
2. Ruggles & Ruggles – National Income Accounts and Economic Analysis
3. D.N. Dwivedi – Macroeconomics: Theory and Policy
4. W. Leontief – Input–Output Economics
5. Richard Stone – Social Accounting Matrices and Their Use

ECO (HS)RBCRM(B): RESEARCH PUBLICATIONS AND ETHICS

Topology of the code	**Major**
Course Duration/LTP (Lectures, Tutorials, Practicals)	**L= 60 hours (4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 90 marks + Internal Assessment 10 marks**
Time	**3 hours**
Level	**400**

Course Objective : : This course aims to develop an understanding of ethical issues related to research. Workshops have been designed to internalize ethical issues through in-depth teaching and discussion.

PedagogyoftheCourseOutcomes:80%Classroomteaching.

20%InternalAssessment.

InstructionsforPaper-setterandcandidates:-

- Themaximummarksforthepaperwillbe 100. Thequestionpaperwillbe of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe**9**questionsinall.

The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20). Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

UNIT- I

Introduction toEthics, concept, scope; Meaning and objectives of research ethics; Academic integrity and honesty in higher education

UNIT- II

Research misconduct: meaning, types, and implications; Fabrication and falsification of data; Data manipulation and data tampering.
Scientific Reading, Cite and Write

UNIT-III

Purpose and importance of citation and referencing; Overview of major style manuals: APA and MLA; In-text citation and reference list preparation; Footnotes, endnotes, and bibliographies
Introduction to Reference Management Tools (RMT)

UNIT-IV

Concept and types of plagiarism, Institutional and legal implications of plagiarism, Features and functionalities of anti-plagiarism software
Detection of plagiarism using online tools, Limitations and ethical use of plagiarism-detection software

Suggested Readings:

1. Bird, A (2006) *Philosophy of Science*, Routledge
2. Israel M. and 1. Hay (2006). *Research Ethics in Social Sciences*, Sage
3. Macintyre, Alasdair (1967) *A Short History of Ethics,* London

ECO (HS) DSE (C) – ECONOMICS OF MONEY AND BANKING

Topologyofthecode	Elective
LeveloftheCourse	400
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Objectives: Money and banking constitute important components towards understanding of economics. A clear understanding of the operations of money and banking and their interaction with the rest of the economy is essential to realize how monetary forces operate through a multitude of channels – market, non-market, institutions and among others, the state. The operation of financial markets and their regulations are to be studied to appreciate their key-role in an economy, especially after the far reaching banking and financial sector reforms in India and elsewhere. The present course is designed to acquaint the students fully with the changing role of financial institutions in the process of growth and development. Accordingly, the paper on 'Economics of Money and Banking' is an optimal integration of monetary theory, banking and non-banking financial institutions, which combine with itself a systematic discussion of the theory, institutions and policy with special reference to India.

PedagogyoftheCourseWork:

80%Classroomteaching.
20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

Money Supply: Theoretical and Empirical Attempts to Define Money. Balance sheet of a Central Bank and Components of Money Supply. Money Creation by the Banking System. High Powered

Money and Money Multiplier. Measures of Money Supply and liquidity in India.
Demand for Money: Interest sensitivity of demand for money- A review of Classical, Keynesian and Monetarist theories of demand for Money. Empirical attempts to estimate demand for money function.

UNIT-II

Term Structure of Interest Rates: The Expectations Theory, Liquidity Premium Theory, Market Segmented Theory, Preferred Habitat Theory, Structure of Interest Rates in India.

Monetary Policy: Targets, Goals and the Trade Offs among Alternate Goals, Lags in Operation, Keynesian and Monetarist Views on Transmission Mechanism, Rules vs. Discretion.

UNIT-III

Central Banking: Main Functions, Policy Tools, Recent Monetary Policy of RBI, money aggregates targeting, interest targeting and inflation targeting approaches of RBI, Autonomy of RBI.

Commercial Banking: Types of commercial banks in India. Banking Sector Reforms, Major Developments in Commercial Banking in India since reforms (including mergers).

UNIT-IV

Non-Banking Financial Institutions: Role, Growth and Structure of Non-Banking Financial Institutions (NBFIs) in India. Types and Control of Non-Banking Financial Companies (NBFCs).
International Monetary System: IMF as provider of International Liquidity. European Monetary System–Features of the Maastricht Treaty and Post Maastricht Treaty Developments. Constituents of international money and capital markets.

Recommended Readings:

Bhole, L.M. (2016). Financial Institutions and Markets. Tata McGraw Hill.

Bofinger, P. (2001). Monetary Policy: Goals, Institutions, Strategy and Instruments. Oxford University Press.

Dennis Geoggrey, E.J. (1981). Monetary Economics. Longman.

Desai, V. (2019). The Indian Financial System and development. Himalaya Publishing House.

Diulio, A. E. (1987). Theory and Problems of Money and Banking. International Edition (Schaum's Outline Series).

Fabozzi, F. J. & Franco (2009). Capital Markets Institutions and Instruments, (4th ed.). New Jersey: Prentice Hall, Modigliani.

Gordon, E. & Natarajan, K. (2019), Financial Markets & Services. Himalaya Publishing House.

Howells, P. & Bain, K. (2002). The Economics of Money, Banking and Finance. Prentice Hall, Pearson Education Ltd.

Khan, M.Y. (2017). Indian Financial System. Tata Mcgraw Hill.

Kulkarni, G. (1999). Modern Monetary Theory. New Delhi: Macmillan.

Mankiw, N.G. (2008). Macro-economics. New York: Macmillan Worth Publishers.

Mohan, R. (2009). Monetary Policy in a Globalized Economy – A Practitioner's View. Oxford University Press. Government of India Economic Survey (Recent Issues).

Reserve Bank of India Various Reports.

OR

ECO (HS) DSE (D) – HEALTH ECONOMICS

Topologyofthecode	Elective
LeveloftheCourse	400
Course Duration /LTP (Lectures,Tutorials,Practical)	L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0
Credits	4
Max.Marks	100Marks=Theory80marks+InternalAssessment 20 marks
Time	3hours

Course Objectives: The course aims to develop a comprehensive understanding of health economics by integrating principles of planning, management, and economic analysis with the structure and functioning of India's health system. It seeks to equip students with analytical tools to evaluate health demand and supply, health financing, insurance markets, and economic efficiency, enabling informed assessment of health policies, programmes, and resource allocation decisions.

PedagogyoftheCourseOutcomes:80%Classroomteaching.

20%InternalAssessment.

InstructionsforPaper-setterandcandidates:-

- Themaximummarksforthepaperwillbe 100. Thequestionpaperwillbe of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe**9**questionsinall.

The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20). Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

UNIT I

Concept and Meaning of Planning, Health Planning, Health Needs and Demands, Resources, Objectives, Targets and Goals, Plan, Pre-Planning, Planning Cycle and its Steps.

Concept of Management, Management Methods and Techniques. Methods based on Behavioural Sciences, Quantitative Methods, Personal, Financial and Material Management.

UNIT II

Health Planning in India, Various Committees (from Bhore Committee to Health for All 2000), Planning Commission, Health Sector Plans, Investments and Achievements during the Five Year Plans,.

Health Systems in India at the Centre, the State and the District Level.

UNIT III

The Relevance of Health Economics, Key Economic Concepts (Elasticity, Marginal Analysis, Opportunity Cost, Efficiency, Equity). Supply and Demand – The Law of demand, Price Elasticity of Demand, The Law of Supply, Equilibrium, Consumer Theory, Production and Market Supply in different market conditions.

Economic Efficiency and Basic Principle of Cost Benefit Analysis, Cost Effectiveness Analysis, Cost Utility Analysis – QALY & DALY.

UNIT IV

Demand-Side Considerations, Demand for Health, The Production of Health, The Demand for Medical Care, Measuring Demand. The Market for Health Insurance, Health Insurance and Market Failure. Supply-Side Considerations, The Market for Health Care Professionals - The Theory of Labour Markets, Input Pricing, Demand for Inputs, Human Capital Investment. Financing of Health Services in India, National Health Accounts.

Suggested Practical Work: Use of Linear Programming in Industry.

Essential Reading

1. Thomas, Richard K, 2010(2ndedition), Health Services Planning, New York, Kluwer Academic Publishers.
2. Folland Sherman, Goodman Allen, 2008 (5th edition), The Economics of Health and Health Care New Delhi, Pearson (Indian Edition).
3. Henderson James W. 2005(3rd edition), Health Economics and Policy, New Delhi, South-Western a part of The Thomson Corporation (Indian Edition)
4. Feldstein Paul J. (2009), Economics of Health Care, DelmerCengage (Indian Edition)
5. Dewar Diane M. 2010 (1st edition), Essentials of Health Economics, Sudbury, Jones and Barlett Publishers.
6. Novick L.F., Morrow C.B., Mays G.P., 2008(2nd edition), Public Health Administration, Principles for Population Based Management, Sudbury, Jones and Barlett Publishers.
7. Whittington Ruth, 2008(1st edition), Introduction to Health Economics: A Beginners Guide, UK, Greenflint.

Reference Books

1. Fallon L.F., Zgodzinski E. J., 2012(3rd edition), Essentials of Public Health Management, Sudbury, Jones and Barlett Publishers.
2. Drummond, M.F., Sculpher, M.J., Torrance, G.W. 2005(1st edition), Methods for the economic evaluation of health care programmes, New York, Oxford University Press
3. Devi V.Renuka, JhanM.Gowhar, 2011(1st edition), Health Economics : Issues and Challenges, New Delhi, Deep and Deep Publications.
4. Issel, L.M., 2009 (2nd edition), Health Program Planning and Evaluation: A practical systematic approach for community health, Sudbury, Jones and Barlett Publishers
5. Gupta, J.P., Sood, A.K., 2005(4th edition), Contemporary public health: policy, planning, management, New Delhi, Apothecaries Foundation.

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